CHAPTER 6

Lower Limb

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CONTENTS

RADIOGRAPHIC ANATOMY

Distal Lower Limb

The bones of the distal lower limb are divided into the foot, lower leg, and distal femur (Fig. 6.1). The ankle and knee joints are also discussed in this chapter. The proximal femur and the hip are included in Chapter 7, along with the pelvic girdle.

FOOT

The bones of the foot are fundamentally similar to the bones of the hand and wrist, which are described in Chapter 4.

The 26 bones of one foot (Fig. 6.2) are divided into three groups as follows:

1. Phalanges (toes or digits)	14
2. Metatarsals (instep)	5
3. Tarsals	7
Total	26

Phalanges—Toes (Digits)

The most distal bones of the foot are the **phalanges,** which make up the toes, or digits. The five digits of each foot are numbered 1 through 5, starting on the medial or big-toe side of the foot. The large toe, or first digit, has only two phalanges, similar to the thumb: the **proximal phalanx** and the **distal phalanx.** Each of the second, third, fourth, and fifth digits has a **middle phalanx,** in addition to a proximal phalanx and a distal phalanx. Because the first digit has two phalanges and digits 2 through 5 have three phalanges apiece, **14 phalanges** are found in each foot.

Similarities to the hand are obvious because there are also 14 phalanges in each hand. However, two noticeable differences exist: the phalanges of the foot are smaller, and their movements are more limited than those of the phalanges of the hand.

When any of the bones or joints of the foot are described, the specific digit and foot should also be identified. For example, referring to the "distal phalanx of the first digit of the right foot" would leave no doubt as to which bone is being described.

The distal phalanges of the second through fifth toes are very small and may be difficult to identify as separate bones on a radiograph.

Metatarsals

The five bones of the instep are the **metatarsal** bones. These are numbered along with the digits, with number 1 on the medial side and number 5 on the lateral side.

Each of the metatarsals consists of three parts. The small, rounded distal part of each metatarsal is the **head.** The centrally located, long, slender portion is termed the **body** (shaft). The expanded, proximal end of each metatarsal is the **base.**

The **base of the fifth metatarsal** is expanded laterally into a prominent rough **tuberosity,** which provides for the attachment of a tendon. The proximal portion of the fifth metatarsal, including this tuberosity, is readily visible on radiographs and is a **common trauma site** for the foot; this area must be well visualized on radiographs.

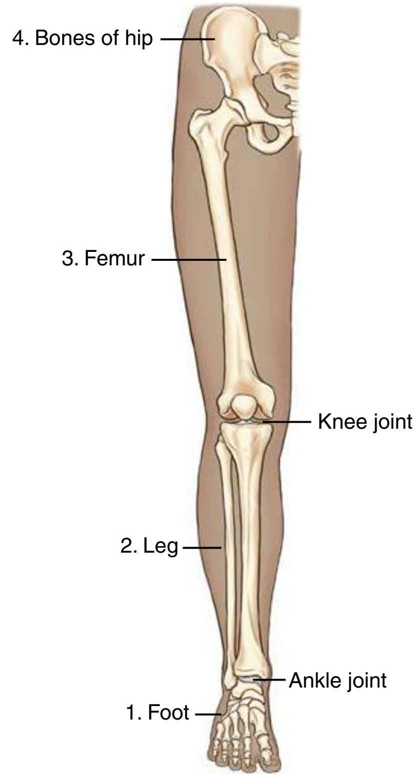


Fig 6.1 Lower limb.

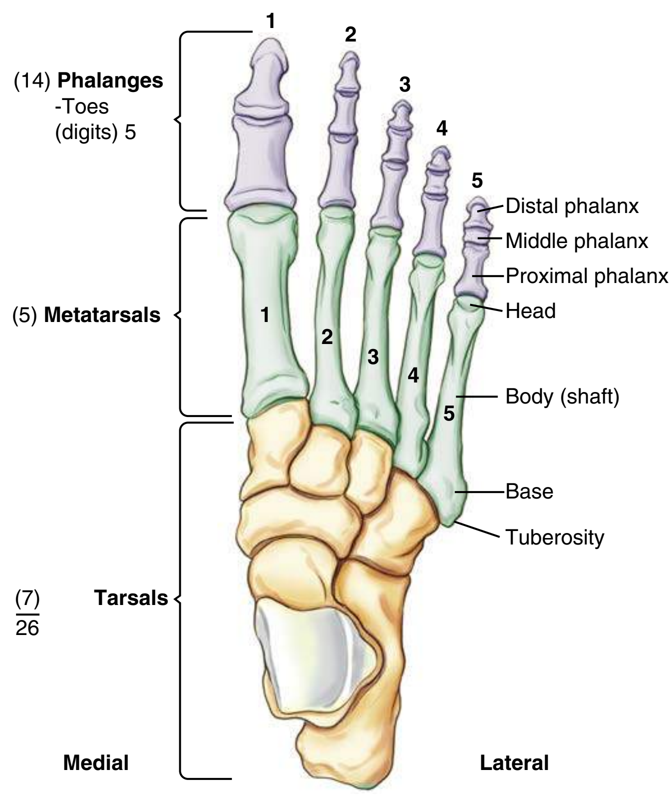


Fig 6.2 Bones of foot.

Joints of Phalanges (Digits) and Metatarsals

Joints of Digits The joints or articulations of the digits of the foot are important to identify because fractures may involve the joint surfaces. Each joint of the foot has a name derived from the two bones on either side of that joint. Between the proximal and distal phalanges of the first digit is the **interphalangeal (IP) joint.**

Because digits 2 through 5 each comprise three bones, these digits also have two joints each. Between the middle and distal phalanges is the **distal interphalangeal (DIP) joint.** Between the proximal and middle phalanges is the **proximal interphalangeal (PIP) joint.**

Joints of Metatarsals Each of the joints at the head of the metatarsal is a **metatarsophalangeal (MTP) joint,** and each of the joints at the base of the metatarsal is a **tarsometatarsal (TMT) joint.** The base of the third metatarsal or the third tarsometatarsal joint is important because this is the centering point or the central ray (CR) location for anteroposterior (AP) and oblique foot projections.

When joints of the foot are described, the name of the joint should be stated first, followed by the digit or metatarsal, and finally the foot. For example, an injury or fracture may be described as near the DIP joint of the fifth digit of the left foot.

Sesamoid Bones Several small, detached bones, called **sesamoid** bones, often are found in the feet and hands. These extra bones, which are embedded in certain tendons, are often present near various joints. In the upper limbs, sesamoid bones are quite small and most often are found on the palmar surface near the metacarpophalangeal joints or occasionally at the interphalangeal joint of the thumb.

In the **lower limbs,** sesamoid bones tend to be larger and more significant radiographically. The largest sesamoid bone in the body is the *patella,* or *kneecap,* as described later in this chapter. The sesamoid bones illustrated in Figs. 6.3 and 6.4 are almost always present on the posterior or **plantar surface at the head of the first metatarsal** near the first MTP joint. Specifically, the sesamoid bone on the medial side of the lower limb is termed the **tibial** sesamoid and the lateral is the **fibular** sesamoid bone. Sesamoid bones also may be found near other joints of the foot. Sesamoid bones are important radiographically because fracturing these small bones is possible. Because of their plantar location, such fractures can be quite painful and may cause discomfort when weight is placed on that foot. Special tangential projections may be necessary to demonstrate a fracture of a sesamoid bone, as shown later in this chapter (p. 231).

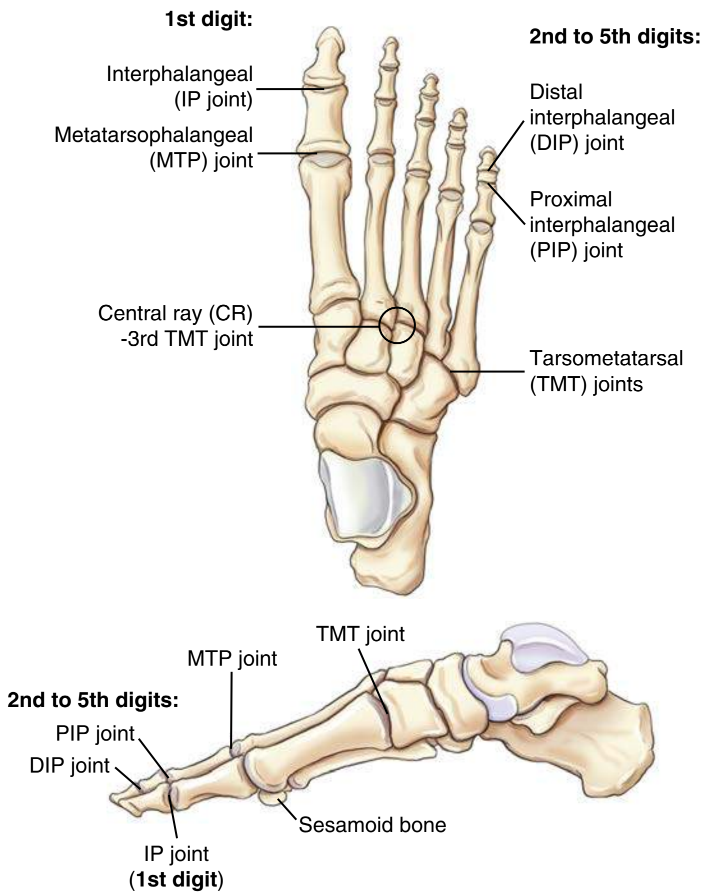


Fig 6.3 Joints of right foot.

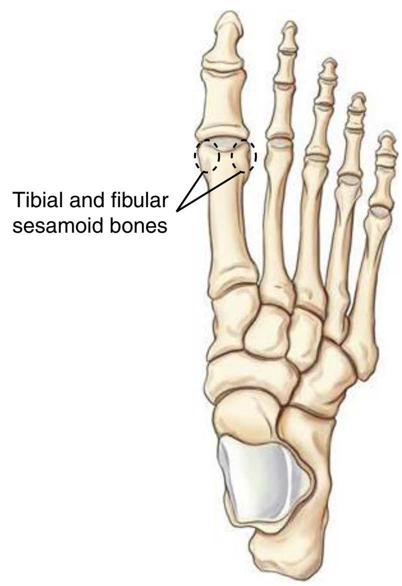


Fig 6.4 Sesamoid bones.

Tarsals

The seven large bones of the proximal foot are called *tarsal bones* (Fig. 6.5). The names of the tarsals can be remembered with the aid of a mnemonic: **C**ome **t**o **C**olorado (the) **n**ext **3** **C**hristmases.

(1) **C**ome	**C**alcaneus (os calcis)
(2) **T**o	**T**alus (astragalus)
(3) **C**olorado	**C**uboid
(4) **N**ext	**N**avicular (scaphoid)
(5, 6, 7) **3** **C**hristmases	First, second, and third **c**uneiforms

The calcaneus, talus, and navicular bones are sometimes known by alternative names: the *os calcis, astragalus,* and *scaphoid*. However, correct usage dictates that the tarsal bone of the foot should be called the *navicular,* and the carpal bone of the wrist, which has a similar shape, should be called the *scaphoid.* (The carpal bone more often has been called the *navicular* rather than the preferred *scaphoid.*)

Similarities to the upper limb are less obvious with the tarsals in that there are only **seven tarsal bones,** compared with **eight carpal bones** in the wrist. Also, the tarsals are larger and less mobile because they provide a basis of support for the body in an erect position, compared with the more mobile carpals of the hand and wrist.

The seven tarsal bones sometimes are referred to as the *ankle bones,* although only one of the tarsals, the talus, is directly involved in the ankle joint. Each of these tarsals is described individually, along with a list of the bones with which they articulate.

Calcaneus The largest and strongest bone of the foot is the *calcaneus (kal-kay′-ne-us).* The posterior portion is often called the *heel bone.* The most posterior-inferior part of the calcaneus contains a process called the tuberosity. The tuberosity can be a common site for bone spurs, which are sharp outgrowths of bone that can be painful on weight bearing.

Certain large tendons, the largest of which is the Achilles tendon, are attached to this rough and striated process, which at its widest point includes two small, rounded processes. The largest of these is labeled the lateral process. The medial process is smaller and less pronounced.

Another ridge of bone that varies in size and shape and is visualized laterally on an axial projection is the peroneal trochlea *(per″-o-ne′-al trok′-le-ah).* Sometimes, in general, this is also called the *trochlear process.* On the medial proximal aspect is a larger, more prominent bony process called the *sustentaculum (sus″-ten-tak′-u-lum) tali,* which literally means a support for the talus.

Articulations The calcaneus articulates with **two** bones: anteriorly with the **cuboid** and superiorly with the **talus.** The superior articulation with the talus forms the important **subtalar** (talocalcaneal) joint. Three specific articular facets appear at this joint with the talus through which the weight of the body is transmitted to the ground in an erect position: the larger **posterior articular facet** and the smaller **anterior** and **middle articular facets.** The middle articular facet is the superior portion of the prominent sustentaculum tali, which provides medial support for this important weight-bearing joint.

The deep depression between the posterior and middle articular facets is called the **calcaneal sulcus** (Fig. 6.6). This depression, combined with a similar groove or depression of the talus, forms an opening for certain ligaments to pass through. This opening in the middle of the subtalar joint is the **sinus tarsi,** or tarsal sinus (Fig. 6.7).

Talus The talus, the second largest tarsal bone, is located between the lower leg and the calcaneus. The weight of the body is transmitted by this bone through the important ankle and talocalcaneal joints.

Articulations The talus articulates with **four** bones: superiorly with the **tibia** and **fibula,** inferiorly with the **calcaneus,** and anteriorly with the **navicular.**

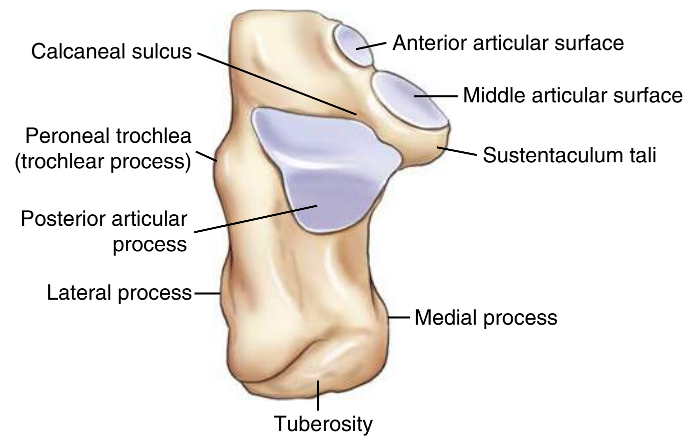


Fig 6.6 Right calcaneus (superior or proximal surface).

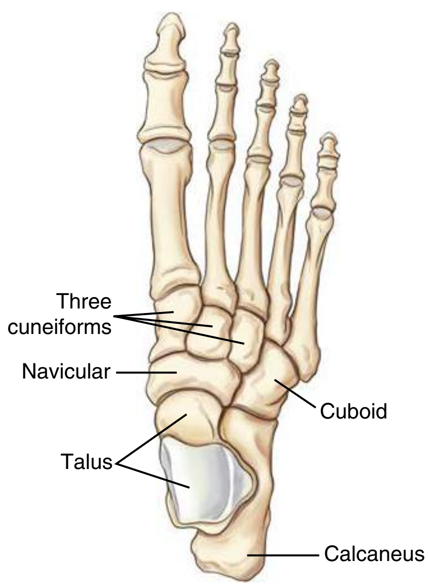


Fig 6.5 Tarsals (7).

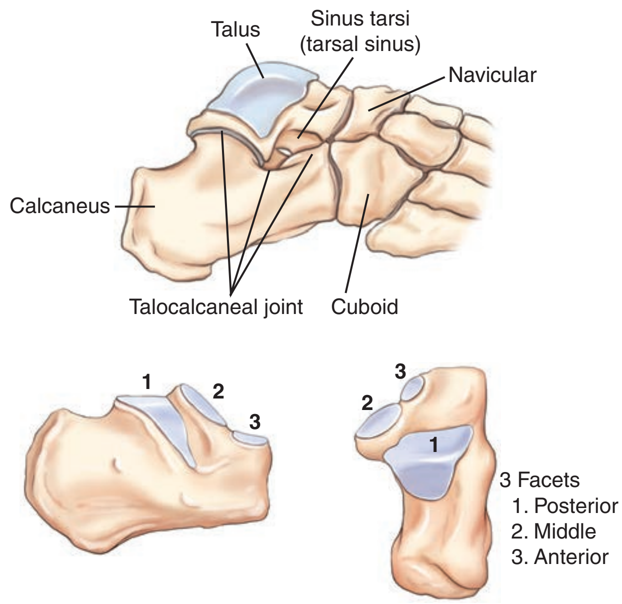


Fig 6.7 Calcaneus and talus (with ankle and subtalar joints).

Navicular

The navicular is a flattened, oval bone located on the medial side of the foot between the talus and the three cuneiforms.

Articulations The **navicular** articulates with **five** bones: posteriorly with the talus, laterally with the cuboid, and anteriorly with the three cuneiforms (Fig. 6.8).

Cuneiforms The three cuneiforms (meaning "wedge shaped") are located on the medial and mid aspects of the foot between the first three metatarsals distally and the navicular proximally. The largest cuneiform, which articulates with the first metatarsal, is the **medial** (first) cuneiform. The **intermediate** (second) cuneiform, which articulates with the second metatarsal, is the smallest of the cuneiforms. The **lateral** (third) cuneiform articulates with the third metatarsal distally and with the cuboid laterally. All three cuneiforms articulate with the navicular proximally.

Articulations The **medial cuneiform** articulates with **four** bones: the navicular proximally, the first and second metatarsals distally, and the intermediate cuneiform laterally.

The **intermediate cuneiform** also articulates with **four** bones: the navicular proximally, the second metatarsal distally, and the medial and lateral cuneiforms on each side.

The **lateral cuneiform** articulates with **six** bones: the navicular proximally; the second, third, and fourth metatarsals distally; the intermediate cuneiform medially; and the cuboid laterally.

Cuboid The cuboid is located on the lateral aspect of the foot, distal to the calcaneus and proximal to the fourth and fifth metatarsals.

Articulations The **cuboid** articulates with **five** bones: the calcaneus proximally, the lateral cuneiform and navicular medially, and the fourth and fifth metatarsals distally.

Arches of Foot

Longitudinal Arch The bones of the foot are arranged in **longitudinal** and **transverse arches,** providing a strong, shock-absorbing support for the weight of the body. The springy, longitudinal arch comprises a medial and a lateral component, with most of the arch located on the medial and mid aspects of the foot.

Transverse Arch The transverse arch is located primarily along the plantar surface of the distal tarsals and the tarsometatarsal joints. The transverse arch is primarily made up of the wedge-shaped cuneiforms, especially the smaller second and third cuneiforms, in combination with the larger first cuneiform and the cuboid (Fig. 6.9).

Box 6.1 presents a summary of the tarsals and articulating bones.

BOX 6.1 SUMMARY OF TARSALS AND ARTICULATING BONES

CALCANEUS (2)[a]
1. Cuboid
2. Talus

TALUS (4)
1. Tibia and fibula
2. Calcaneus
3. Navicular

NAVICULAR (5)
1. Talus
2. Cuboid
3. Three cuneiforms

MEDIAL CUNEIFORM (4)
1. Navicular
2. First and second metatarsals
3. Intermediate cuneiform

INTERMEDIATE CUNEIFORM (4)
1. Navicular
2. Second metatarsal
3. Medial and lateral cuneiforms

LATERAL CUNEIFORM (6)
1. Navicular
2. Second, third, and fourth metatarsals
3. Intermediate cuneiform
4. Cuboid

CUBOID (5)
1. Calcaneus
2. Lateral cuneiform
3. Navicular
4. Fourth and fifth metatarsals

[a] Numbers in parentheses indicate total number of bones with which each of these tarsals articulates.

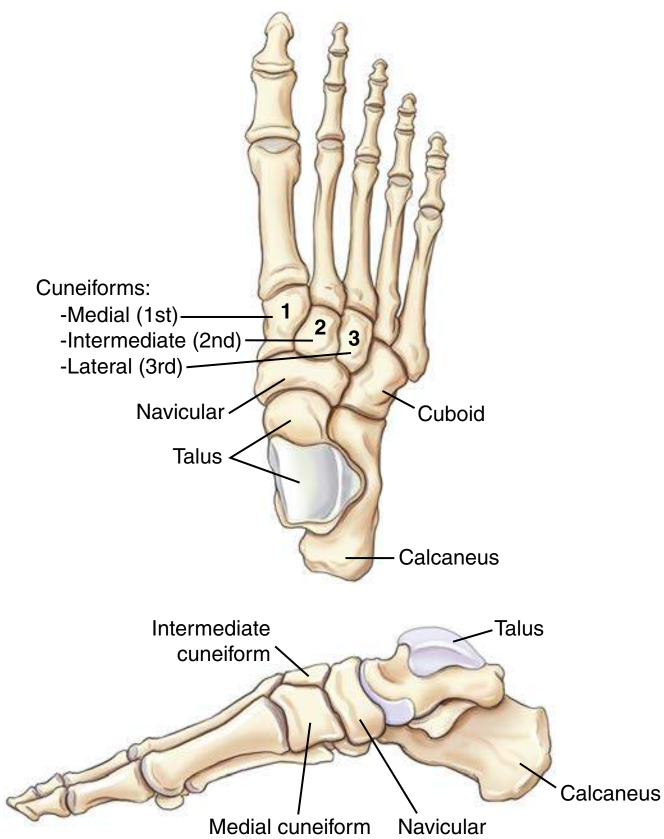


Fig 6.8 Navicular, cuneiforms (3), and cuboid.

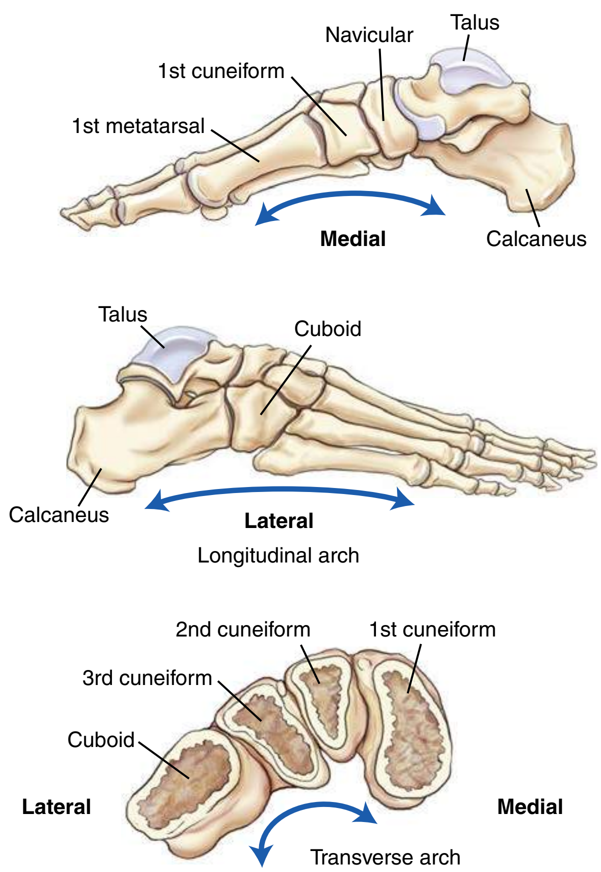


Fig 6.9 Arches and tarsal relationships.

ANKLE JOINT

Frontal View

The **ankle joint** is formed by three bones: the two long bones of the lower leg, the **tibia** and **fibula,** and one tarsal bone, the **talus.** The expanded distal end of the slender fibula, which extends well down alongside the talus, is called the **lateral malleolus.**

The distal end of the larger and stronger tibia has a broad articular surface for articulation with the similarly shaped broad upper surface of the talus. The medial elongated process of the tibia that extends down alongside the medial talus is called the **medial malleolus.**

The inferior portions of the tibia and fibula form a deep "socket," or three-sided opening, called a **mortise,** into which the superior talus fits. However, the entire three-part joint space of the ankle mortise is **not seen** on a true frontal view (AP projection) because of overlapping of portions of the distal fibula and tibia by the talus. This overlapping is caused by the more posterior position of the distal fibula, as is shown on these drawings. A 15° internally rotated AP oblique projection, called the **mortise position,**[1] is performed (see Fig. 6.15) to demonstrate the mortise of the joint, which should have an even space over the entire talar surface.

The **anterior tubercle** is an expanded process at the distal anterior and lateral tibia that has been shown to articulate with the superolateral talus, while partially overlapping the fibula anteriorly (Figs. 6.10 and 6.11).

The **distal tibial joint surface** that forms the roof of the ankle mortise joint is called the **tibial plafond** (ceiling). Certain types of fractures of the ankle in children and youth involve the distal tibial epiphysis and the tibial plafond.

Lateral View

The ankle joint, seen in a true lateral position in Fig. 6.11, demonstrates that the **distal fibula is located about ⅜ inch (1 cm) posterior in relation to the distal tibia.** This relationship becomes important in evaluation for a **true lateral** radiograph of the lower leg, ankle, or foot. A common mistake in positioning a lateral ankle is to rotate the ankle slightly so that the medial and lateral malleoli are directly superimposed; however, this results in a partially oblique ankle, as these drawings illustrate. A true lateral requires the **lateral malleolus** to be **about (⅜) inch (1 cm) posterior** to the medial malleolus. The lateral malleolus extends **about ⅜ inch (1 cm) more distal** than its counterpart, the medial malleolus (best seen on frontal view, Fig. 6.10).

Axial View

An axial view of the inferior margin of the distal tibia and fibula is shown in Fig. 6.12; this visualizes an "end-on" view of the ankle joint looking from the bottom up, demonstrating the concave inferior surface of the tibia (tibial plafond). Also demonstrated are the relative positions of the **lateral** and **medial malleoli** of the fibula and tibia. The smaller **fibula** is shown to be **more posterior.** A horizontal plane drawn through the midportions of the two malleoli would be approximately **15° to 20°** from the coronal plane (the true side-to-side plane of the body). This positioning line is termed the **intermalleolar plane**. The lower leg and ankle must be rotated 15° to 20° to bring the intermalleolar plane parallel to the coronal plane. This relationship of the distal tibia and fibula becomes important in positioning for various views of the ankle joint or ankle mortise, as described in the positioning sections of this chapter.

Joint Structure

The ankle joint is a **synovial joint** of the saddle (**sellar) type** with flexion and extension (dorsiflexion and plantar flexion) movements only. This joint requires strong collateral ligaments that extend from the medial and lateral malleoli to the calcaneus and talus. Lateral stress can result in a "sprained" ankle with stretched or torn collateral ligaments and torn muscle tendons leading to an increase in parts of the mortise joint space. AP stress views of the ankle can be performed to evaluate the stability of the mortise joint space.

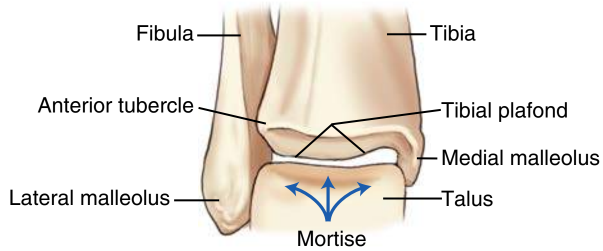


Fig 6.10 Right ankle joint—frontal view.

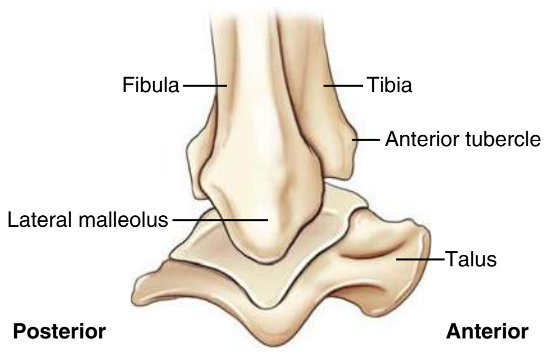


Fig 6.11 Right ankle joint—true lateral view.

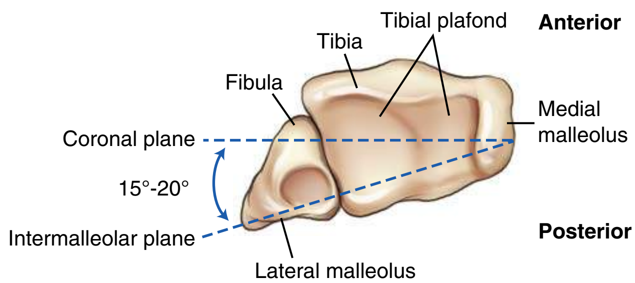


Fig 6.12 Ankle joint—axial view.

REVIEW EXERCISE WITH RADIOGRAPHS

Three common projections of the foot and ankle are shown with labels for an anatomy review of the bones and joints. A good review exercise is to cover up the answers that are listed here and identify all the labeled parts before checking the answers.

Lateral Left Foot (Fig. 6.13)

A. Tibia
B. Calcaneus
C. Tuberosity of calcaneus
D. Cuboid
E. Fifth metatarsal tuberosity
F. Superimposed cuneiforms
G. Navicular
H. Subtalar joint
I. Talus

Oblique Right Foot (Fig. 6.14)

A. Interphalangeal joint of first digit of right foot
B. Proximal phalanx of first digit of right foot
C. Metatarsophalangeal joint of first digit of right foot
D. Head of first metatarsal
E. Body of first metatarsal
F. Base of first metatarsal
G. Second or intermediate cuneiform (partially superimposed over first or medial cuneiform)
H. Navicular
I. Talus
J. Tuberosity of calcaneus
K. Third or lateral cuneiform
L. Cuboid
M. Tuberosity of the base of the fifth metatarsal
N. Fifth metatarsophalangeal joint of right foot
O. Proximal phalanx of fifth digit of right foot

AP Mortise View Right Ankle (Fig. 6.15)

A. Fibula
B. Lateral malleolus
C. "Open" mortise joint of ankle
D. Talus
E. Medial malleolus
F. Tibial epiphyseal plate (epiphyseal fusion site)

Lateral Right Ankle (Fig. 6.16)

A. Fibula
B. Calcaneus
C. Cuboid
D. Tuberosity at base of fifth metatarsal
E. Navicular
F. Talus
G. Sinus tarsi
H. Anterior tubercle
I. Tibia

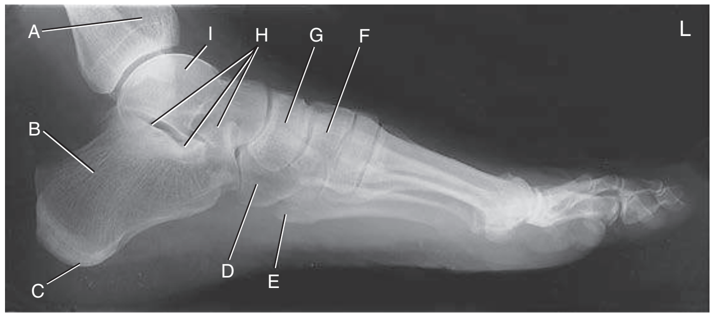


Fig 6.13 Lateral left foot.

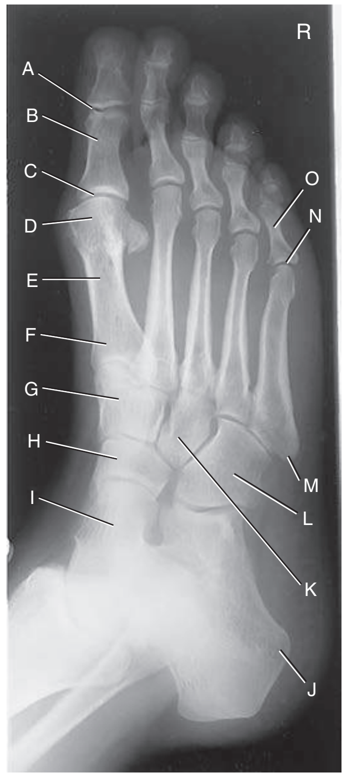


Fig 6.14 AP medial oblique right foot.

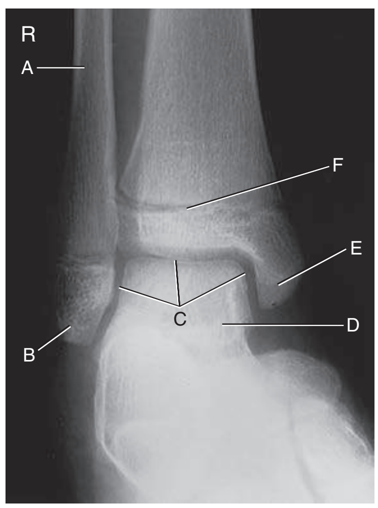


Fig 6.15 AP right ankle (mortise view—15° medial oblique).

Fig 6.16 Lateral right ankle.

LOWER LEG—TIBIA AND FIBULA

The second group of bones of the lower limb to be studied in this chapter consists of the two bones of the lower leg: the **tibia** and **fibula** (Fig. 6.17).

Tibia

The tibia, as one of the larger bones of the body, is the weight-bearing bone of the lower leg. The tibia can be felt easily through the skin in the anteromedial part of the lower leg. It is made up of three parts: the central **body** (shaft) and **two extremities.**

Proximal Extremity The **medial** and **lateral condyles** are the two large processes that make up the medial and lateral aspects of the proximal tibia.

The **intercondylar eminence** (also known as the **tibial spine**) includes two small pointed prominences, called the **medial** and **lateral intercondylar tubercles,** which are located on the superior surface of the tibial head between the two condyles.

The upper articular surface of the condyles includes two smooth concave **articular facets,** commonly called the **tibial plateau,** which articulate with the femur. As can be seen on the lateral view, the **articular facets making up the tibial plateau slope posteriorly from 10° to 20°** in relation to the long axis of the tibia[2] (Fig. 6.18). This is an important anatomic consideration because when an AP knee is positioned, the CR must be angled as needed in relation to the image receptor (IR) and the tabletop to be parallel to the tibial plateau. This CR angle is essential in demonstrating an "open" joint space on an AP knee projection.

The **tibial tuberosity** on the proximal extremity of the tibia is a rough-textured prominence located on the midanterior surface of the tibia just distal to the condyles. This tuberosity is the distal attachment of the patellar tendon, which connects to the large muscle of the anterior thigh. Sometimes in young persons the tibial tuberosity separates from the body of the tibia, a condition known as *Osgood-Schlatter disease* (see Clinical Indications, p. 226).

Body The **body** (shaft) is the long portion of the tibia between the two extremities. Along the anterior surface of the body, extending from the tibial tuberosity to the medial malleolus, is a sharp ridge called the **anterior crest** or **border.** This sharp anterior crest is just under the skin surface and often is referred to as the *shin* or *shin bone.*

Distal Extremity The distal extremity of the tibia is smaller than the proximal extremity and ends in a short pyramid-shaped process called the **medial malleolus,** which is easily palpated on the medial aspect of the ankle.

The lateral aspect of the distal extremity of the tibia forms a flattened, triangular **fibular notch** for articulation with the distal fibula.

Fibula

The smaller fibula is located **laterally** and **posteriorly** to the larger tibia. The fibula articulates with the tibia proximally and the tibia and talus distally. The proximal extremity of the fibula is expanded into a **head,** which articulates with the lateral aspect of the posteroinferior surface of the lateral condyle of the tibia. The extreme proximal aspect of the head is pointed and is known as the **apex** of the head of the fibula. The tapered area just below the head is the **neck** of the fibula.

The **body** (shaft) is the long, slender portion of the fibula between the two extremities. The enlarged distal end of the fibula can be felt as a distinct "bump" on the lateral aspect of the ankle joint and, as described earlier, is called the **lateral malleolus.**

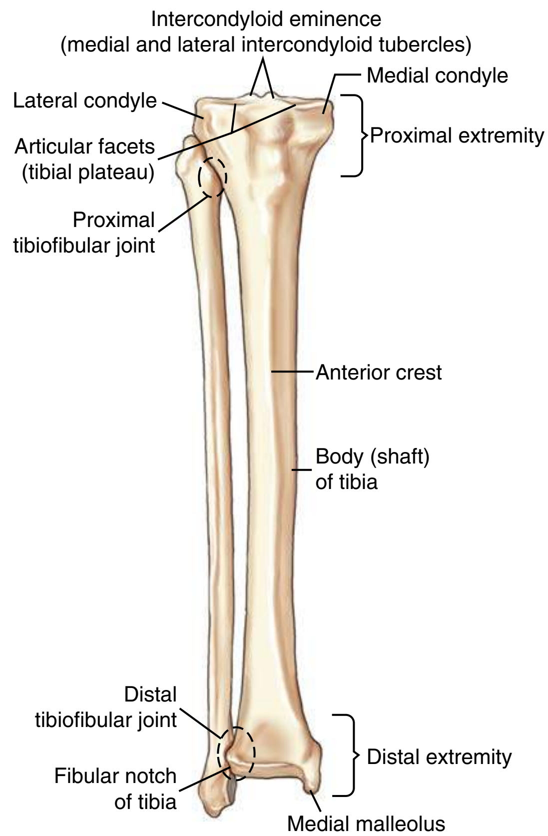


Fig 6.17 Tibia—anterior view.

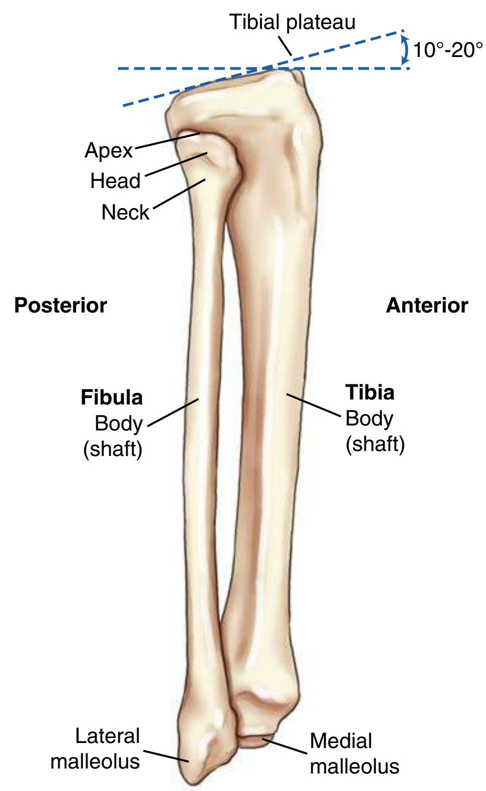


Fig 6.18 Tibia and fibula—lateral view.

Midfemur and Distal Femur—Anterior View Similar to all long bones, the body or shaft of the femur is the slender, elongated portion of the bone. The distal femur viewed anteriorly demonstrates the position of the patella or kneecap (Fig. 6.19). The **patella,** which is the largest sesamoid bone in the body, is located anteriorly to the distal femur. The most distal part of the patella is **superior** or **proximal** to the actual knee joint by approximately ½ inch (1.25 cm) in this position with the lower leg fully extended. This relationship becomes important in positioning for the knee joint.

The **patellar surface** is the smooth, shallow, triangular depression at the distal portion of the anterior femur that extends up under the lower part of the patella, as seen in Fig. 6.19. This depression sometimes is referred to as the **intercondylar sulcus.** (*Sulcus* means a groove or depression.) Some literature also refers to this depression as the **trochlear groove.** (*Trochlea* means pulley or pulley-shaped structure in reference to the medial and lateral condyles.) All three of these terms should be recognized as referring to this smooth, shallow depression.

The patella itself most often is superior to the patellar surface with the leg fully extended. However, as the leg is flexed, the patella, which is attached to large muscle tendons, moves distally or downward over the patellar surface. This is best shown on the lateral knee drawing (see Fig. 6.21).

Midfemur and Distal Femur—Posterior View The posterior view of the distal femur best demonstrates the two large, rounded condyles that are separated distally and posteriorly by the deep **intercondylar fossa** or notch, above which is the popliteal surface (see Fig. 6.20; also Fig. 6.21).

The rounded distal portions of the **medial** and **lateral condyles** contain smooth articular surfaces for articulation with the tibia. The **medial condyle extends lower or more distally** than the lateral condyle when the femoral shaft is vertical, as in Fig. 6.20. This explains why the **CR must be angled 5° to 7° cephalad for a lateral knee** to cause the two condyles to be directly superimposed when the femur is parallel to the IR. The explanation for this is apparent in Fig. 6.19, which demonstrates that in an erect anatomic position, wherein the distal femoral condyles are parallel to the floor at the knee joint, the femoral shaft is at an angle of approximately 10° from vertical for an average adult. The range is 5° to 15°.[3] This angle would be greater on a person of short stature and a wider pelvis. The angle would be less on a person of tall stature with a narrow pelvis. In general, this angle is greater on a woman than on a man.

A distinguishing difference between the medial and lateral condyles is the presence of the **adductor tubercle,** a slightly raised area that receives the tendon of an adductor muscle. This tubercle is present on the **posterolateral aspect of the medial condyle.** It is best seen on a slightly rotated lateral view of the distal femur and knee. The presence of this adductor tubercle on the medial condyle is important in critiquing a lateral knee for rotation. It allows the viewer to determine whether the knee is under-rotated or over-rotated to correct a positioning error when the knee is not in a true lateral position (see the radiograph in Fig. 6.33).

The **medial** and **lateral epicondyles,** which can be palpated, are rough prominences for attachments of the medial and lateral collateral ligaments and are located on the outermost portions of the condyles. The medial epicondyle, along with the adductor tubercle, is the more prominent of the two.

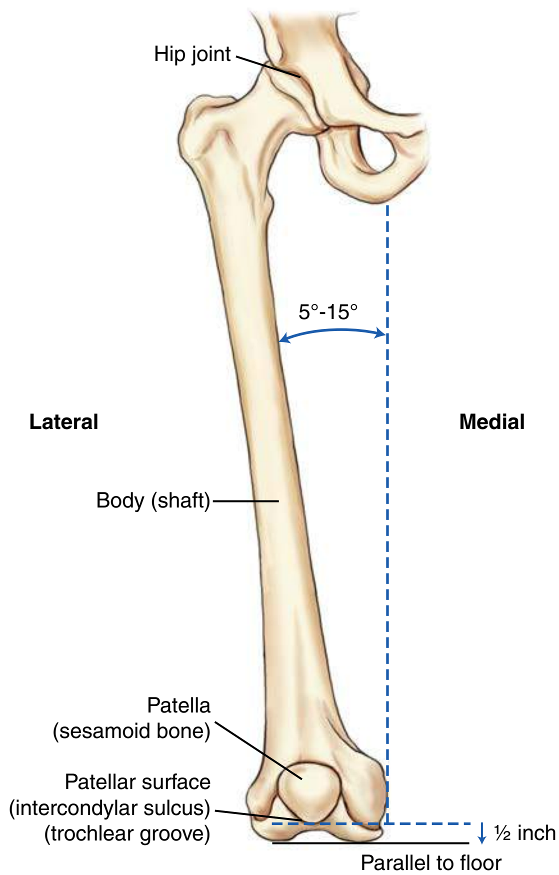


Fig 6.19 Femur—anterior view.

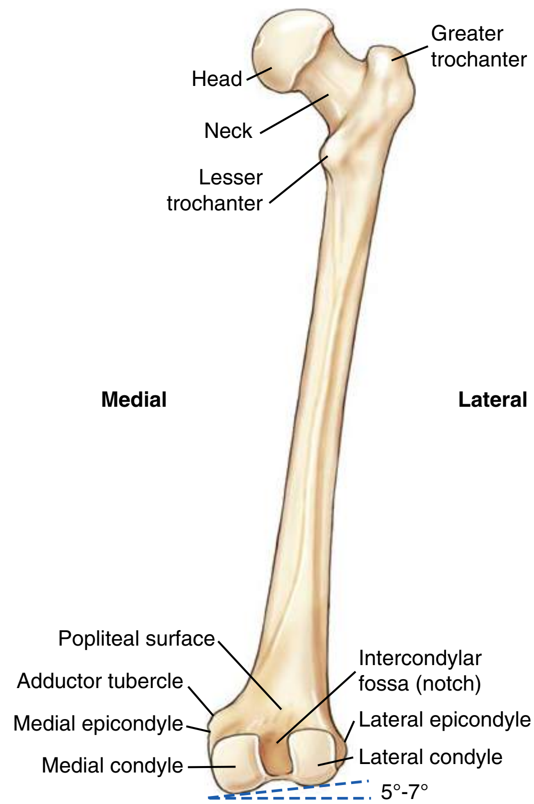


Fig 6.20 Femur—posterior view.

Distal Femur and Patella (Lateral View) The lateral view in Fig. 6.21 shows the relationship of the patella to the **patellar surface** of the distal femur. The patella, as a large sesamoid bone, is embedded in the tendon of the large quadriceps femoris muscle. As the lower leg is flexed, the patella moves downward and is drawn inward into the intercondylar groove or sulcus. A partial flexion of almost 45°, as shown in Fig. 6.21, demonstrates the patella being pulled only partially downward. With 90° flexion, the patella would move down farther over the distal portion of the femur. This movement and the relationship of the patella to the distal femur become important in positioning for the knee joint and for the tangential projection of the patellofemoral (femoropatellar) joint (articulation between patella and distal femur).

The posterior surface of the distal femur just proximal to the intercondylar fossa is called the **popliteal surface,** over which popliteal blood vessels and nerves pass.

Distal Femur and Patella (Axial View) The axial or end-on view of the distal femur demonstrates the relationship of the patella to the **patellar surface** (intercondylar sulcus or trochlear groove) of the distal femur. The patellofemoral joint space is visualized in this axial view (Fig. 6.22). Other parts of the distal femur are also well visualized.

The **intercondylar fossa** (notch) is shown to be very deep on the posterior aspect of the femur. The **epicondyles** are seen as rough prominences on the outermost tips of the large **medial** and **lateral condyles.**

PATELLA

The **patella** (kneecap) is a flat triangular bone approximately 2 inches (5 cm) in diameter (Fig. 6.23). The patella appears to be upside down because its pointed **apex** is located along the **inferior border,** and its **base** is the **superior** or **upper border.** The outer or **anterior surface** is convex and rough, and the inner or **posterior surface** is smooth and oval shaped for articulation with the femur. The patella serves to protect the anterior aspect of the knee joint and acts as a pivot to increase the leverage of the large quadriceps femoris muscle, the tendon of which attaches to the tibial tuberosity of the lower leg. The patella is loose and movable in its more superior position when the leg is extended and the quadriceps muscles are relaxed. However, as the leg is flexed and the muscles tighten, it moves distally and becomes locked into position. The patella articulates only with the femur, not with the tibia.

KNEE JOINT

The knee joint proper is a large complex joint that primarily involves the **femorotibial joint** between the two condyles of the **femur** and the corresponding condyles of the **tibia.** The **patellofemoral joint** is also part of the knee joint, wherein the patella articulates with the anterior surface of the distal femur.

Proximal Tibiofibular Joint and Major Knee Ligaments

The proximal fibula is not part of the knee joint because it does not articulate with any aspect of the femur, even though the **fibular (lateral) collateral ligament (LCL)** extends from the femur to the lateral proximal fibula, as shown in Fig. 6.24. However, the head of the fibula does articulate with the lateral condyle of the tibia, to which it is attached by this ligament.

Additional major knee ligaments shown on this posterior view are the tibial (medial) collateral ligament (MCL), located medially, and the major posterior and anterior cruciate *(kroo′-she-at)* ligaments (PCL and ACL), located within the knee joint capsule (Fig. 6.25). (The abbreviations ACL, PCL, LCL, and MCL are commonly used to refer to these four ligaments.[2]) The knee joint is highly dependent on these two important pairs of major ligaments for stability.

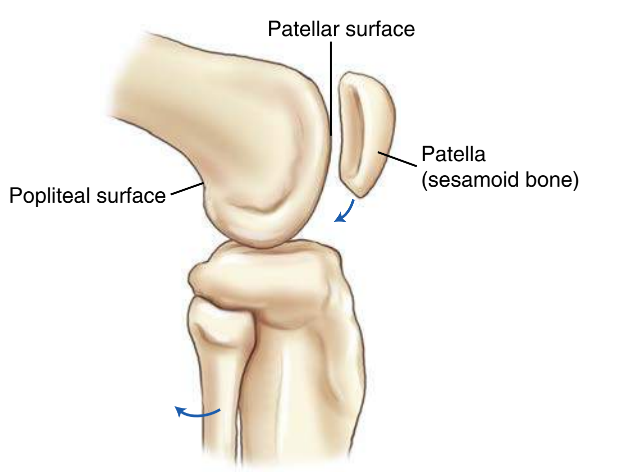


Fig 6.21 Distal femur and patella—lateral view.

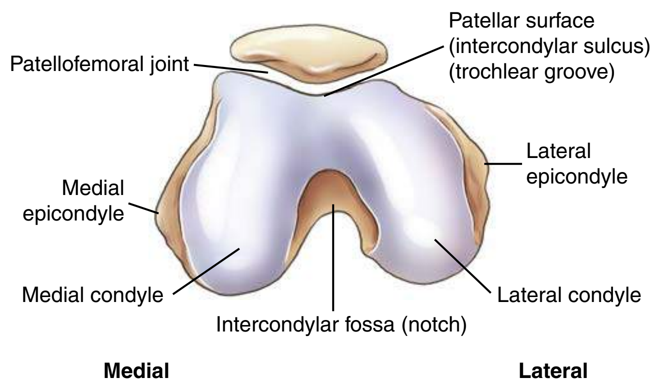


Fig 6.22 Distal femur and patella—axial view.

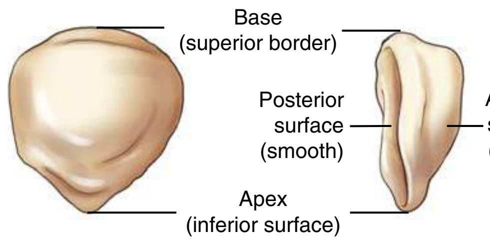


Fig 6.23 Patella.

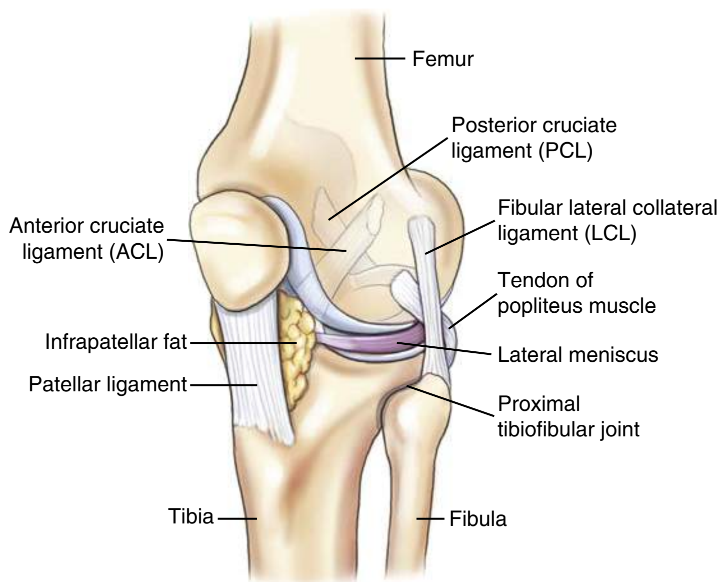


Fig 6.24 Knee joint and proximal tibiofibular joint—anterior oblique view.

The two **collateral ligaments** are strong bands at the sides of the knee that **prevent adduction and abduction** movements at the knee. The two **cruciate ligaments** are strong, rounded cords that cross each other as they attach to the respective anterior and posterior aspects of the intercondylar eminence of the tibia. They stabilize the knee joint by **preventing anterior or posterior movement** within the knee joint.

In addition to these two major pairs of ligaments, an anteriorly located **patellar ligament** and various minor ligaments help to maintain the integrity of the knee joint (Fig. 6.26). The patellar ligament is shown as part of the tendon of insertion of the large quadriceps femoris muscle, extending over the patella to the tibial tuberosity. The **infrapatellar fat pad,** posterior to this ligament, aids in protecting the anterior aspect of the knee joint.

Synovial Membrane and Cavity

The articular cavity of the knee joint is the largest joint space of the human body. The total knee joint is a synovial type enclosed in an **articular capsule,** or **bursa.** It is a complex, saclike structure filled with a lubricating-type synovial fluid. This is demonstrated in the arthrogram radiograph, wherein a combination of negative and positive contrast media has been injected into the articular capsule or bursa (Fig. 6.27).

The articular cavity or bursa of the knee joint extends upward under and superior to the patella, identified as the **suprapatellar bursa** (see Fig. 6.26). Distal to the patella, the **infrapatellar bursa** is separated by a large **infrapatellar fat pad,** which can be identified on radiographs. The spaces posterior and distal to the femur also can be seen and are filled with negative contrast media on the lateral arthrogram radiograph.

Menisci (Articular Disks)

The **medial** and **lateral menisci** *(me-nis′-ci)* are crescent-shaped fibrocartilage disks between the articular facets of the tibia (tibial plateau) and the femoral condyles (Fig. 6.28). They are thicker at their external margins, tapering to a very thin center portion. They act as shock absorbers to reduce some of the direct impact and stress that occur at the knee joint. The synovial membrane and the menisci produce synovial fluid, which lubricates the articulating ends of the femur and tibia that are covered with a tough, slick hyaline membrane.

Knee Trauma

The knee has great potential for traumatic injury, especially in activities such as skiing or snowboarding or in contact sports such as football or basketball. A tear of the tibial MCL frequently is associated with a tear of the ACL and a tear of the medial meniscus. Patients with these injuries typically come to the imaging department for magnetic resonance imaging (MRI) to visualize these soft tissue structures or for knee arthrography.

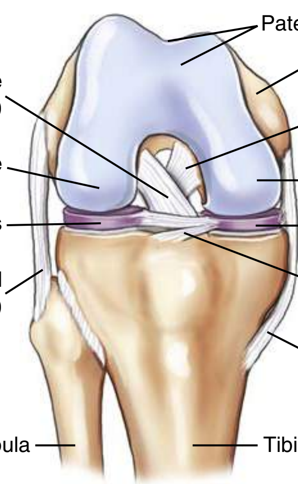


Fig 6.25 Right knee joint (flexed)—anterior view.

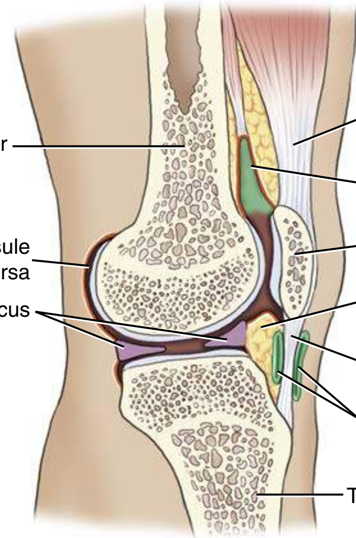


Fig 6.26 Sagittal section of knee joint.

Fig 6.27 Lateral knee arthrogram radiograph (demonstrates articular capsule or bursa as outlined by a combination of negative and positive contrast media).

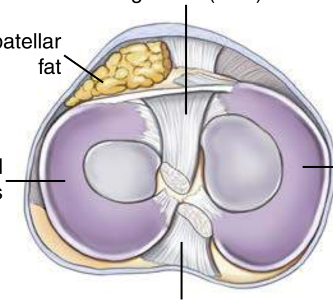


Fig 6.28 Superior view of articular surface of tibia (shows menisci and cruciate ligament attachments).

REVIEW EXERCISE WITH RADIOGRAPHS

Common projections of the lower leg, knee, and patella are shown with labels for an anatomy review.

AP Lower Leg (Fig. 6.29)

A. Medial condyle of the tibia
B. Body or shaft of tibia
C. Medial malleolus
D. Lateral malleolus
E. Body or shaft of fibula
F. Neck of fibula
G. Head of fibula
H. Apex (styloid process) of head of fibula
I. Lateral condyle of tibia
J. Intercondylar eminence (tibial spine)

Lateral Lower Leg (Fig. 6.30)

A. Intercondylar eminence (tibial spine)
B. Tibial tuberosity
C. Body or shaft of tibia
D. Body or shaft of fibula
E. Medial malleolus
F. Lateral malleolus

AP Knee (Fig. 6.31)

A. Medial and lateral intercondylar tubercles; extensions of intercondylar eminence (tibial spine)
B. Lateral epicondyle of femur
C. Lateral condyle of femur
D. Lateral condyle of tibia
E. Articular facets of tibia (tibial plateau)
F. Medial condyle of tibia
G. Medial condyle of femur
H. Medial epicondyle of femur
I. Patella (seen through femur)

Lateral Knee (Fig. 6.32)

A. Base of patella
B. Apex of patella
C. Tibial tuberosity
D. Neck of fibula
E. Head of fibula
F. Apex (styloid process) of head of fibula
G. Superimposed medial and lateral condyles
H. Patellar surface (intercondylar sulcus or trochlear groove)

Rotated Lateral Knee (*Fig. 6.33*) Projection demonstrates some rotation.

I. Adductor tubercle
J. Lateral condyle
K. Medial condyle

Tangential Projection (Patellofemoral Joint) (Fig. 6.34)

A. Patella
B. Patellofemoral (femoropatellar) joint
C. Lateral condyle
D. Patellar surface (intercondylar sulcus, trochlear groove)
E. Medial condyle

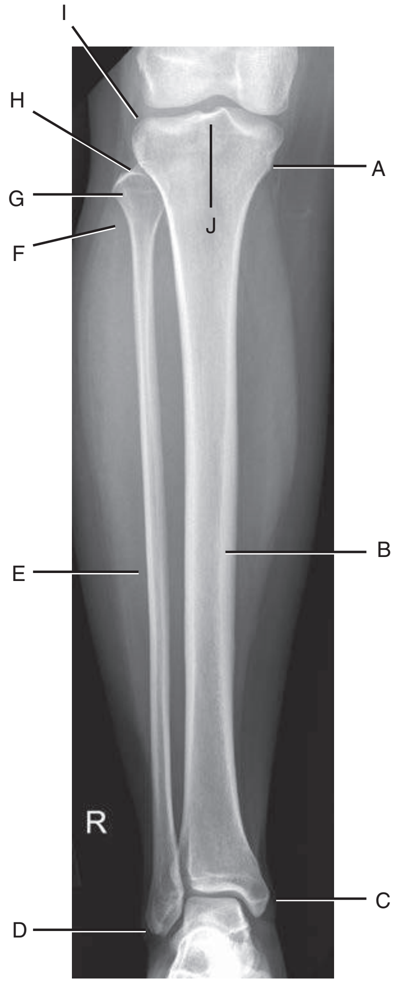


Fig 6.29 AP right tibia and fibula. (Courtesy Joss Wertz, DO.)

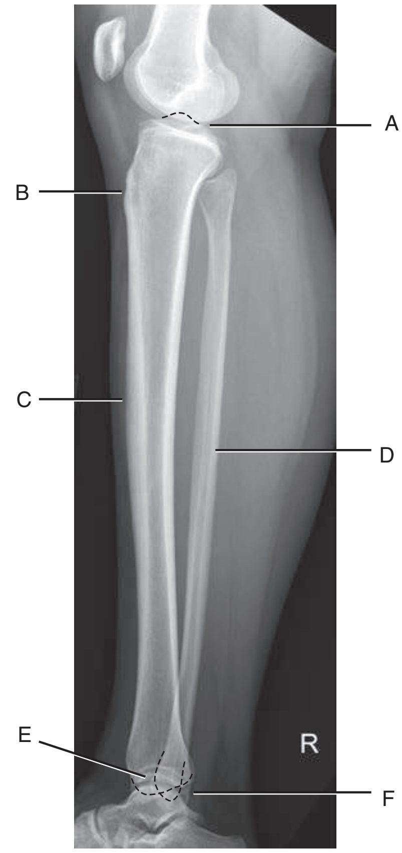


Fig 6.30 Lateral right tibia and fibula. (Courtesy Joss Wertz, DO.)

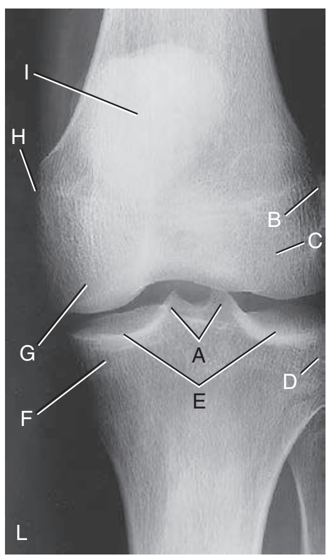


Fig 6.31 AP left knee.

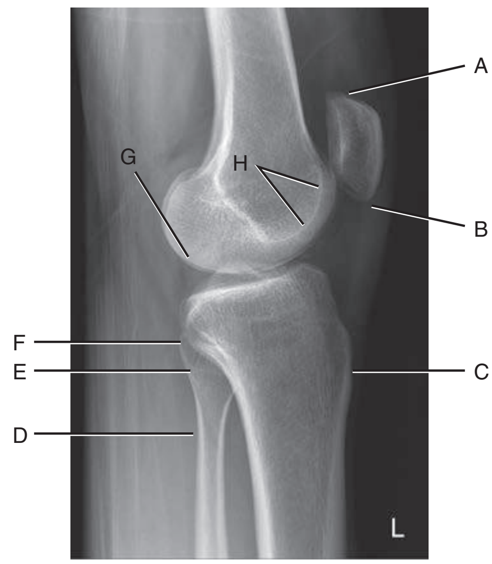


Fig 6.32 Lateral left knee—true lateral. (Courtesy Joss Wertz, DO.)

Fig 6.33 Rotated lateral knee (medial condyle more posterior).

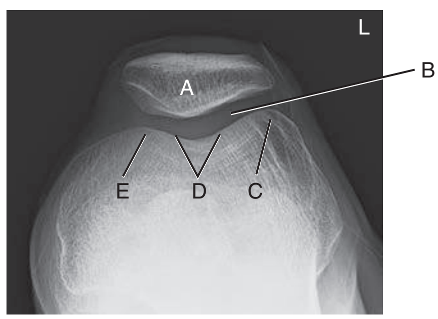


Fig 6.34 Tangential projection (patellofemoral joint). (Courtesy Joss Wertz, DO.)

CLASSIFICATION OF JOINTS

The joints or articulations of the lower limb (Fig. 6.35) all (with one exception) are classified as **synovial joints** and are characterized by a fibrous-type capsule that contains synovial fluid. They also are (with one exception) **diarthrodial,** or freely movable.

The single exception to the synovial joint is the **distal tibiofibular joint,** which is classified as a **fibrous joint** with fibrous interconnections between the surfaces of the tibia and fibula. It is of the **syndesmosis type** and is only slightly movable, or **amphiarthrodial.** However, the most distal part of this joint is smooth and is lined with a synovial membrane that is continuous with the ankle joint.

Box 6.2 summarizes the foot, ankle, lower leg, and knee joints.

SURFACES AND PROJECTIONS OF THE FOOT

Surfaces

The surfaces of the foot are sometimes confusing because the top or anterior surface of the foot is called the **dorsum.** *Dorsal* usually refers to the posterior part of the body. *Dorsum,* in this case, comes from the term **dorsum pedis,** which refers to the upper surface, or the surface opposite the sole of the foot.

The sole of the foot is the **posterior** surface or **plantar surface.** These terms are used to describe common projections of the foot.

Projections

The **AP projection** of the foot is the same as a **dorsoplantar (DP) projection.** The less common **posteroanterior (PA) projection** can also be called a **plantodorsal (PD) projection** (Fig. 6.36). Technologists should be familiar with each of these projection terms and should know which projection they represent.

MOTIONS OF THE FOOT AND ANKLE

Other potentially confusing terms involving the ankle and intertarsal joints are **dorsiflexion, plantar flexion, inversion,** and **eversion** (Fig. 6.37). To decrease the angle (flex) between the dorsum pedis and the anterior part of the lower leg is to **dorsiflex** at the ankle joint. Extending the ankle joint or pointing the foot and toe downward with respect to the normal position is called *plantar flexion.*

Inversion, or **varus,** is an inward turning or bending of the ankle and subtalar (talocalcaneal) joints, and **eversion,** or **valgus,** is an outward turning or bending. The lower leg does not rotate during inversion or eversion. Most sprained ankles result from an accidental forced inversion or eversion.

BOX 6.2 SUMMARY OF FOOT, ANKLE, LOWER LEG, AND KNEE JOINTS

ALL JOINTS OF LOWER LIMB EXCEPT DISTAL TIBIOFIBULAR

Classification: Synovial (articular capsule containing synovial fluid)
Mobility Type: Diarthrodial (freely movable)
Movement Types

1. Interphalangeal joints	**Ginglymus (hinge):** Flexion and extension movements
2. Metatarsophalangeal joints	**Modified ellipsoidal (condyloid):** Flexion, extension, abduction, and adduction (circumduction similar to metacarpophalangeal joints of hand is generally not possible)
3. Tarsometatarsal joints	**Plane (gliding):** Limited gliding movement
4. Intertarsal joints	**Plane (gliding):** Subtalar in combination with some other intertarsal joints provides for gliding and rotation; results in inversion and eversion of foot
5. Ankle joint	**Saddle (sellar):** Alignment between talus and lateral and medial malleolus creates a saddle type of joint. Dorsiflexion and plantar flexion only (side-to-side movements occur only with stretched or torn ligaments)
6. Knee joints	
Femorotibial	**Bicondylar:** Flexion and extension and some gliding and rotational movements when knee is partially flexed
Patellofemoral	**Saddle (sellar):** Considered a saddle type because of its shape and relationship of patella to anterior, distal femur
7. Proximal tibiofibular joint	**Plane (gliding):** Limited gliding movement between lateral condyle and head of fibula

DISTAL TIBIOFIBULAR

Classification: Fibrous
Mobility Type: Amphiarthrodial (slightly movable) of syndesmosis type

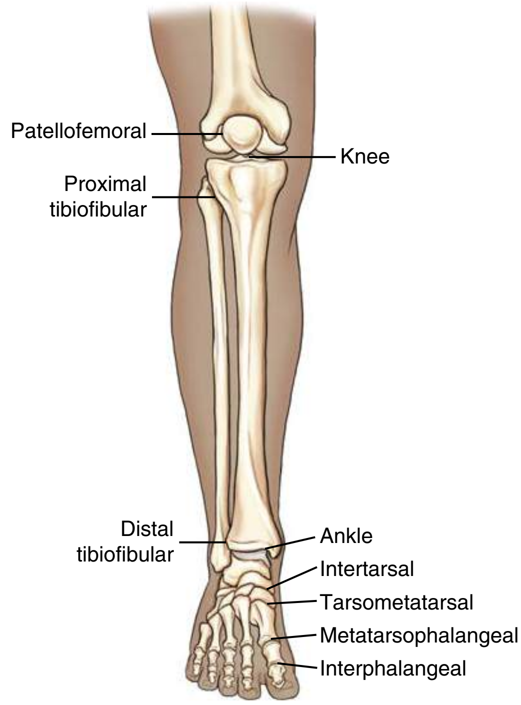


Fig 6.35 Joints of lower limb.

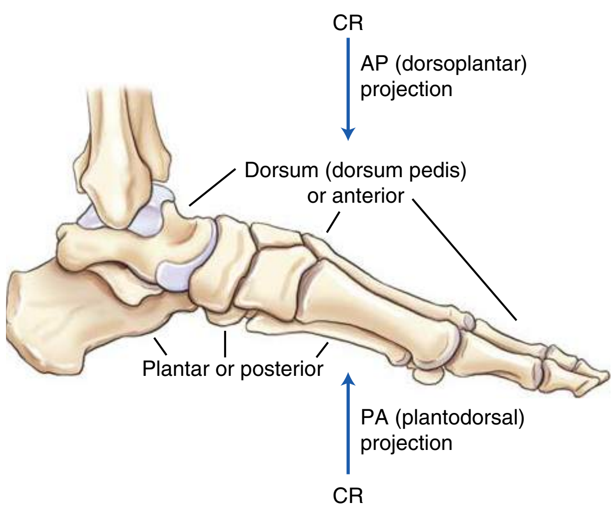


Fig 6.36 Surfaces and projections of foot.

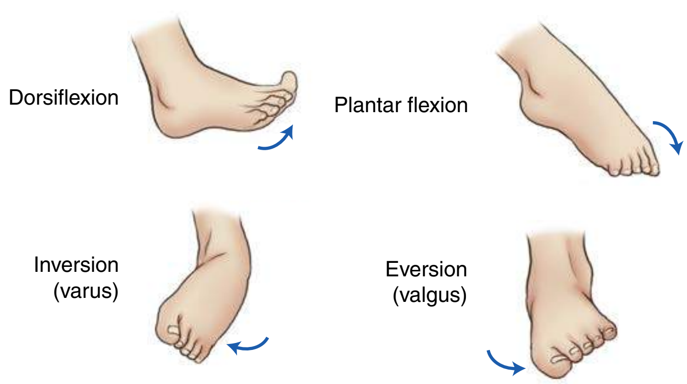


Fig 6.37 Motions of foot and ankle.

RADIOGRAPHIC POSITIONING

Positioning Considerations

Radiographic examinations involving the lower limb below the knee generally are done on a tabletop, as shown in Fig. 6.38. Patients with severe trauma or patients who are difficult to move can be radiographed directly on the cart.

DISTANCE

A common minimum source–image receptor distance (SID) is 40 inches (100 cm). When you are radiographing with IRs directly on the tabletop, to maintain a constant SID, increase the tube height compared with radiographs taken with the IR in the bucky tray. This difference is generally 3 to 4 inches (8 to 10 cm) for floating-type tabletops. The same minimum 40-inch (100-cm) SID should be used when you are radiographing directly on the cart, unless exposure factors are adjusted to compensate for a change in SID.

SHIELDING

Shielding of radiation-sensitive regions is important for examinations of the lower limb because of these regions' proximity to the divergent x-ray beam and scatter radiation. Red bone marrow in the hips and gonadal tissues are two of the key radiation-sensitive regions. A lead vinyl-covered shield should be draped over the patient's gonadal area. Although the gonadal rule states that this should be done on patients of reproductive age, when the gonads lie within or close to the primary field, providing gonadal shielding for all patients is good practice.

COLLIMATION

The collimation rule should be followed: collimation borders should be visible on all four sides if the IR is large enough to allow this without cutting off essential anatomy. A general rule concerning IR size is to use the smallest IR size possible for the specific part that is being radiographed. However, four-sided collimation is generally possible, even with a minimum-sized IR, for most if not all radiographic examinations of the lower limb.

With film-screen radiography and sometimes with CR, two or more projections may be taken on one IR for some examinations, such as for the toes, foot, ankle, or lower leg. Close collimation of the part that is being radiographed is required, and lead masking should be used to cover the parts of the IR not in the collimation field.

With **digital imaging,** multiple exposures on the same imaging plate are not suggested; however, when this is done, lead masking should be used to protect aspects of the IR not within the collimation field. The reason for this is to prevent unwanted exposure from scatter radiation from reaching the hypersensitive IR plate.

Four-sided collimation permits checking of radiographs for accuracy of centering and positioning by placing a large imaginary X from the four corners of the collimation field. The center point of the X indicates the CR location.

GENERAL POSITIONING

A general positioning rule that is especially applicable to both the upper and the lower limbs is **always to place the long axis of the part that is being radiographed parallel to the long axis of the IR.** If more than one projection is taken on the same IR, the part should be parallel to the long axis of the part of the IR being used. Also, **all body parts should be oriented in the same direction** when two or more projections are taken on the same IR.

An exception to this rule is the lower leg of an adult. This limb generally can be placed diagonally to include both the knee and the ankle joints.

CORRECT CENTERING

Accurate centering and alignment of the body part to the IR and correct CR location are especially important for examinations of the upper and lower limbs, in which shape and size distortion must be avoided and the narrow joint spaces clearly demonstrated. In general, the part being radiographed should be parallel to the plane of the IR, and the CR should be 90° or perpendicular and should be directed to the correct centering point, as indicated on each positioning page. (Exceptions to the 90° or perpendicular CR do occur, as indicated in the following pages.)

Multiple Exposures Per Imaging Plate

Placing multiple images on the same digital imaging plate (IP) is not commonly performed. Most experts would recommend that one exposure be placed centered to the IP for computed radiography and digital radiography imaging systems. But if multiple images are placed on the same IP, careful collimation and lead masking must be used to prevent pre-exposure or fogging of other images.

EXPOSURE FACTORS

The principal exposure factors for radiographs of the lower limbs are as follows:

1. Low-to-medium kVp (50–85)
2. Short exposure time
3. Small focal spot
4. Adequate milliamperes (mAs) for sufficient density (brightness)

Correctly exposed radiographs of the lower limbs generally should visualize both soft tissue margins and fine bony trabecular markings of the bones being radiographed.

IMAGE RECEPTORS

For examinations distal to the knee, IRs without grids are generally used. High resolution IRs may be used for extremities to obtain better detail.

Grids

A general rule is that grids should be used with body parts that measure more than 10 cm. (Some references suggest a grid for body parts greater than13 cm.) This rule places the average knee (measuring 9 to 13 cm) at a size for which either a nongrid or a grid technique may be used, depending on patient size and departmental preferences. This text recommends a nongrid technique on smaller patients with knees measuring 10 cm or less and a grid for larger patients with knees measuring more than 10 cm, especially on the AP knee. Anything proximal to the knee, such as the midfemur or distal femur, requires the use of a grid. When grids are used, you may choose either the moving bucky or a fine-lined portable grid.

Fig 6.38 Tabletop mediolateral projection of lower limb demonstrating *(A)* correct CR location and *(B)* collimation.

Special Patient Considerations

PEDIATRIC APPLICATIONS

Pediatric patients should be addressed in language they are able to understand. Parents are often helpful in positioning younger children in nontrauma situations. If parents are allowed to stay in the room, they must be provided appropriate shielding, and females must be asked their current pregnancy status according to facility policy. Immobilization is needed in many cases to assist in holding the limb in the proper position. Sponges and tape are useful, but sandbags should be used with caution because of their weight. Accurate part measurement is critical in setting technical factors.

Generally, exposure factors must be decreased because of reduced tissue quantity and density (brightness). Shorter exposure times, along with the highest mA possible, help to eliminate motion on the radiograph.

GERIATRIC APPLICATIONS

Older patients must be handled carefully when they are being moved, and radiography of the lower limbs is no exception. Look for telltale signs of hip fracture (i.e., foot in extreme external rotation position). Routine positioning maneuvers may have to be adjusted to accommodate potential pathology and lack of joint flexibility. Positioning aids and supports should be used to enhance patient comfort and assist in immobilizing the limb in the correct position.

Exposure factors may require adjustment because of underlying pathologic conditions such as osteoarthritis or osteoporosis. Shorter exposure time and higher mA are desirable for reducing the possibility of imaging involuntary or voluntary motion.

BARIATRIC PATIENT CONSIDERATIONS

Properly dressing patients for radiographic examinations is important to maintain radiographic quality. It is especially important in digital imaging and with bariatric patients. The increased sensitivity of digital imaging can cause clothing and other artifacts to appear on the image. Also, tight-fitting clothing on bariatric patients can interfere with radiographic quality. For example, increased compression due to improper clothing and added soft tissue can make it difficult to visualize fad pad signs indicative of fractures.

When imaging bariatric patients, proper alignment of the central ray with the body part remains the same. The increase in soft tissue does not change the position of the osseous anatomy. Increases in soft tissue may warrant changes in exposure factors. These changes may include an increase in kVp to improve penetration through additionally thick tissue. mA and time may also be increased, but sparingly to avoid excessive skin exposure. Care should also be taken to decrease scatter radiation to the IR due to the increased amount of tissue. The use of a grid for anatomic structures over 10 cm can be used to eliminate the demonstration of scatter.

Although most factors remain the same when imaging bariatric patients, there are factors to be considered. The largest consideration involves imaging of the knee. An increased cephalad CR angle may be required to visualize an open joint space when a patient has a greater thickness of the lower torso. Also, bariatric patients may require modifications to conventional imaging positions for the ease and comfort of the patient. For example, tissue may interfere with performing a tangential view of the patella using the Merchant method, requiring the use of the inferosuperior projection. Bariatric patients may also have difficulty internally rotating for a medial oblique knee, requiring instead the CR and IR to be adjusted to compensate.

PLACEMENT OF MARKERS AND PATIENT IDENTIFICATION INFORMATION

At the top of each of the following positioning pages is a small rectangular diagram that shows the correct IR size and placement (portrait or landscape). When film-screen technology is used, a suggested corner placement for the patient identification blocker is shown for each IR. However, this is only a suggested location because the location of the blocker changes with each manufacturer. The important consideration is **always to place it in the location that is least likely to superimpose anatomy of interest** for that projection. This concern is eliminated when computed radiography or digital radiography applications are used. The size and location of multiple projections on one IR are also shown.

When final radiographs are evaluated, as part of the evaluation criteria, right (R) and left (L) markers should always be visible on the lateral margin of the collimation field **on at least one projection on each IR** without superimposing any anatomy of interest. If film-screen systems are used, the patient identification information should always be checked to see whether it is legible and to ensure that it is not superimposing essential anatomy.

INCREASED EXPOSURE WITH CAST

A lower limb with a cast requires an increase in exposure. The thickness of the cast and the body part and the type of cast affect the increase in exposure required. See Table 6.1 for a recommended conversion guide for casts.

Digital Imaging Considerations

Following is a summary of guidelines that should be followed when digital imaging technology (computed radiography or digital radiography) is used for the lower limbs:

1. **Four-sided collimation:** Collimate to the area of interest with a minimum of two collimation parallel borders clearly demonstrated in the image. Four-sided collimation is always preferred if the study permits it.
2. **Accurate centering:** It is important that the body part and the central ray be centered to the IR.
3. **Grid use with cassette-less systems:** Anatomy thickness and kVp range are deciding factors if a grid is to be used. With these systems, it may be impractical and difficult to remove the grid, so the grid is commonly left in place even for smaller body parts measuring 10 cm or less (i.e., some upper and lower limb examinations). If the grid is left in place, ensure that the CR is centered to the grid.
4. **Exposure factors:** Patient exposure should follow the ALARA (as low as reasonably achievable) principle, and the lowest exposure factors required to obtain a diagnostic image should be used. This includes the highest kVp and the lowest mAs that would result in desirable image quality. The kVp may need to be higher than that used for analog (film-screen) imaging for larger body parts, with 50 kVp as the general minimum used on any procedure (exception is mammography).
5. **Post-processing evaluation of exposure indicator:** The exposure indicator value on the final processed image must be checked to verify that the exposure factors used were in the correct range to ensure optimal quality with the least radiation to the patient. If the index is outside of the acceptable range, the technologist must adjust the kVp or mAs or both accordingly for any repeat exposures.

TABLE 6.1 **CAST CONVERSION CHART**

TYPE OF CAST	INCREASE IN EXPOSURE
Small to medium plaster cast	Increase 5 to 7 kVp
Large plaster cast	Increase 8 to 10 kVp
Fiberglass cast	Increase 3 to 4 kVp

6

Alternative Modalities and Procedures

ARTHROGRAPHY

Arthrography sometimes is used to image large diarthrodial joints such as the knee. This procedure requires the use of a contrast medium injected into the joint capsule under sterile conditions. Disease or traumatic damage to the menisci, ligaments, and articular cartilage may be evaluated with arthrography (see Chapter 19).

COMPUTED TOMOGRAPHY (CT)

CT often is used on the lower limbs to evaluate soft tissue involvement of lesions. The cross-sectional images are also excellent for determining the extent of fractures and for evaluating bone mineralization.

MAGNETIC RESONANCE IMAGING (MRI)

MRI may be used to image the lower limbs when soft tissue injuries are suspected. The knee is the most often evaluated portion of the lower limb, and MRI is invaluable in detecting ligament damage or meniscal tears of the joint capsule. MRI also may be used to evaluate lesions in the skeletal system.

BONE DENSITOMETRY

Bone densitometry may be used to evaluate loss of bone in geriatric patients or in patients with a lytic (bone-destroying) type of bone disease (see Chapter 20 for more information on bone density measurement procedures).

NUCLEAR MEDICINE (NM)

NM uses radioisotopes injected into the bloodstream. These isotopes are absorbed in great concentration in areas where pathologic conditions exist. Nuclear medicine bone scans are particularly useful in showing osteomyelitis and metastatic bone lesions.

Clinical Indications

Radiographers should be familiar with common pathologic indications that relate to the lower limb, as follows:

Bone cysts are benign, neoplastic bone lesions filled with clear fluid that most often occur near the knee joint in children and adolescents. Generally, these are not detected on radiographs until a pathologic fracture occurs. When bone cysts are detected on radiographs, they appear as lucent areas with a thin cortex and sharp boundaries.

Chondromalacia patellae (commonly known as *runner's knee*) involves a softening of the cartilage under the patella, which results in erosion of this cartilage, causing pain and tenderness in this area. Cyclists and runners are vulnerable to this condition.

Chondrosarcomas are malignant tumors of the cartilage that usually occur in the pelvis and long bones of men older than 45 years.

Enchondroma is a slow-growing **benign cartilaginous tumor** that most often is found in small bones of the hands and feet in adolescents and young adults. Generally, these are well-defined, radiolucent-appearing tumors with a thin cortex, and they often lead to pathologic fracture with only minimal trauma.

Ewing sarcoma is a common **primary malignant bone tumor** that arises from bone marrow in children and young adults. Symptoms are similar to those of osteomyelitis, with low-grade fever and pain. Bone has stratified new bone formation, resulting in an "onion peel" look on radiographs. Ewing sarcoma generally occurs in the diaphysis of long bones. The prognosis is poor by the time it is evident on radiographs.

Exostosis (osteochondroma) is a benign, neoplastic bone lesion that is caused by consolidated overproduction of bone at a joint (usually the knee). The tumor grows parallel to the bone and away from the adjacent joint. Tumor growth stops as soon as the epiphyseal plates close. Pain is an associated symptom if the tumor is large enough to irritate surrounding soft tissues.

Fractures are breaks in the structure of bone caused by a force (direct or indirect). The types of fracture are named according to the extent of fracture, direction of fracture lines, alignment of bone fragments, and integrity of overlying skin (see Chapter 15 for fracture types and descriptions).

Gout is a form of arthritis that may be hereditary in which uric acid appears in excessive quantities in the blood and may be deposited in the joints and other tissues; common initial attacks occur in the **first MTP joint** of the foot. Later attacks may occur in other joints, such as the first metacarpophalangeal (MCP) joint of the hand, but generally these are not evident radiographically until more advanced conditions develop. Most cases occur in men, and first attacks rarely occur before the age of 30.

Joint effusions occur as accumulated fluid (synovial or hemorrhagic) in the joint cavity. These are signs of an underlying condition (e.g., fracture, dislocation, soft tissue damage).

The **Lisfranc ligament** is a large band that spans the articulation of the medial cuneiform and the first and second metatarsal base. Because no transverse ligament exists between the first and second metatarsal bases, this region of the foot is prone to stress injury caused by motor vehicle crashes, twisting falls, and falls from high places. Athletes often may acquire a Lisfranc injury that is due to high stress placed on the midfoot. **Lisfranc joint injuries** range from sprains to fracture-dislocations of the bases of the first and second metatarsals. A moderate sprain of the Lisfranc ligament is characterized by an abnormal separation between the first and second metatarsals. A small avulsion fracture may indicate a more severe injury. Lisfranc joint injuries may be missed if weight-bearing AP and lateral foot projections are not performed.

Multiple myeloma is the most common type of **primary cancerous bone tumor.** Generally, these tumors affect persons 40 to 70 years old. As the name implies, they occur in various parts of the body. Because this tumor arises from bone marrow or marrow plasma cells, it is not a truly exclusive bony tumor. Multiple myelomas are highly malignant and usually are fatal within a few years. The typical radiographic appearance consists of multiple "punched-out" osteolytic (loss of calcium in bone) lesions scattered throughout the affected bones.

Osgood-Schlatter disease, which involves inflammation of the bone and cartilage of the anterior proximal tibia, is most common in boys 10 to 15 years old. The cause is believed to be an injury that occurs when the large patellar tendon detaches part of the **tibial tuberosity** to which it is attached. Severe cases may require immobilization by plaster cast.

Osteoarthritis, also called **degenerative joint disease (DJD),** is a noninflammatory joint disease that is characterized by gradual deterioration of the articular cartilage with hypertrophic (enlargement or overgrown) bone formation. This is the most common type of arthritis and is considered part of the normal aging process.

Osteoclastomas (giant cell tumors) are benign lesions that typically occur in the long bones of young adults; they usually occur in the proximal tibia or distal femur after epiphyseal closure. These tumors appear on radiographs as large "bubbles" separated by thin strips of bone.

Osteogenic sarcomas (osteosarcomas) are **highly malignant primary bone tumors** that occur from childhood to young adulthood (peak age, 20 years). The neoplasm usually is seen in long bones and may cause gross destruction of bone.

Osteoid osteomas are **benign bone lesions** that usually occur in teenagers or young adults. Symptoms include localized pain that typically worsens at night but is relieved by over-the-counter anti-inflammatory or pain medications. The tibia and the femur are the most likely locations of these lesions.

Osteomalacia (rickets) literally means "bone softening." This disease is caused by lack of bone mineralization secondary to a deficiency of calcium, phosphorus, or vitamin D in the diet or an inability to absorb these minerals. Because of the softness of the bones, bowing defects in weight-bearing parts often result. This disease is known as *rickets* in children and commonly results in bowing of the tibia.

Paget disease (osteitis deformans) is one of the most common diseases of the skeleton. It is most common in midlife and is twice as common in men as in women. It is a non-neoplastic bone disease that disrupts new bone growth, resulting in overproduction of very dense yet soft bone. Bone destruction creates lytic or lucent areas; this is followed by reconstruction of bone, by which sclerotic or dense areas are created. The result is a very characteristic radiographic appearance that sometimes is described as cotton wool. Lesions typically occur in the skull, pelvis, femurs, tibias, vertebrae, clavicles, and ribs. Long bones generally bow or fracture because of softening of the bone; the associated joint may develop arthritic changes. The **pelvis** is the most common initial site of this disease.

Reiter syndrome affects the sacroiliac joints and lower limbs of young men; the radiographic hallmark is a specific area of bony erosion at the Achilles tendon insertion on the **posterosuperior margin of the calcaneus.** Involvement is usually bilateral, and arthritis, urethritis, and conjunctivitis are characteristic of this syndrome. Reiter syndrome is caused by a previous infection of the gastrointestinal tract, such as salmonella, or by a sexually transmitted infection.

See Table 6.2 for a summary of clinical indications.

Routine, Alternate, and Special Projections

Routine, alternate, and special projections for the toes, foot, ankle, lower leg, and knee are demonstrated and described on the following pages.

TABLE 6.2 **SUMMARY OF CLINICAL INDICATIONS**

CONDITION OR DISEASE	MOST COMMON RADIOGRAPHIC EXAMINATION	POSSIBLE RADIOGRAPHIC APPEARANCE	EXPOSURE FACTOR ADJUSTMENT[a]
Bone cyst	AP and lateral of affected limb	Well-circumscribed lucency	None
Chondromalacia patellae	AP and lateral knee, tangential (axial) of patellofemoral joint	Pathology of patellofemoral joint space, possible misalignment of patella	None
Chondrosarcoma	AP and lateral of affected limb, CT, MRI	Bone destruction with calcifications in cartilaginous tumor	None
Enchondroma (benign cartilaginous tumor)	AP and lateral of affected limb	Well-defined radiolucent tumor with thin cortex (often results in pathologic fracture with minimal trauma)	None
Ewing sarcoma (malignant bone tumor)	AP and lateral of affected limb, CT, MRI	Ill-defined area of bone destruction with surrounding "onion peel" (layers of periosteal reaction)	None
Exostosis (osteochondroma)	AP and lateral of affected limb	Projection of bone with cartilaginous cap; grows parallel to shaft and away from nearest joint	None
Gout (a form of arthritis)	AP (oblique) and lateral of affected part (most common initially in MTP joint of foot)	Uric acid deposits in joint space; destruction of joint space	None
Lisfranc joint injury	Weight-bearing AP and lateral and 30° medial oblique projections, CT, MRI	Abnormal separation or avulsion fracture between base of first and second metatarsals and cuneiforms	Slight increase in exposure factors to penetrate tarsal region of foot
Multiple myeloma (most common primary cancerous bone tumor)	AP and lateral of affected part	Multiple "punched-out" osteolytic lesions throughout affected bone	None
Osgood-Schlatter disease	AP and lateral of knee	Fragmentation or detachment of tibial tuberosity by patellar tendon	None
Osteoarthritis (degenerative joint disease)	AP, oblique, and lateral of affected part	Narrowed, irregular joint spaces with sclerotic articular surfaces and spurs	Advanced stage may require slight decrease (−)
Osteoclastoma (giant cell tumor)	AP and lateral of affected part	Large radiolucent lesions with thin strips of bone between	None
Osteogenic sarcoma (primary bone tumor)	AP and lateral of affected part, CT, MRI	Extensively destructive lesion with irregular periosteal reaction; classic appearance is sunburst pattern that is diffuse periosteal reaction	None
Osteoid osteoma (benign bone lesions)	AP and lateral of affected part	Small, round-to-oval density with lucent center	None
Osteomalacia (rickets)	AP and lateral of affected limb	Decreased bone density, bowing deformity in weight-bearing limbs	Loss of bone matrix requires decrease (−)
Paget disease (osteitis deformans)	AP and lateral of affected parts	Mixed areas of sclerotic and cortical thickening and lytic or radiolucent lesions; cotton wool appearance	Extensive sclerotic areas may require increase (+)
Reiter syndrome	AP and lateral of affected part	Asymmetric erosion of joint spaces; calcaneus erosion, usually bilateral	None

[a]Dependent on stage or severity of disease or condition.

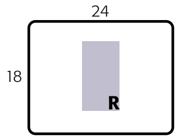
6

AP PROJECTION: TOES

Clinical Indications

- Fractures or dislocations of the phalanges of the digits in question
- Pathologies such as osteoarthritis and gouty arthritis (gout), especially in the first digit

Toes
ROUTINE
- **AP**
- **Oblique**
- **Lateral**



Technical Factors

- Minimum SID—40 inches (100 cm)
- IR size—8 × 10 inches (18 × 24 cm), landscape
- Nongrid
- kVp range: 50–65

NOTE: Some departmental routines include centering and collimation for AP toes to include all the toes and distal metatarsals. Most routines involve centering to the toe of interest with closer collimation to include only one digit on each side of injury.

Shielding Shield radiosensitive tissues outside region of interest.

Patient Position Place patient supine or seated on table; knee should be flexed with plantar surface of foot resting on IR.

Part Position

- Center and align long axis of digit to CR and long axis of portion of IR being exposed.
- Ensure that MTP joint of digit in question is centered to CR.

CR

- Angle CR **10° to 15° toward calcaneus** (CR perpendicular to phalanges) (Fig. 6.39).
- If a **15° wedge** is placed under the foot for parallel part-film alignment, the CR is **perpendicular** to the IR (Fig. 6.40).
- Center CR to **MTP joint** in question.

Recommended Collimation Collimate on four sides to area of interest. On side margins, include a minimum of at least part of one digit on each side of the digit in question.

Computed Radiography or Digital Radiography Close collimation is important over unexposed portions of IR to prevent fogging from scatter radiation.

Evaluation Criteria

Anatomy Demonstrated: • Digits of interest and a minimum of the distal half of metatarsals should be included (Figs. 6.41 and 6.42).

Position: • Individual digits should be separated with no overlapping of soft tissues. • Long axis of foot is aligned to long axis of portion of IR being exposed. • **No rotation** is present if shafts of the phalanges and distal metatarsals appear equally concave on both sides. • Rotation appears as one side being more concave than the other. • Side with increased concavity has been rolled away from IR.[4] • IP and MTP joint spaces are open. Incorrect CR angulation or insufficient elevation of forefoot may distort or close joint spaces.[4] • Collimation to **area of interest.**

Exposure: • **No motion** as evidenced by sharply defined cortical margins of the bone and detailed bony trabeculae. • Optimal contrast and density (brightness) allow visualization of bony cortical margins and trabeculae and soft tissue structures.

Fig 6.39 Second digit (CR, 10° to 15°).

Fig 6.40 AP second digit with wedge (CR perpendicular).

Fig 6.41 AP second digit.

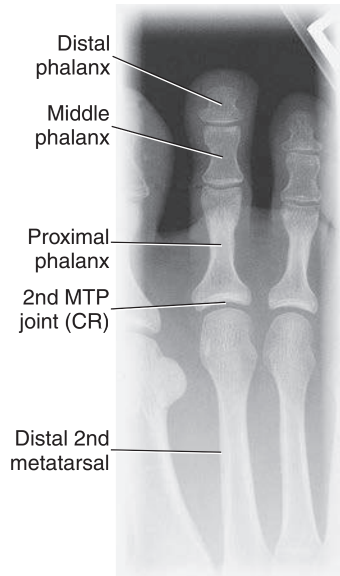
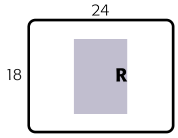


Fig 6.42 AP second digit.

AP OBLIQUE PROJECTION—MEDIAL OR LATERAL ROTATION: TOES

Toes
ROUTINE
- **AP**
- **Oblique**
- **Lateral**

Clinical Indications
- Fractures or dislocations of the phalanges of the digits in question
- Pathologies such as osteoarthritis and gouty arthritis (gout), especially in the first digit



Technical Factors
- Minimum SID—40 inches (100 cm).
- IR size—8 × 10 inches (18 × 24 cm), landscape
- Nongrid
- kVp range: 50–60

Shielding Shield radiosensitive tissues outside region of interest.

Patient Position Place patient supine or seated on table; knee should be flexed with plantar surface of foot resting on IR.

Part Position
- Center and align long axis of digit to CR and long axis of portion of IR being exposed.
- Ensure that MTP joint of digit in question is centered to CR.
- Rotate the leg and foot 30° to 45° medially for the first, second, and third digits (Fig. 6.43) and laterally for the fourth and fifth digits (Fig. 6.44). (See oblique foot projections for degree of obliquity.)
- Use 45° radiolucent support under elevated portion of foot to prevent motion.

CR
- CR **perpendicular** to IR, directed to MTP joint in question

Recommended Collimation Collimate on four sides to include phalanges and a minimum of distal half of metatarsals. On side margins, include a minimum of one digit on each side of digit in question.

Computed Radiography or Digital Radiography Close collimation is important over unexposed portions of IR to prevent fogging from scatter radiation.

Evaluation Criteria
Anatomy Demonstrated: • Digits in question and distal half of metatarsals should be included without overlap (superimposition) (Figs. 6.45 and 6.46).
Position: • Long axis of foot is aligned to long axis of portion of IR being exposed. • Correct obliquity should be evident by increased concavity on one side of shafts and by overlapping of soft tissues of digits. • Heads of metatarsals should appear directly side by side with no (or only minimal) overlapping.[4] • Collimation to **area of interest.**
Exposure: • No motion as evidenced by sharply defined cortical margins of bone and detailed bony trabeculae. • Optimal contrast and density (brightness) allow visualization of bony cortical margins and trabeculae and soft tissue structures.

Fig 6.43 Medial oblique rotation—first digit.

Fig 6.44 Lateral oblique rotation—fourth digit.

Fig 6.45 Medial oblique—second digit.

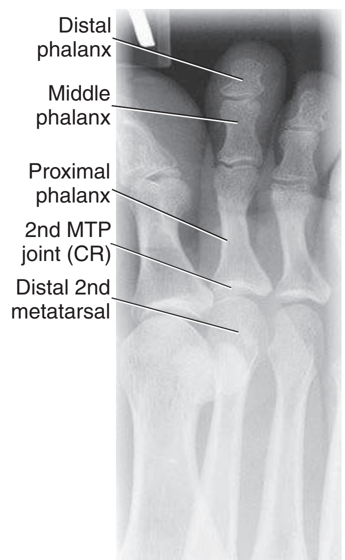


Fig 6.46 Medial oblique—second digit.

6

LATERAL—MEDIOLATERAL OR LATEROMEDIAL PROJECTIONS: TOES

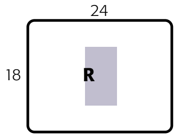
Clinical Indications

- Fractures or dislocations of the phalanges of the digits in question
- Pathologies such as osteoarthritis and gouty arthritis (gout), especially in the first digit

Toes
ROUTINE
- **AP**
- **Oblique**
- **Lateral**

Technical Factors

- Minimum SID—40 inches (100 cm).
- IR size—8 × 10 inches (18 × 24 cm), landscape
- Nongrid
- kVp range: 50–60



Shielding Shield radiosensitive tissues outside region of interest.

Patient and Part Position

- Rotate affected leg and foot medially (lateromedial) for first, second, and third digits (Figs. 6.47 and 6.48) and laterally (mediolateral) for fourth and fifth digits (Fig. 6.49).
- Adjust IR to center and align long axis of toe in question to CR and to long axis of portion of IR being exposed.
- Ensure that interphalangeal joint or proximal interphalangeal joint in question is centered to CR.
- Use tape, gauze, or tongue blade to flex and separate unaffected toes to prevent superimposition.

CR

- CR perpendicular to IR
- CR directed to interphalangeal joint for first digit and to proximal interphalangeal joint for second to fifth digits

Recommended Collimation Collimate closely on four sides to affected digit.

Computed Radiography or Digital Radiography Close collimation is important over unexposed portions of IR to prevent fogging from scatter radiation.

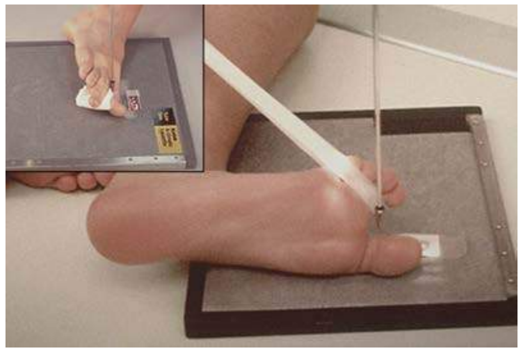


Fig 6.47 Lateromedial—first digit.

Fig 6.48 Lateromedial—second digit.

Fig 6.49 Mediolateral—fourth digit.

Evaluation Criteria

Anatomy Demonstrated: • Phalanges of digit in question should be seen in lateral position free of superimposition by other digits, if possible (Figs. 6.50 and 6.51). • (When total separation of toes is impossible, especially third to fifth digits, the distal phalanx at least should be separated, and the proximal phalanx should be visualized through superimposed structures.)

Position: • Long axis of digit is aligned to long axis of portion of IR being used. • True lateral of digit demonstrates increased concavity on anterior surface of the distal phalanx and posterior surface of the proximal phalanx. • Opposing surface of each phalanx appears straighter.[4] • Collimation to area of interest.

Exposure: • No motion as evidenced by sharply defined cortical margins of bone and detailed bony trabeculae. • Optimal contrast and density (brightness) allow visualization of bony cortical margins and trabeculae and soft tissue structures.

Fig 6.50 Lateromedial—second digit.

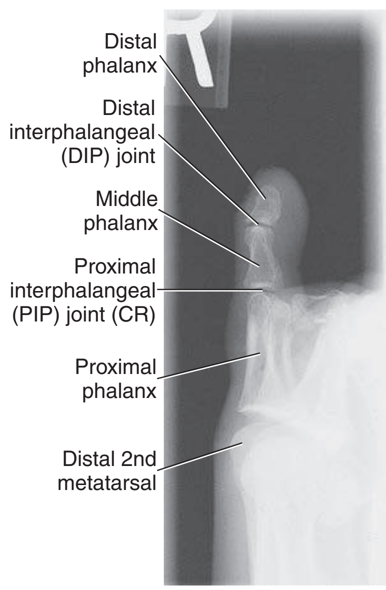


Fig 6.51 Lateromedial—second digit.

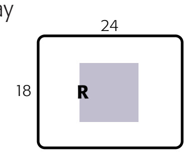
TANGENTIAL PROJECTION: TOES—SESAMOIDS

Toes
SPECIAL
- Sesamoids (tangential)

Clinical Indications

- This projection provides a profile image of the sesamoid bones at the first MTP joint for evaluation of extent of injury

NOTE: A lateral of the first digit in dorsiflexion also may be taken to visualize these sesamoids.



Technical Factors

- Minimum SID—40 inches (100 cm)
- IR size—8 × 10 inches (18 × 24 cm), landscape
- Nongrid
- kVp range: 50–60

Shielding Shield radiosensitive tissues outside region of interest.

Patient Position Place patient prone; provide a pillow for patient's head and a small sponge or folded towel under lower leg for comfort.

Part Position

- Dorsiflex the foot so that the plantar surface of the foot forms a **15° to 20° angle** from vertical (Figs. 6.52 and 6.53).
- Dorsiflex the first digit (great toe) and rest on IR to maintain position.
- Ensure that long axis of foot is not rotated; place sandbags or other support on both sides of foot to prevent movement.

NOTE: This is an uncomfortable and often painful position; do not keep patient in this position longer than necessary.

CR

- CR **perpendicular** to IR, directed tangentially to posterior aspect of **first MTP joint** (depending on amount of dorsiflexion of foot, may need to angle CR slightly for a true tangential projection)

Recommended Collimation Collimate closely to area of interest. Include at least the first, second, and third distal metatarsals for possible sesamoids but with CR at first MTP joint.

Evaluation Criteria

Anatomy Demonstrated: • Sesamoids should be seen in profile free of superimposition. • A minimum of the first three distal metatarsals should be included in collimation field for possible sesamoids, with the center of the four-sided collimation field (CR) at the posterior portion of the first MTP joint (Figs. 6.54 and 6.55).

Position: • **Borders** of posterior margins of first to third distal metatarsals are seen in profile, indicating correct dorsiflexion of foot. • Centering and CR angulation are correct if the sesamoids are free of any bony superimposition and open space is demonstrated between sesamoids and first metatarsal.

Exposure: • **No motion** as evidenced by sharp bony cortical margins and detailed trabeculae. • Optimal contrast and density (brightness) allow visualization of bony cortical margins and trabeculae and soft tissue structures without sesamoids appearing overexposed.

Alternative Projection If the patient cannot tolerate the previously described prone position, this radiograph may be performed in a reverse projection with the patient supine and the use of a long strip of gauze for the patient to hold the toes as shown. The CR would be directed tangential to the posterior aspect of the first MTP joint. Use support to prevent motion. However, this is not a desirable projection because of the increased object–image receptor distance (OID) with accompanying magnification and loss of definition.

Fig 6.52 Tangential projection—patient prone.

Fig 6.53 Alternative projection—patient supine.

Fig 6.54 Tangential projection. (Courtesy Joss Wertz, DO.)

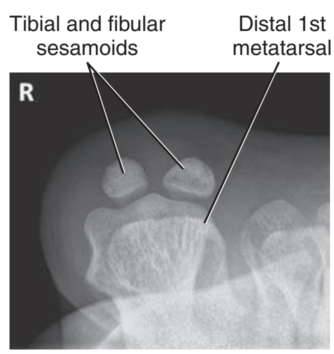


Fig 6.55 Tangential projection. (Courtesy Joss Wertz, DO.)

AP PROJECTION: FOOT

DORSOPLANTAR PROJECTION

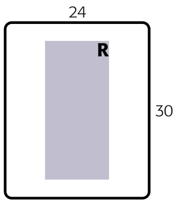
Clinical Indications

- Location and extent of fractures and fragment alignments, joint space abnormalities, soft tissue effusions
- Location of opaque foreign bodies

Foot
ROUTINE
- AP
- Oblique
- Lateral

Technical Factors

- Minimum SID—40 inches (100 cm)
- IR size—10 × 12 inches (24 × 30 cm), portrait
- Nongrid
- kVp range: 55–65



Shielding Shield radiosensitive tissues outside region of interest.

Patient Position Place patient supine; provide a pillow for patient's head; flex knee and place plantar surface (sole) of affected foot flat on IR.

Part Position

- Extend (plantar flex) foot but maintain plantar surface resting flat and firmly on IR (Fig. 6.56).
- Align and center long axis of foot to CR and to long axis of portion of IR being exposed. (Use sandbags if necessary to prevent IR from slipping on tabletop.)
- If immobilization is needed, flex opposite knee also and rest against affected knee for support.

CR

- Angle CR **10° posteriorly** (toward heel) with CR perpendicular to metatarsals (see **NOTE**).
- Direct CR to base of third metatarsal.

Recommended Collimation Collimate to outer margins of foot on four sides.

Computed Radiography or Digital Radiography Close collimation is important over unexposed portions of IR to prevent fogging from scatter radiation.

NOTE: A high arch requires a greater angle (15°) and a low arch nearer 5° to be perpendicular to the metatarsals. For a foreign body, CR should be perpendicular to IR with no CR angle.

Fig 6.56 AP foot—CR 10°.

Evaluation Criteria

Anatomy Demonstrated: • Entire foot should be demonstrated, including all phalanges and metatarsals and navicular, cuneiforms, and cuboids (Figs. 6.57 and 6.58).
Position: • Long axis of foot should be aligned to long axis of portion of IR being exposed. • **No rotation** as evidenced by nearly equal distance between second through fifth metatarsals. • Bases of first and second metatarsals generally are separated, but bases of second to fifth metatarsals appear to overlap. • Intertarsal joint space between first and second cuneiforms should be demonstrated. • Collimation to **area of interest.**
Exposure: • Optimal density (brightness) and contrast with **no motion** should visualize sharp borders and trabecular markings of distal phalanges and tarsals distal to talus. • Higher kVp technique used for more uniform densities between phalanges and tarsals. • Sesamoid bones (if present) should be seen through head of first metatarsal.

Fig 6.57 AP foot.

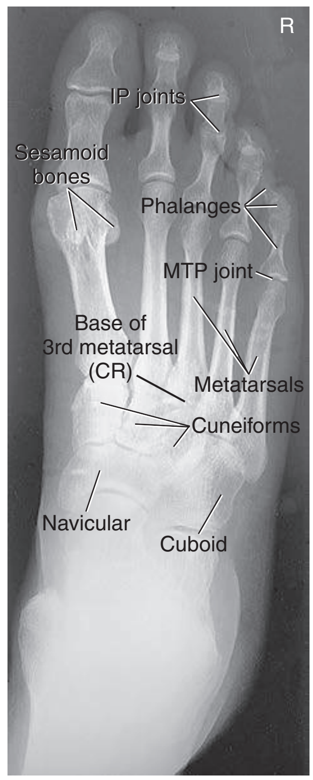


Fig 6.58 AP foot.

AP OBLIQUE PROJECTION—MEDIAL ROTATION: FOOT

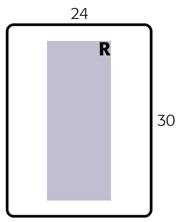
Clinical Indications

- Location and extent of fractures and fragment alignments, joint space abnormalities, soft tissue effusions
- Location of opaque foreign bodies

Foot
ROUTINE
- AP
- Oblique
- Lateral

Technical Factors

- Minimum SID—40 inches (100 cm)
- IR size—10 × 12 inches (24 × 30 cm), portrait
- Nongrid
- kVp range: 60–70



Shielding Shield radiosensitive tissues outside region of interest.

Patient Position Place patient supine or sitting; flex knee, with plantar surface of foot on table; turn body slightly away from side in question.

Part Position

- Align and center long axis of foot to CR and to long axis of portion of IR being exposed.
- Rotate foot **medially** to place **plantar surface 30° to 40° to plane of IR** (see NOTE). The general plane of the dorsum of the foot should be parallel to IR and perpendicular to CR (Fig. 6.59).
- Use 45° radiolucent support block to prevent motion. Use sandbags if necessary to prevent IR from slipping on tabletop.

CR

- CR perpendicular to IR, directed to base of third metatarsal

Recommended Collimation Collimate to outer margins of skin on four sides.

NOTE: Some references suggest only a 30° oblique routinely. This text recommends greater obliquity, 40°, to demonstrate tarsals and proximal metatarsals best relatively free of superimposition for the foot with an average transverse arch.

Evaluation Criteria (Medial Oblique)
Anatomy Demonstrated: • Entire foot should be demonstrated from distal phalanges to posterior calcaneus and proximal talus (Figs. 6.61 and 6.62).
Position: • Long axis of foot should be aligned to long axis of IR. • Correct obliquity is demonstrated when third through fifth metatarsals are free of superimposition. • First and second metatarsals also should be free of superimposition except for base area. • Tuberosity at base of fifth metatarsal is seen in profile and is well visualized. • Joint spaces around cuboid and the sinus tarsi are open and well demonstrated when foot is positioned obliquely correctly. • Collimation to **area of interest.**
Exposure: • Optimal density (brightness) and contrast with **no motion** should visualize sharp borders and trabecular markings of phalanges, metatarsals, and tarsals.

Optional Lateral Oblique (Fig. 6.60)

- Rotate the foot laterally 30° (less obliquity is required because of the natural arch of the foot).
- A lateral oblique best demonstrates the space between first and second metatarsals and between first and second cuneiforms. The navicular also is well visualized on the lateral oblique.

Fig 6.59 30° to 40° AP medial oblique.

Fig 6.60 30° AP lateral oblique.

Fig 6.61 40° AP medial oblique.

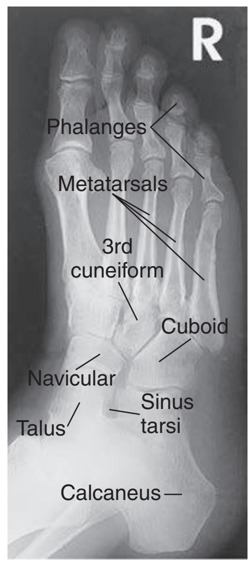


Fig 6.62 40° medial oblique.

6

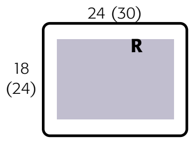
LATERAL—MEDIOLATERAL OR LATEROMEDIAL PROJECTIONS: FOOT

Clinical Indications

- Location and degree of anterior or posterior displacement of fracture fragments, joint abnormalities, and soft tissue effusions
- Location of opaque foreign bodies

Foot
ROUTINE
- **AP**
- **Oblique**
- **Lateral**

Technical Factors



- Minimum SID—40 inches (100 cm)
- IR size—8 × 10 inches (18 × 24 cm), for smaller foot, or 10 × 12 inches (24 × 30 cm) for larger foot, portrait
- Nongrid
- kVp range: 60–70

Shielding Shield radiosensitive tissues outside region of interest.

Patient Position Place patient in lateral recumbent position; provide pillow for patient's head.

Part Position (Mediolateral Projection)

- Flex knee of affected limb about 45°; place opposite leg **behind** the injured limb to prevent over-rotation of affected leg.
- Carefully dorsiflex foot if possible to assist in positioning for a true lateral foot and ankle (Fig. 6.63).
- Place support under leg and knee as needed so that **plantar surface is perpendicular to IR.** Do not **over-rotate** foot.
- Align long axis of foot to long axis of IR (unless diagonal placement is needed to include entire foot).
- Center mid area of base of metatarsals to CR.

CR

- CR **perpendicular** to IR, directed to **medial cuneiform** (at level of base of third metatarsal)

Recommended Collimation Collimate to the outer skin margins of the foot to include approximately 1 inch (2.5 cm) proximal to ankle joint.

Digital Imaging Systems Close collimation is important over unexposed portions of IR to prevent fogging from scatter radiation. Close collimation and lead masking are important over unused portions of IP to prevent fogging from scatter radiation to the hypersensitive IP or IR.

Evaluation Criteria

Anatomy Demonstrated: • Entire foot should be demonstrated, with a minimum of 1 inch (2.5 cm) of distal tibia-fibula. • Heads of metatarsals are superimposed with the tuberosity of the fifth metatarsal seen in profile (Figs. 6.65 and 6.66).
Position: • Long axis of the foot should be aligned to the long axis of IR. • True lateral position is achieved when tibiotalar joint is open, distal fibula is superimposed by the posterior tibia, and distal metatarsals are superimposed. • Collimation to **area of interest.**
Exposure: • Optimal density (brightness) and contrast should visualize borders of superimposed tarsals and metatarsals. • **No motion;** cortical margins and trabecular markings of calcaneus and nonsuperimposed portions of other tarsals should appear sharply defined.

Alternative Lateromedial Projection A lateromedial projection may be taken as an alternative lateral. This position can be more uncomfortable or painful for the patient, but it may be easier to achieve a true lateral (Fig. 6.64).

Fig 6.63 Mediolateral projection.

Fig 6.64 Alternative lateromedial.

Fig 6.65 Mediolateral foot.

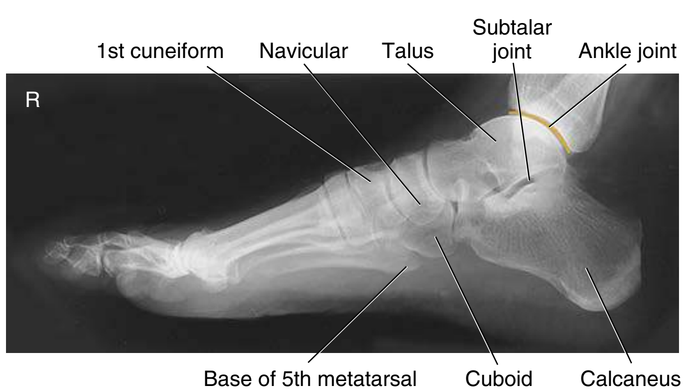


Fig 6.66 Mediolateral foot.

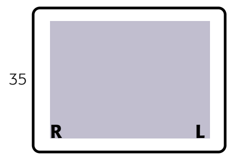
AP WEIGHT-BEARING PROJECTIONS: FOOT

Foot
SPECIAL
- AP and lateral (weight-bearing)

Clinical Indications

- Demonstrate the bones of the feet to show the condition of the longitudinal arches under the full weight of the body
- May demonstrate injury to structural ligaments of the foot such as a Lisfranc joint injury

NOTE: Bilateral projections of both feet often are taken for comparison. Some AP routines include separate projections of each foot taken with CR centered to individual foot.



AP both feet

Technical Factors

- Minimum SID—40 inches (100 cm)
- IR size—10 × 12 inches (24 × 30 cm), 14 × 17 inches (35 × 43 cm) for bilateral study, landscape
- Nongrid
- kVp range: 60–70

Shielding Shield radiosensitive tissues outside region of interest.

Patient Position

AP

- Place patient erect, with full weight evenly distributed on *both* feet.
- Feet should be directed straight ahead, parallel to each other (Fig. 6.67).

CR

- Angle CR 15° posteriorly to midpoint between feet at level of base of metatarsals.

Recommended Collimation Collimate to outer skin margins of the feet.

Evaluation Criteria

Anatomy Demonstrated: • For **AP,** projection shows bilateral feet from soft tissue surrounding phalanges to distal portion of talus (Fig. 6.68).

Position: • For **AP,** proper angulation is demonstrated by open tarsometatarsal joint spaces and visualization of joint between first and second cuneiforms. • Metatarsal bases should be at center of the collimated field (CR) with four-sided collimation, including the soft tissue surrounding the feet.

Exposure: • Optimal density (brightness) and contrast should visualize soft tissue and bony borders of superimposed tarsals and metatarsals. • Adequate penetration of midfoot region. • Bony trabecular markings should be sharp.

Fig 6.67 AP—bilateral feet (projection taken on digital IR).

Fig 6.68 AP weight-bearing—bilateral feet. (Courtesy Joss Wertz, DO.)

LATERAL WEIGHT-BEARING PROJECTIONS: FOOT

Clinical Indications

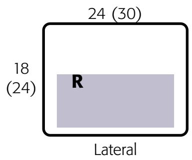
- Demonstrate the bones of the feet to show the condition of the longitudinal arches under the full weight of the body
- May demonstrate injury to structural ligaments of the foot such as a Lisfranc joint injury

Foot
SPECIAL
- AP and lateral (weight-bearing)

NOTE: Bilateral projections of both feet often are taken for comparison. Some AP routines include separate projections of each foot taken with CR centered to individual foot.

Technical Factors

- Minimum SID—40 inches (100 cm)
- IR size—10 × 12 inches (24 × 30 cm), 14 × 17 inches (35 × 43 cm) for bilateral study, landscape
- Nongrid
- kVp range: 60–70



Shielding Shield radiosensitive tissues outside region of interest.

Patient Position

- Have patient stand erect, with weight placed on affected foot (Fig. 6.69).
- Have patient stand on wood blocks placed on a step stool or the foot rest attached to the table. You also may use a special wooden box with a slot for the IR. (It has to be high enough from the floor to get the x-ray tube down into a horizontal beam position.)
- Provide some support for patient to hold onto for security.

Part Position

- Align long axis of foot to long axis of IR.
- Change IR and turn patient for lateral of other foot for comparison after first lateral has been taken.

CR

- Direct CR horizontally to level of base of third metatarsal.

Recommended Collimation Collimate to margins of feet.

Fig 6.69 Lateral weight-bearing—right foot (projection taken on digital IR).

Fig 6.70 Weight-bearing lateral.

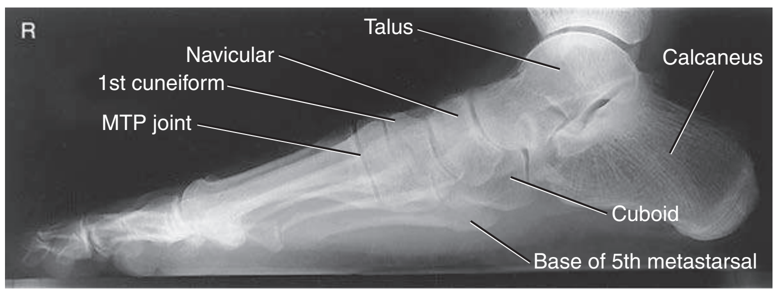


Fig 6.71 Weight-bearing lateral.

Evaluation Criteria

Anatomy Demonstrated: • For **lateral,** entire foot should be demonstrated, along with a minimum of 1 inch (2.5 cm) of distal tibia-fibula. • Distal fibula should be seen superimposed over posterior half of the tibia, and plantar surfaces of heads of metatarsals should be superimposed if no rotation is present. • The longitudinal arch of the foot must be demonstrated in its entirety (Figs. 6.70 and 6.71).

Position: • For **lateral,** center of collimated field (CR) should be to level of base of third metatarsal. • Four-sided collimation should include all surrounding soft tissue from the phalanges to the calcaneus and from the dorsum to the plantar surface of the foot with approximately 1 inch (2.5 cm) of the distal tibia-fibula demonstrated.

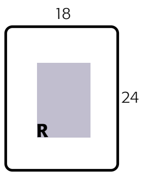
Exposure: • Optimal density (brightness) and contrast should visualize borders of superimposed tarsals and metatarsals. • **No motion;** cortical margins and trabecular markings of calcaneus and nonsuperimposed portions of other tarsals should appear sharply defined.

PLANTODORSAL (AXIAL) PROJECTION: CALCANEUS

Calcaneus
ROUTINE
- Plantodorsal (axial)
- Lateral

Clinical Indications
- Pathologies or fractures with medial or lateral displacement

Technical Factors
- Minimum SID—40 inches (100 cm)
- IR size—8 × 10 inches (18 × 24 cm), portrait
- Nongrid
- kVp range: 65–75 (increase by 8 to 10 kVp over other foot projections)



Shielding Shield radiosensitive tissues outside region of interest.

Patient Position Place patient supine or seated on table with leg fully extended.

Part Position
- Center and align ankle joint to CR and to portion of IR being exposed.
- Dorsiflex foot so that plantar surface is near perpendicular to IR (Fig. 6.72).

CR
- Direct CR to **base of third metatarsal** to emerge at a level just distal to lateral malleolus.
- Angle CR **40° cephalad from long axis of foot** (which also would be 40° from vertical *if* long axis of foot is perpendicular to IR). (See NOTE.)

Recommended Collimation Collimate closely to region of calcaneus.

Digital Imaging Systems Close collimation is important over unexposed portions of IR to prevent fogging from scatter radiation.

NOTE: CR angulation must be increased if long axis of plantar surface of foot is not perpendicular to IR.

Fig 6.72 Plantodorsal (axial) projection of calcaneus.

Fig 6.73 Plantodorsal (axial) projection.

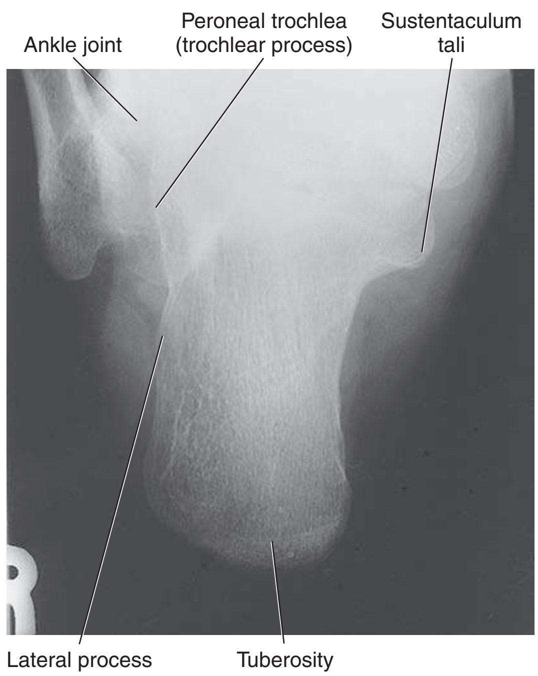


Fig 6.74 Plantodorsal (axial) projection.

Evaluation Criteria

Anatomy Demonstrated: • Entire calcaneus should be visualized from tuberosity posteriorly to talocalcaneal joint anteriorly (Figs. 6.73 and 6.74).

Position: • **No rotation;** a portion of the sustentaculum tali should appear in profile medially. • With the foot in proper 90° flexion, correct alignment and angulation of CR are evidenced by open talocalcaneal joint space, no distortion of the calcaneal tuberosity, and adequate elongation of the calcaneus. • Collimation to **area of interest.**

Exposure: • Optimal density (brightness) and contrast with no motion demonstrate sharp bony margins and trabecular markings and at least faintly visualize talocalcaneal joint without overexposing distal tuberosity area.

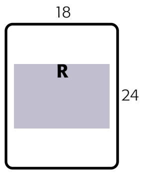
LATERAL—MEDIOLATERAL PROJECTION: CALCANEUS

Clinical Indications

- Bony lesions involving calcaneus, talus, and talocalcaneal joint
- Demonstrate extent and alignment of fractures

Calcaneus
ROUTINE
• Plantar dorsal
• Lateral

Technical Factors



- Minimum SID—40 inches (100 cm)
- IR size—8 × 10 inches (18 × 24 cm), portrait
- Nongrid
- kVp range: 60–75

Shielding Shield radiosensitive tissues outside region of interest.

Patient Position Place patient in lateral recumbent position, affected side down. Provide a pillow for patient's head. Flex knee of affected limb about 45°; place opposite leg behind injured limb.

Part Position

- Center calcaneus to CR and to unmasked portion of IR, with long axis of foot parallel to plane of IR (Fig. 6.75).
- Place support under knee and leg as needed to place plantar surface perpendicular to IR.
- Position ankle and foot for a **true lateral,** which places the lateral malleolus approximately 1 cm posterior to the medial malleolus.
- Dorsiflex foot so that plantar surface is at right angle to leg.

CR

- CR perpendicular to IR, directed to a point 1 inch (2.5 cm) inferior to medial malleolus

Recommended Collimation Collimate to outer skin margins to include ankle joint proximally and entire calcaneus.

Digital Imaging Systems Close collimation is important over unexposed portions of IR to prevent fogging from scatter radiation.

Fig 6.75 Mediolateral calcaneus.

Fig 6.76 Mediolateral calcaneus.

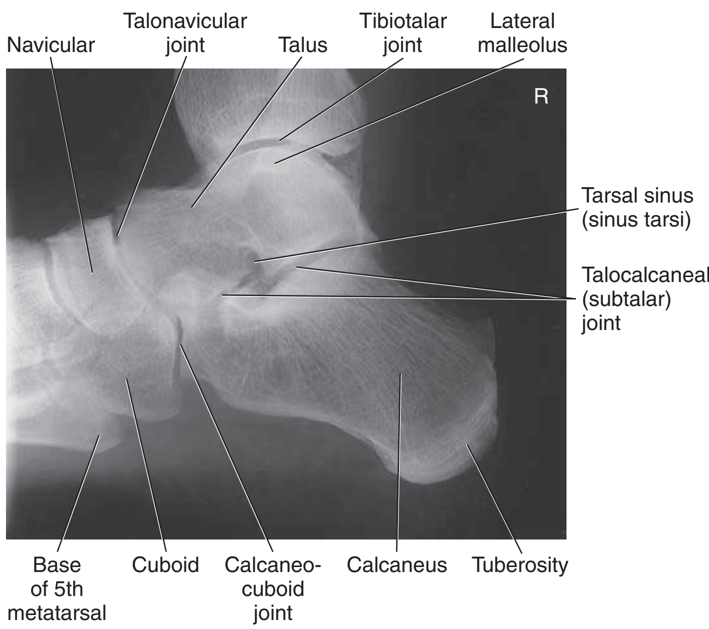


Fig 6.77 Mediolateral calcaneus.

Evaluation Criteria

Anatomy Demonstrated: • Calcaneus is demonstrated in profile with talus and distal tibia-fibula demonstrated superiorly and navicular and open joint space of the calcaneus and cuboid demonstrated distally (Figs. 6.76 and 6.77).

Position: • **No rotation** as evidenced by superimposed superior portions of the talus, open talocalcaneal joint, and lateral malleolus superimposed over posterior half of the tibia and talus. • Tarsal sinus and calcaneocuboid joint space should appear open. • Four-sided collimation should include ankle joint proximally and talonavicular joint and base of fifth metatarsal anteriorly.

Exposure: • Optimal exposure visualizes some soft tissue and more dense portions of calcaneus and talus. • Outline of the distal fibula should be faintly visible through the talus. • Trabecular markings appear clear and sharp, indicating no motion.

AP PROJECTION: ANKLE

Ankle
ROUTINE
- AP
- AP mortise (15°)
- Lateral

SPECIAL
- Oblique (45°)
- AP stress

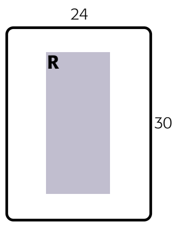
Clinical Indications

- Bony lesions or diseases involving the ankle joint, distal tibia and fibula, proximal talus, and proximal fifth metatarsal

The lateral portion of the ankle joint space should not appear open on this projection—see AP Mortise Projection.

Technical Factors

- Minimum SID—40 inches (100 cm)
- IR size—10 × 12 inches (24 × 30 cm), portrait
- Nongrid
- kVp range: 60–75



Shielding Shield radiosensitive tissues outside region of interest.

Patient Position Place patient in the supine position; place pillow under patient's head; legs should be fully extended.

Part Position

- Center and align ankle joint to CR and to long axis of portion of IR being exposed (Fig. 6.78).
- Do not force dorsiflexion of the foot; allow it to remain in its natural position (see NOTE 1).
- Adjust the **foot and ankle** for a **true AP projection.** Ensure that the entire lower leg is not rotated. The intermalleolar line should not be parallel to IR (see NOTE 2).

CR

- CR perpendicular to IR, directed to a point midway between malleoli

Recommended Collimation Collimate to lateral skin margins; include proximal one-half of metatarsals and distal tibia-fibula.

Digital Imaging Systems Close collimation is important over unexposed portions of IR to prevent fogging from scatter radiation.

NOTE 1: Forced dorsiflexion of the foot can be painful and may cause additional injury.

NOTE 2: The malleoli are *not* the same distance from the IR in the anatomic position with a true AP projection. (The lateral malleolus is approximately 15° more posterior.) The lateral portion of the mortise joint should *not* appear open. If this portion of the ankle joint does appear open on a true AP, it may suggest spread of the ankle mortise from ruptured ligaments.[1]

Fig 6.78 AP ankle.

Fig 6.79 AP ankle. (Courtesy E. Frank, RT[R], FASRT.)

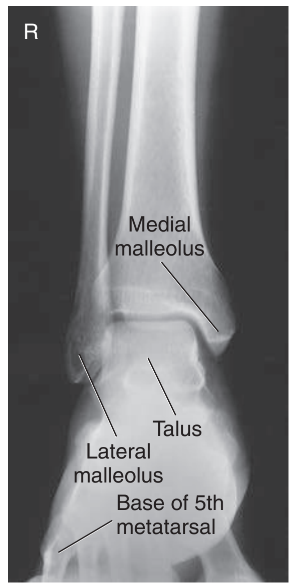


Fig 6.80 AP ankle. (Courtesy E. Frank, RT[R], FASRT.)

Evaluation Criteria

Anatomy Demonstrated: • Distal one-third of tibia-fibula, lateral and medial malleoli, and talus and proximal half of metatarsals should be demonstrated (Figs. 6.79 and 6.80).

Position: • Long axis of the leg should be aligned to collimation field and to IR. • **No rotation** if the medial mortise joint is open and the lateral mortise is closed. • Some superimposition of the distal fibula by the distal tibia and talus exists. • Four-sided collimation should include the distal one-third of the lower leg to the proximal half of the metatarsals. • All surrounding soft tissue also should be included.

Exposure: • Optimal exposure with **no motion** demonstrates clear bony margins and trabecular markings. • Talus must be penetrated enough to demonstrate the cortical margins and trabeculae of the bone. • Soft tissue structures also must be visible.

6

AP MORTISE PROJECTION—15° TO 20° MEDIAL ROTATION: ANKLE

Ankle
ROUTINE
- AP
- AP mortise (15°)
- Lateral

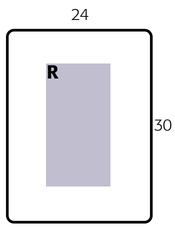
SPECIAL
- Oblique (45°)
- AP stress

Clinical Indications

- Evaluation of pathology involving the entire ankle mortise[1] and the proximal fifth metatarsal, a common fracture site. This is a common projection taken during open reduction surgery of the ankle (see NOTE).

Technical Factors

- Minimum SID—40 inches (100 cm)
- IR size—10 × 12 inches (24 × 30 cm), portrait
- Nongrid
- kVp range: 60–75



Shielding Shield radiosensitive tissues outside region of interest.

Patient Position Place patient in the supine position; place pillow under patient's head; legs should be fully extended.

Part Position

- Center and align ankle joint to CR and to long axis of portion of IR being exposed (Fig. 6.81).
- Do not dorsiflex foot; allow foot to remain in natural extended (plantar flexed) position (allows for visualization of base of fifth metatarsal, a common fracture site).[4]
- Internally rotate entire leg and foot approximately 15° to 20° until intermalleolar line is parallel to IR.
- Place support against foot if needed to prevent motion.

CR

- CR perpendicular to IR, directed midway between malleoli

Recommended Collimation Collimate to lateral skin margins, including proximal metatarsals and distal tibia-fibula.

Digital Imaging Systems Close collimation is important over unexposed portions of IR to prevent fogging from scatter radiation.

NOTE: This position should *not* be a substitute for either the AP projection or the oblique ankle position but rather should be a separate projection of the ankle that is taken routinely when potential trauma or sprains of the ankle joint are involved.[1]

Fig 6.81 Mortise projection, demonstrating 15° to 20° medial rotation of lower leg and foot.

Fig 6.82 Mortise projection.

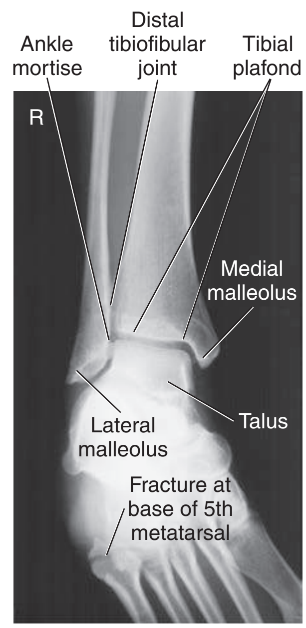


Fig 6.83 Mortise projection.

Evaluation Criteria

Anatomy Demonstrated: • Distal one-third of tibia and fibula, tibial plafond involving the epiphysis if present, lateral and medial malleoli, talus, and proximal half of the metatarsals should be demonstrated. • Entire ankle mortise should be open and well visualized (3- to 4-mm space over entire talar surface is normal; an extra 2 mm of widening is abnormal)[2] (Figs. 6.82 and 6.83).

Position: • Proper obliquity for the mortise joint is evidenced by demonstration of open lateral and medial mortise joints with malleoli demonstrated in profile. • Only minimal superimposition should exist at distal tibiofibular joint. • Collimation to **area of interest.**

Exposure: • **No motion** as demonstrated by sharp bony outlines and trabecular markings. • Optimal exposure should demonstrate soft tissue structures and sufficient density (brightness) for talus and distal tibia and fibula.

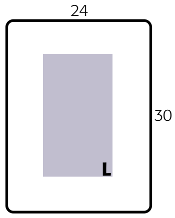
AP OBLIQUE PROJECTION—45° MEDIAL ROTATION: ANKLE

Ankle
ROUTINE
- AP
- AP mortise (15°)
- Lateral

SPECIAL
- Oblique (45°)
- AP stress

Clinical Indications
- Pathologies including possible fractures involving distal tibiofibular joint
- Fractures of distal fibula and lateral malleolus and base of the fifth metatarsal

Technical Factors
- Minimum SID—40 inches (100 cm)
- IR size—10 × 12 inches (24 × 30 cm), portrait
- Nongrid
- kVp range: 60–75



Shielding Shield radiosensitive tissues outside region of interest.

Patient Position Place patient in the supine position; place pillow under patient's head; legs should be fully extended (small sandbag or other support under knee increases comfort of patient).

Part Position
- Center and align ankle joint to CR and to long axis of portion of IR being exposed (Fig. 6.84).
- If patient's condition allows, dorsiflex the foot if needed so that the plantar surface is at least 80° to 85° from IR (10° to 15° from vertical). (See NOTE 1.)
- **Rotate leg and foot** medially 45°.

CR
- CR perpendicular to IR, directed to a point midway between malleoli

Recommended Collimation Collimate to include distal tibia and fibula to midmetatarsal area (see NOTE 2).

Digital Imaging Systems Close collimation is important over unexposed portions of IR to prevent fogging from scatter radiation.

NOTE 1: If the foot is extended or plantar flexed more than 10° or 15° from vertical, the calcaneus is superimposed over the lateral malleolus on this 45° oblique, obscuring an important area of interest.

NOTE 2: The base of the fifth metatarsal (a common fracture site) may be demonstrated in this projection and should be included in the collimation field. If fracture of the fifth metatarsal is suspected, a foot series should be ordered by physician.

Fig 6.84 45° AP medial oblique.

Fig 6.85 45° AP medial oblique. (Courtesy E. Frank, RT[R], FASRT.)

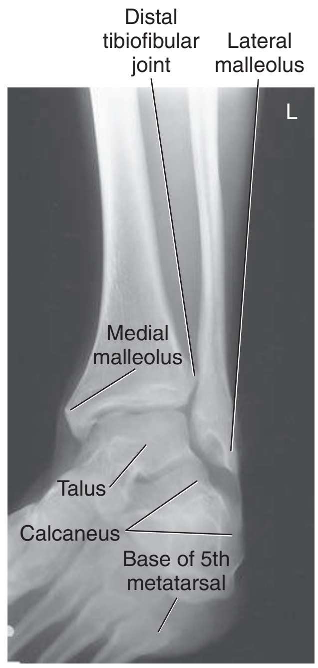


Fig 6.86 45° AP medial oblique. (Courtesy E. Frank, RT[R], FASRT.)

Evaluation Criteria
Anatomy Demonstrated: • Distal one-third of lower leg, malleoli, talus, and proximal half of metatarsals should be seen (Figs. 6.85 and 6.86).
Position: • A 45° medial oblique demonstrates distal tibiofibular joint open, with no or only minimal overlap on the average person. • Lateral malleolus and talus joint should show no or only slight superimposition, but medial malleolus and talus are partially superimposed. • Ankle joint should be in center of four-sided collimated field with distal one-third of lower leg to proximal half of metatarsals and surrounding soft tissues included.
Exposure: • Bony cortical margins and trabecular patterns should be sharply defined on image if no motion is present. • The talus should be sufficiently penetrated to demonstrate trabeculae; soft tissue structures also must be evident.

LATERAL—MEDIOLATERAL (OR LATEROMEDIAL) PROJECTION: ANKLE

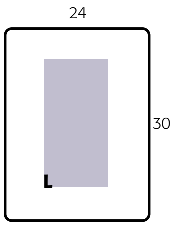
Ankle

ROUTINE
- AP
- AP mortise (15°)
- Lateral

SPECIAL
- Oblique (45°)
- AP stress

Clinical Indications
- Projection is useful in the evaluation of fractures, dislocations, and joint effusions associated with other joint pathologies

Technical Factors
- Minimum SID—40 inches (100 cm)
- IR size—10 × 12 inches (24 × 30 cm), portrait
- Nongrid
- kVp range: 60–75



Shielding Shield radiosensitive tissues outside region of interest.

Patient Position Place patient in the lateral recumbent position, affected side down; provide a pillow for patient's head; flex knee of affected limb approximately 45°; place opposite leg behind injured limb to prevent over-rotation.

Part Position (Mediolateral Projection)
- Center and align ankle joint to CR and to long axis of portion of IR being exposed (Fig. 6.87).
- Place support under knee as needed to place leg and foot in **true lateral position.**
- Dorsiflex foot so that plantar surface is at a right angle to leg or as far as patient can tolerate; do *not* force. (This helps maintain a true lateral position.)

CR
- CR perpendicular to IR, directed to medial malleolus

Recommended Collimation Collimate to include distal tibia and fibula to midmetatarsal area.

Digital Imaging Systems Close collimation is important over unexposed portions of IR to prevent fogging from scatter radiation.

Alternative Lateromedial Projection This lateral may be performed rather than the more common mediolateral projection (Fig. 6.88). (This position is more uncomfortable for the patient but may make it easier to achieve a true lateral position.)

Evaluation Criteria

Anatomy Demonstrated: • Distal one-third of tibia and fibula with the distal fibula superimposed by the distal tibia, talus, and calcaneus appear in lateral profile. • Tuberosity of fifth metatarsal, navicular, and cuboid also are visualized (Figs. 6.89 and 6.90).

Position: • **No rotation** is evidenced by distal fibula being superimposed over the posterior half of tibia. • Tibiotalar joint is open with uniform joint space. • Collimation field should include distal one-third of lower leg, calcaneus, tuberosity of fifth metatarsal, and surrounding soft tissue structures. • Collimation to **area of interest.**

Exposure: • **No motion,** as evidenced by sharp bony margins and trabecular patterns. • Lateral malleolus should be seen through the distal tibia and talus, and soft tissue must be demonstrated for evaluation of joint effusion.

Fig 6.87 Mediolateral ankle.

Fig 6.88 Alternative lateromedial ankle.

Fig 6.89 Mediolateral ankle.

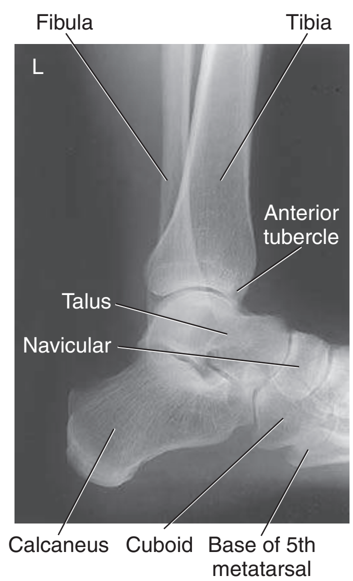


Fig 6.90 Mediolateral ankle.

AP STRESS PROJECTIONS: ANKLE

INVERSION AND EVERSION POSITIONS

WARNING: Proceed with utmost care with injured patient.

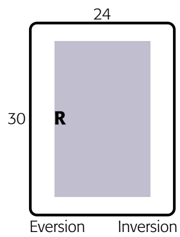
Clinical Indications

- Pathology involving ankle joint separation secondary to ligament tear or rupture

Ankle
SPECIAL
- **Oblique (45°)**
- **AP stress**

Technical Factors

- Minimum SID—40 inches (100 cm)
- IR size—10 × 12 inches (24 × 30 cm), portrait
- Nongrid
- kVp range: 60–75



Shielding Shield radiosensitive tissues outside region of interest. Supply lead gloves and a lead apron for the individual who is applying stress if stress positions are handheld during exposures.

Patient Position Place patient in supine position; place pillow under patient's head; leg should be fully extended, with support under knee.

Part Position

- Center and align ankle joint to CR and to long axis of portion of IR being exposed.
- Dorsiflex the foot as near the right angle to the lower leg as possible.
- Stress is applied with leg and ankle in position for a **true AP** with no rotation, wherein the entire plantar surface is turned medially for inversion and laterally for eversion (Figs. 6.91 and 6.92) (see NOTE).

CR

- CR perpendicular to IR, directed to a point midway between malleoli

Recommended Collimation Collimate to lateral skin margins, including proximal metatarsals and distal tibia-fibula.

Digital Imaging Systems Close collimation is important over unexposed portions of IR to prevent fogging from scatter radiation.

NOTE: A physician or another health professional must be present to hold the foot and ankle in these stress views (or to strap into position with weights), or patient must hold this position with long gauze looped around ball of foot. If this is too painful for the patient, a local anesthetic may be injected by the physician.

Fig 6.91 AP ankle—inversion stress.

Fig 6.92 AP ankle—eversion stress.

Fig 6.93 Inversion stress.

Fig 6.94 Eversion stress.

Evaluation Criteria

Anatomy Demonstrated and Position: • Ankle joint for evaluation of joint separation and ligament tear or rupture is shown. • Appearance of joint space may vary greatly depending on the severity of ligament damage. • Collimation to **area of interest** (Figs. 6.93 and 6.94).

Exposure: • **No motion,** as evidenced by sharp bony margins and trabecular patterns. • Optimal exposure should visualize soft tissue, lateral and medial malleoli, talus, and distal tibia and fibula.

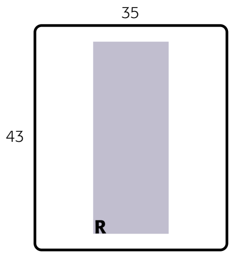
AP PROJECTION: LOWER LEG (TIBIA AND FIBULA)

6

Lower Leg
- AP
- Lateral

Clinical Indications
- Pathologies involving fractures, foreign bodies, or lesions of the bone

Technical Factors
- Minimum SID—40 inches (100 cm); may increase to 44 to 48 inches (110 to 120 cm) to reduce divergence of x-ray beam and to include more of body part
- IR size—14 × 17 inches (35 × 43 cm), portrait (or diagonal, which requires 44-inch [110-cm] minimum SID)
- Nongrid (unless lower leg measures >10 cm)
- kVp range: 70–80
- To make best use of anode heel effect, place knee at cathode end of x-ray beam



Shielding Shield radiosensitive tissues outside region of interest.

Patient Position Place patient in the supine position; provide a pillow for patient's head; entire leg should be fully extended.

Part Position
- Adjust pelvis, knee, and leg into true AP with no rotation (Fig. 6.95).
- Place sandbag against foot if needed for stabilization, and dorsiflex foot to 90° to lower leg if possible.
- Ensure that both ankle and knee joints are 1 to 2 inches (2.5 to 5 cm) from ends of IR (so that divergent rays do not project either joint off IR).
- If limb is too long, place the lower leg diagonally (corner to corner) on one 14 × 17 inch (35 × 43 cm) IR to ensure that both joints are included. (Also, if needed, a second smaller IR may be taken of the joint farthest from the injury site.)

CR
- CR perpendicular to IR, directed to midpoint of lower leg

Recommended Collimation Collimate on both sides to skin margins, with full collimation at ends of IR borders to include maximum knee and ankle joints.

Evaluation Criteria
Anatomy Demonstrated: • Entire tibia and fibula must include ankle and knee joints on this projection (or two if needed). • The exception is an alternative routine on follow-up examinations (Figs. 6.96 and 6.97).
Position: • No rotation as evidenced by demonstration of femoral and tibial condyles in profile with intercondylar eminence centered within intercondylar fossa. • Some overlap of the fibula and tibia is visible at both proximal and distal ends. • Collimation to **area of interest.**
Exposure: • Correct use of anode heel effect results in an image with nearer equal density at both ends of IR. • **No motion** is present, as evidenced by sharp cortical margins and trabecular patterns. • Contrast and density (brightness) should be optimum to visualize soft tissue and bony trabecular markings at both ends of tibia.

Alternative Follow-Up Examination Routine The routine for follow-up examinations of long bones in some departments is to include only the joint that is nearest the site of injury and to place this joint a minimum of 2 inches (5 cm) from the end of the IR for better demonstration of the joint. **However, for initial examinations, it is important, especially when the injury site is in the distal leg, to include the proximal tibiofibular joint area** because it is common to have a second fracture at this site. For very large patients, a second AP projection of the knee and proximal lower leg may be needed on a smaller IR.

Fig 6.95 AP lower leg—include both joints.

Fig 6.96 AP lower leg—both joints. (Courtesy J. Sanderson, RT.)

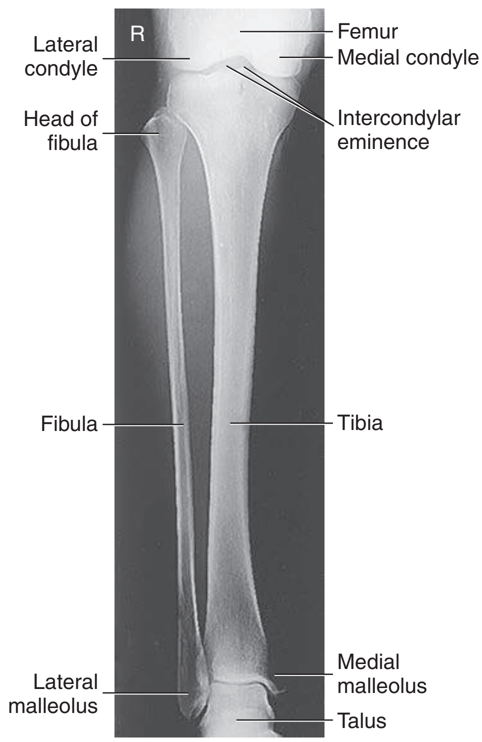


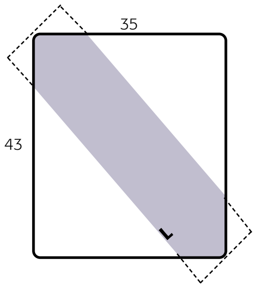
Fig 6.97 AP lower leg—both joints. (Courtesy J. Sanderson, RT.)

LATERAL—MEDIOLATERAL PROJECTION: LOWER LEG (TIBIA AND FIBULA)

Lower Leg
- AP
- Lateral

Clinical Indications
- Localization of lesions and foreign bodies and determination of extent
- Alignment of fractures demonstrated

Technical Factors
- Minimum SID—40 inches (100 cm); may increase to 44 to 48 inches (110 to 120 cm) to reduce divergence of x-ray beam and to include more of body part
- IR size—14 × 17 inches (35 × 43 cm), portrait (or diagonal, which requires 44-inch [110-cm] minimum SID)
- Nongrid (unless lower leg measures >10 cm)
- kVp range: 65–80
- To make best use of anode heel effect, place knee at cathode end of x-ray beam



Diagonal placement

Shielding Shield all radiosensitive tissues outside region of interest.

Patient Position Place patient in the lateral recumbent position, injured side down; the opposite leg may be placed behind the affected leg and supported with a pillow or sandbags.

Part Position
- Ensure that leg is in true lateral position (plane of patella should be perpendicular to IR) (Fig. 6.98).
- Ensure that both ankle and knee joints are 1 to 2 inches (2.5 to 5 cm) from ends of IR so that divergent rays do not project either joint off IR.
- If limb is too long, place the lower leg diagonally (corner to corner) on one 14 × 17 inch (35 × 43 cm) IR to ensure that both joints are included (Fig. 6.98, *inset*). (Also, if needed, a second, smaller IR may be taken of the joint furthest from the injury site.)

CR
- CR perpendicular to IR, directed to midpoint of lower leg

Evaluation Criteria
Anatomy Demonstrated: • Entire tibia and fibula must include ankle and knee joints on this projection (or two if needed). • Exception is alternative routine on follow-up examinations (Fig. 6.99).
Position: • True lateral of tibia and fibula **without rotation** demonstrates tibial tuberosity in profile, a portion of the proximal head of the fibula superimposed by the tibia, and outlines of the distal fibula seen through posterior half of the tibia. • Posterior borders of femoral condyles should appear superimposed. • Collimation to **area of interest.**
Exposure: • **No motion** is present, as evidenced by sharp cortical margins and trabecular patterns. • Correct use of the anode heel effect results in near-equal density at both ends of the image. • Contrast and density (brightness) should be optimum to visualize soft tissue and bony trabecular markings.

Recommended Collimation Collimate on both sides to skin margins, with full collimation at ends to include maximum knee and ankle joints.

Alternative Follow-Up Examination Routine The routine for follow-up examinations of long bones in some departments is to include only the one joint nearest the site of injury and to place this joint a minimum of 2 inches (5 cm) from the end of the IR for better demonstration of the joint. However, for initial examinations, it is especially important when the injury site is in the distal lower leg to include the proximal tibiofibular joint area because it is common to have a second fracture at this site.

Horizontal Beam (Cross-Table) Lateral If patient cannot be turned, this image can be taken cross-table with IR placed on edge between lower legs. Place a support under injured leg to center the lower leg to IR, and direct horizontal beam from lateral side of patient.

Fig 6.98 Mediolateral lower leg—include both joints.

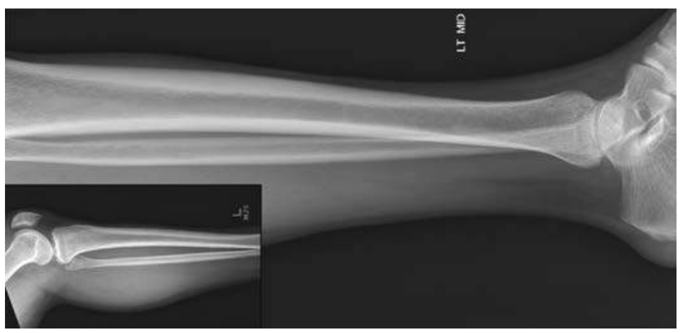


Fig 6.99 Mediolateral lower leg projection. Inset shows proximal lower leg to show both joints are included.

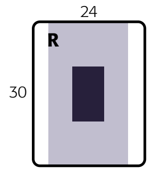
6

AP PROJECTION: KNEE

Knee
ROUTINE
- AP
- Oblique (medial and lateral)
- Lateral

Clinical Indications
- Fractures, lesions, or bony changes related to degenerative joint disease involving the distal femur, proximal tibia and fibula, patella, and knee joint

Technical Factors
- Minimum SID—40 inches (100 cm)
- IR size—10 × 12 inches (24 × 30 cm), portrait
- Grid or bucky, >10 cm
- Nongrid, tabletop, <10 cm
- kVp range: 65–80



Shielding Shield radiosensitive tissues outside region of interest.

Patient Position Place patient in supine position with no rotation of pelvis; provide pillow for patient's head; leg should be fully extended.

Part Position
- Align and center leg and knee to CR and to midline of table or IR (Fig. 6.100).
- Rotate leg internally 3° to 5° for true AP knee (or until **interepicondylar line is parallel** to plane of IR).
- Place sandbags by foot and ankle to stabilize if needed.

CR
- Align CR **parallel to articular facets (tibial plateau);** for average-sized patient, CR is perpendicular to IR (see NOTE).
- Direct CR to a point **½ inch (1.25 cm)** distal to apex of patella.

Recommended Collimation Collimate on both sides to skin margins at ends to IR borders.

NOTE: A suggested guideline for determining that CR is parallel to articular facets (tibial plateau) for open joint space is to measure distance from anterior superior iliac spines (ASIS) to tabletop to determine CR angle as follows[5]:
- <19 cm: **5° caudad** (thin thighs and buttocks)
- 19 to 24 cm: **0° angle** (average thighs and buttocks)
- >24 cm: **5° cephalad** (thick thighs and buttocks)

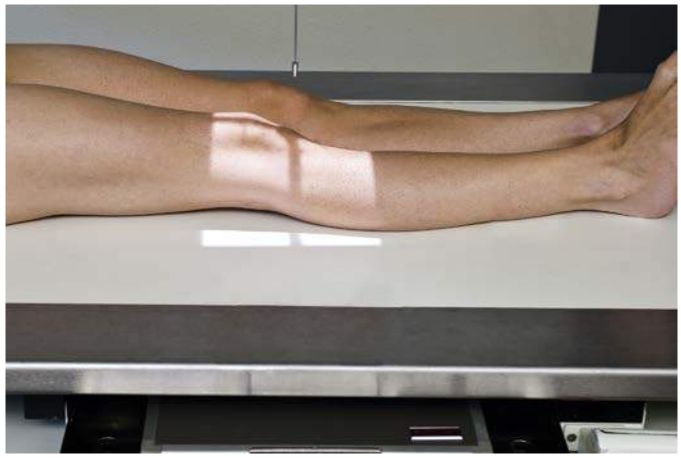


Fig 6.100 AP knee—CR perpendicular to IR (average patient—19 to 24 cm).

Fig 6.101 AP knee—0° CR. (Courtesy Joss Wertz, DO.)

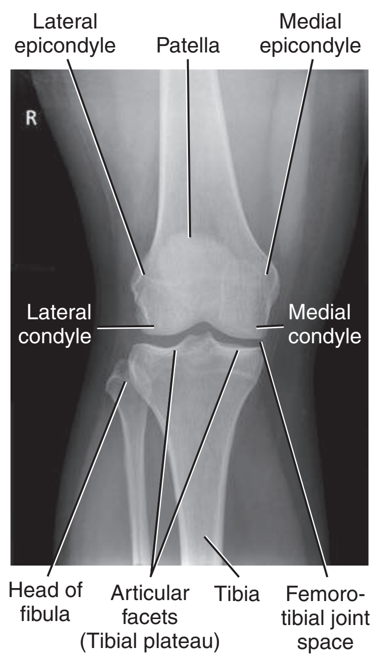


Fig 6.102 AP knee—0° CR. (Courtesy Joss Wertz, DO.)

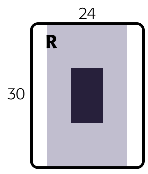
Evaluation Criteria
Anatomy Demonstrated: • Distal femur and proximal tibia and fibula are shown. • Femorotibial joint space should be open, with the articular facets of the tibia seen on end with only minimal surface area visualized (Figs. 6.101 and 6.102).
Position: • **No rotation,** as evidenced by symmetric appearance of femoral and tibial condyles and the joint space. • The approximate medial half of the fibular head should be superimposed by tibia. • The intercondylar eminence is seen in the center of intercondylar fossa. • Center of collimation field (CR) should be to the midknee joint space.
Exposure: • Optimal exposure visualizes the outline of the patella through the distal femur, and the fibular head and neck do not appear overexposed. • **No motion** should occur; trabecular markings of all bones should be visible and appear sharp. • Soft tissue detail should be visible.

AP OBLIQUE PROJECTION—MEDIAL (INTERNAL) ROTATION: KNEE

Knee
ROUTINE
- AP
- Oblique (medial and lateral)
- Lateral

Clinical Indications

- Pathology involving the proximal tibiofibular and femorotibial (knee) joint articulations
- Fractures, lesions, and bony changes related to degenerative joint disease, especially on the anterior and medial or posterior and lateral portions of knee



NOTE: A common departmental routine is to include *both* medial and lateral rotation oblique projections for the knee. If only one oblique is routine, it is most commonly the medial rotation oblique.

Technical Factors

- Minimum SID—40 inches (100 cm)
- IR size—10 × 12 inches (24 × 30 cm), portrait
- Grid or bucky, >10 cm
- Nongrid, tabletop, <10 cm
- kVp range: 65–80

Shielding Shield radiosensitive tissues outside region of interest.

Patient Position Place patient in semisupine position with entire body and leg rotated partially away from side of interest; place support under elevated hip; provide a pillow for patient's head.

Part Position

- Align and center leg and knee to CR and to midline of table or IR.
- Rotate entire leg **internally 45°.** (Interepicondylar line should be 45° to plane of IR.) (Fig. 6.103)
- If needed, stabilize foot and ankle in this position with sandbags.

CR

- Angle CR 0° on average patient (see AP Knee, p. 246).
- Direct CR to midpoint of knee at a level ½ inch (1.25 cm) distal to apex of patella.

Recommended Collimation Collimate on both sides to skin margins, with full collimation at ends to IR borders to include maximum femur and tibia-fibula.

NOTE: The terms *medial (internal) oblique* and *lateral (external) oblique positions* refer to the direction of rotation of the anterior or patellar surface of the knee. This is true for descriptions of AP or PA oblique projections.

Evaluation Criteria

Anatomy Demonstrated: • Distal femur and proximal tibia and fibula with the patella superimposing the medial femoral condyle are shown. • **Lateral condyles** of the femur and tibia are well demonstrated, and the medial and lateral knee joint spaces appear unequal (Figs. 6.104 and 6.105).
Position: • The proper amount of part obliquity demonstrates the proximal tibiofibular articulation open with the lateral condyles of the femur and tibia seen in profile. • The head and neck of the fibula are visualized without superimposition, and approximately half of the patella should be seen free of superimposition by the femur. The center of the collimated field is to the **femorotibial (knee) joint space.**
Exposure: • Optimal exposure with **no motion** should visualize soft tissue in the knee joint area, and trabecular markings of all bones should appear clear and sharp. • Head and neck area of fibula should not appear overexposed.

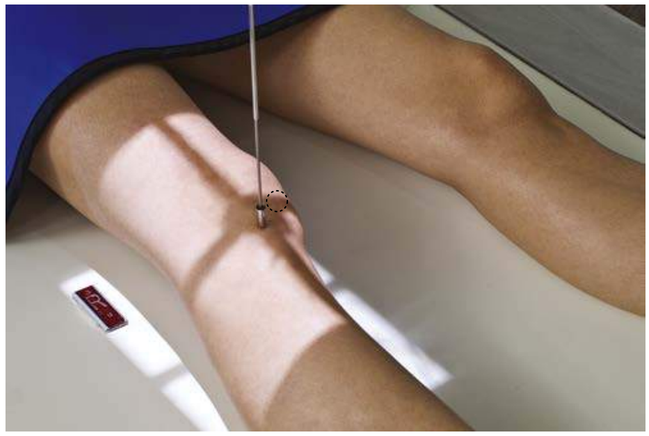


Fig 6.103 AP 45° medial oblique.

Fig 6.104 AP medial oblique.

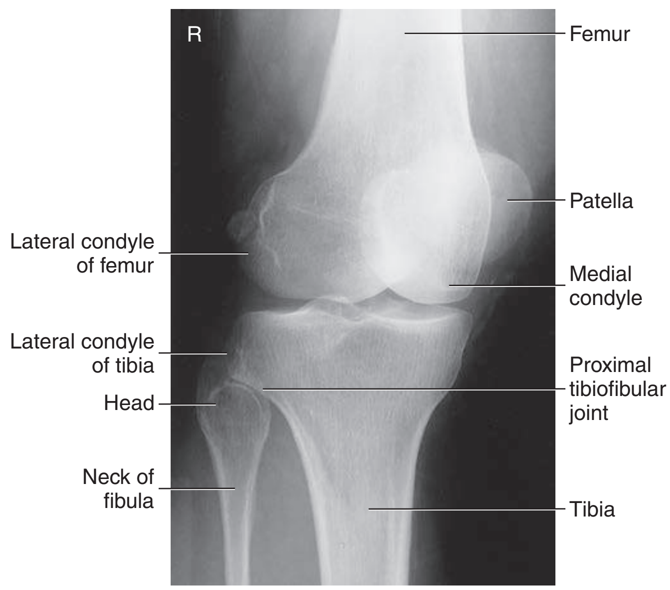


Fig 6.105 AP medial oblique.

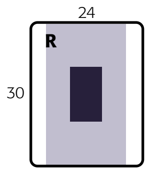
AP OBLIQUE PROJECTION—LATERAL (EXTERNAL) ROTATION: KNEE

6

Knee
ROUTINE
- AP
- Oblique (medial and lateral)
- Lateral

Clinical Indications

- Pathology involving femorotibial (knee) articulation
- Fractures, lesions, and bony changes related to degenerative joint disease, especially on anterior and lateral or posterior and medial aspects of knee



NOTE: A common departmental routine is to include *both* medial and lateral rotation oblique projections of the knee. If only one oblique is routine, it is most commonly the medial (internal) rotation oblique.

Technical Factors

- Minimum SID—40 inches (100 cm)
- IR size—10 × 12 inches (24 × 30 cm), portrait/grid or bucky, >10 cm
- Nongrid, tabletop, <10 cm
- kVp range: 65–80

Shielding Shield radiosensitive tissues outside region of interest.

Patient Position Place patient in semisupine position, with entire body and leg rotated partially away from side of interest; place support under elevated hip; give pillow for head.

Part Position

- Align and center leg and knee to CR and to midline of table or IR.
- Rotate entire leg **externally 45°** (interepicondylar line should be 45° to plane of IR) (Fig. 6.106).
- If needed, stabilize foot and ankle in this position with sandbags.

CR

- Angle CR 0° on average patient (see AP Knee Projection, p. 246).
- Direct CR to midpoint of knee at a level ½ inch (1.25 cm) distal to apex of patella.

Recommended Collimation Collimate on both sides to skin margins, with full collimation at ends to IR borders to include maximum femur and tibia-fibula.

NOTE: The terms *medial (internal) oblique* and *lateral (external) oblique positions* refer to the direction of rotation of the anterior or patellar surface of the knee. This is true for descriptions of either AP or PA oblique projections.

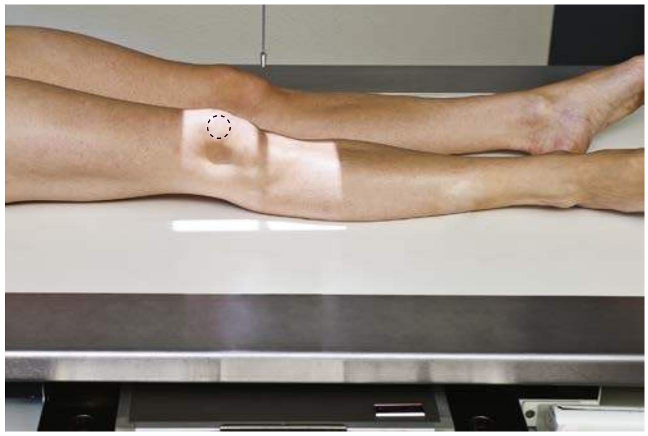


Fig 6.106 AP lateral oblique.

Fig 6.107 AP lateral oblique. (Courtesy Joss Wertz, DO.)

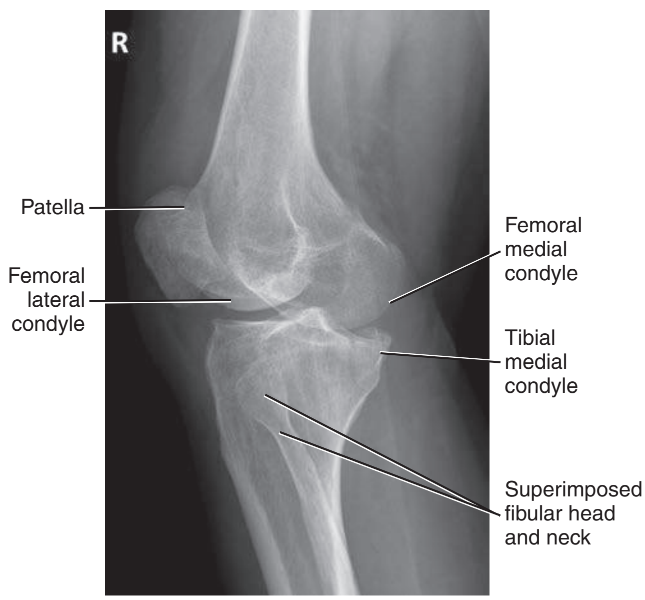


Fig 6.108 AP lateral oblique. (Courtesy Joss Wertz, DO.)

Evaluation Criteria

Anatomy Demonstrated: • Distal femur and proximal tibia and fibula, with the patella superimposing the lateral femoral condyle, are shown. • Medial condyles of the femur and tibia are demonstrated in profile (Figs. 6.107 and 6.108).

Position: • The proper amount of part obliquity demonstrates the proximal fibula superimposed by the proximal tibia, the medial condyles of the femur, and the tibia seen in profile. • Approximately half of patella should be seen free of superimposition by the femur. • Femorotibial (knee) joint space is the center of the collimated field.

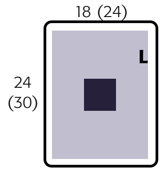
Exposure: • Optimal exposure should visualize soft tissue in knee joint area, and trabecular markings of all bones should appear clear and sharp, indicating **no motion.** • Technique should be sufficient to demonstrate the head and neck area of the fibula through the superimposed tibia.

LATERAL—MEDIOLATERAL PROJECTION: KNEE

Knee
ROUTINE
- AP
- Oblique
- **Lateral**

Clinical Indications
- Fractures, lesions, and joint space abnormalities

Technical Factors
- Minimum SID—40 inches (100 cm)
- IR size—8 × 10 inches (18 × 24 cm) or 10 × 12 inches (24 × 30 cm), portrait
- Grid or bucky, >10 cm
- Nongrid, tabletop, <10 cm
- kVp range: 65–80



Shielding Shield radiosensitive tissues outside region of interest.

Patient Position This position may be taken in the lateral recumbent position or as a horizontal beam lateral.

Lateral Recumbent Position This projection is designed for patients who are able to flex the knee 20° to 30°. Take radiograph with patient in lateral recumbent position, affected side down; provide pillow for patient's head; provide support for knee of opposite limb placed behind knee being examined to prevent over-rotation (Fig. 6.109).

Horizontal Beam Projection This lateromedial beam projection is ideal for a patient who is unable to flex the knee because of pain or trauma. With the patient in the prone position, use a horizontal beam with IR placed beside knee. Place support under knee to avoid obscuring posterior soft tissue structures (see Fig. 6.109, *inset*).

Part Position
- Adjust rotation of body and leg until knee is in **true lateral** position (femoral epicondyles directly superimposed and plane of patella perpendicular to plane of IR).
- Flex knee **20° to 30°** for lateral recumbent projection (see NOTE 1).
- Align and center leg and knee to CR and to midline of table or IR.

CR
- Angle CR **5° to 7° cephalad** for lateral recumbent projection (see NOTES 2 and 3).
- Direct CR to a point **1 inch (2.5 cm) distal** to medial epicondyle.

Evaluation Criteria
Anatomy Demonstrated: • Distal femur, proximal tibia and fibula, and patella are shown in lateral profile. • Patellofemoral and knee joints should be open (Figs. 6.110 and 6.111).
Position: • Over-rotation or under-rotation can be determined by identifying the adductor tubercle on the medial condyle, if visible (see Fig. 6.33), and by the amount of superimposition of the fibular head by the tibia (over-rotation, less superimposition of fibular head; under-rotation, more superimposition). • **True lateral** position of knee without rotation demonstrates **posterior borders** of the femoral condyles directly superimposed. • Patella should be seen in profile with patellofemoral joint space open. • The 5° to 10° cephalad angle of CR should result in direct superimposition of the distal borders of the condyles. • Knee joint is in center of collimated field.
Exposure: • Optimal exposure with **no motion** visualizes important soft tissue detail, including fat pad region anterior to knee joint and sharp trabecular markings.

Recommended Collimation Collimate on both sides to skin margins, with full collimation at ends to IR borders to include maximum femur, tibia, and fibula.

NOTE 1: Additional flexion tightens muscles and tendons that may obscure important diagnostic information in the joint space. The patella is drawn into the intercondylar sulcus, also obscuring soft tissue detail from effusion or fat pad displacement. Additional flexion may result in fragment separation of patellar fractures, if present.

NOTE 2: Angle CR 7° to 10° on a short patient with a wide pelvis and about 5° on a tall, male patient with a narrow pelvis for lateral recumbent projection.

NOTE 3: If the ankle and lower leg can be elevated to the same plane as the long axis of the femur, a perpendicular CR can be used.

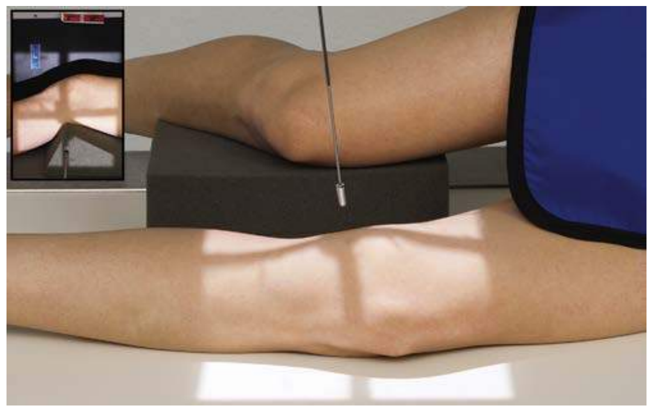


Fig 6.109 Mediolateral knee. *Inset,* Lateromedial—horizontal beam.

Fig 6.110 Mediolateral knee. (Courtesy Joss Wertz, DO.)

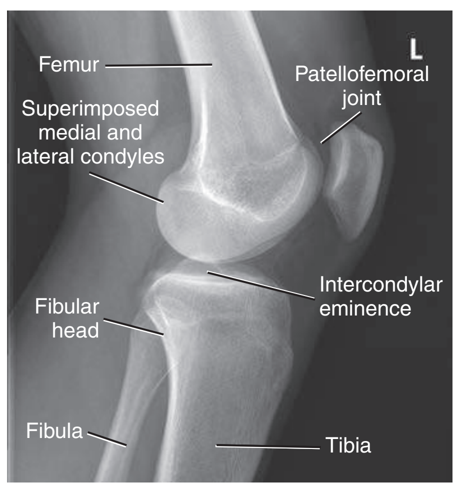


Fig 6.111 Mediolateral knee. (Courtesy Joss Wertz, DO.)

6

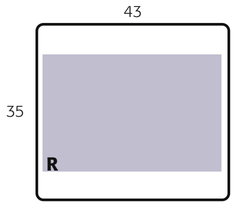
AP WEIGHT-BEARING BILATERAL KNEE PROJECTION: KNEE

Knee
SPECIAL
- **AP bilateral weight-bearing**

Clinical Indications

- Femorotibial joint spaces of the knees demonstrated for possible cartilage degeneration or other knee joint pathologies
- Bilateral knees included on same exposure for comparison

NOTE: This projection most commonly is taken AP but may be taken PA with a cephalic CR angle rather than caudal as with an AP. (This may be easier for patients who are unable to straighten their knee joints fully, such as patients with arthritic conditions or with certain neuromuscular disorders involving the lower limbs.)



Technical Factors

- Minimum SID—40 inches (100 cm)
- IR size—14 × 17 inches (35 × 43 cm), landscape
- Grid
- kVp range: 70–80

Shielding Shield radiosensitive tissues outside region of interest.

Patient and Part Position

- Position patient erect and standing on attached step or on step stool to place patient high enough for horizontal beam x-ray tube.
- Position feet straight ahead with weight evenly distributed on both feet; provide support handles for patient stability.
- Align and center bilateral legs and knees to CR and to midline of table and IR; IR height is adjusted to CR (Fig. 6.112).

CR

- CR perpendicular to IR (average-sized patient), or 5° to 10° caudad on thin patient, directed to midpoint between knee joints at a level ½ inch (1.25 cm) below apex of patellae.

Recommended Collimation Collimate to bilateral knee joint region, including some distal femurs and proximal tibia for alignment purposes.

Evaluation Criteria

Anatomy Demonstrated: • Distal femur, proximal tibia, and fibula and femorotibial joint spaces are demonstrated bilaterally (Fig. 6.113).

Position: • **No rotation** of both knees is evident by symmetric appearance of femoral and tibial condyles. • Approximately one-half of proximal fibula is superimposed by tibia. • Collimation field should be centered to knee joint spaces and should include sufficient femur and tibia to determine long axes of these long bones for alignment.

Exposure: • Optimal exposure should visualize faint outlines of patellae through femora. • Soft tissue should be visible, and trabecular markings of all bones should appear clear and sharp, indicating **no motion.**

Alternative PA If requested, an alternative PA may be performed with patient facing the table or IR holder, knees flexed at approximately 20°, feet straight ahead, and thighs against tabletop or IR holder. Direct CR **10° caudad** (parallel to tibial plateaus) to **level of knee joints** for PA projection.

NOTE: CR angle should be parallel to articular facets (tibial plateau) for best visualization of open knee joint spaces. See AP Knee Projection, p. 246, for correct CR angle.

Fig 6.112 AP bilateral weight-bearing—CR perpendicular to IR.

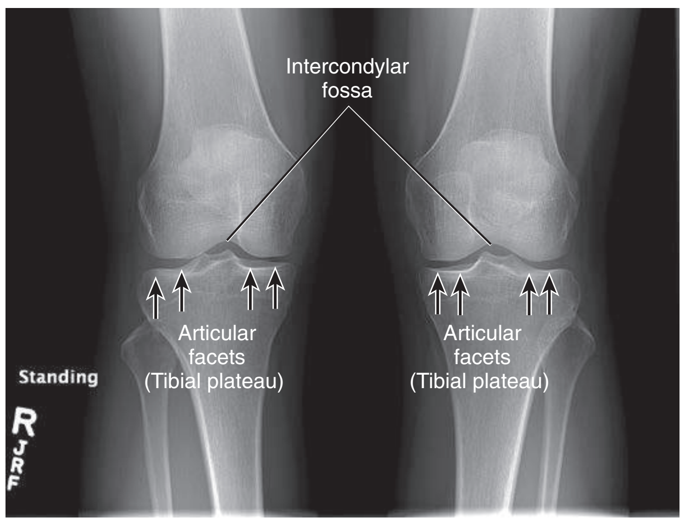
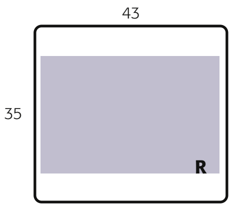


Fig 6.113 AP bilateral weight-bearing—CR 10° caudad. (Courtesy Joss Wertz, DO.)

PA AXIAL WEIGHT-BEARING BILATERAL KNEE PROJECTION: KNEE

ROSENBERG METHOD

Knee
SPECIAL
- AP bilateral weight-bearing
- PA axial bilateral weight-bearing



Clinical Indications

- Femorotibial joint spaces of the knees demonstrated for possible cartilage degeneration or other knee joint pathologies
- Knee joint spaces and intercondylar fossa demonstrated
- Bilateral knees included on same exposure for comparison

Technical Factors

- Minimum SID—40 inches (100 cm)
- IR size—14 × 17 inches (35 × 43 cm), landscape
- Grid
- kVp range: 70–80

Shielding Shield radiosensitive tissues outside region of interest.

Patient and Part Position

- Position patient erect, standing on attached step of x-ray table or on step stool if the upright bucky is used so that patient is placed high enough for 10° caudad angle.
- Position feet straight ahead with weight evenly distributed on both feet and knees flexed to 45°; have patient use bucky device for support, with patella touching the upright bucky (Fig. 6.114).
- Align and center bilateral legs and knees to CR and to midline of upright bucky and IR; IR height is adjusted to CR.

CR

- CR angled 10° caudad and centered directly to midpoint between knee joints at level ½ inch (1.25 cm) below apex of patellae when a bilateral study is performed (Fig. 6.115); alternatively, CR centered directly to midpoint of knee joint at level ½ inch (1.25 cm) below apex of patella when a unilateral study is performed.

Recommended Collimation Collimate to bilateral knee joint region, including some distal femurs and proximal tibia for alignment purposes.

Alternative Unilateral Projection If requested, this examination may be performed unilaterally with patient facing the upright bucky or IR holder, knees flexed to 45°, and feet straight ahead. The patient should put full weight on the affected extremity. This requires the patient to balance with minimal pressure placed on the contralateral side. Direct CR **10° caudad** (parallel to tibial plateau) to **level of knee joint** for this PA unilateral projection.

Fig 6.114 Rosenberg method. Position for standing 45° PA flexion weight-bearing for bilateral knees.

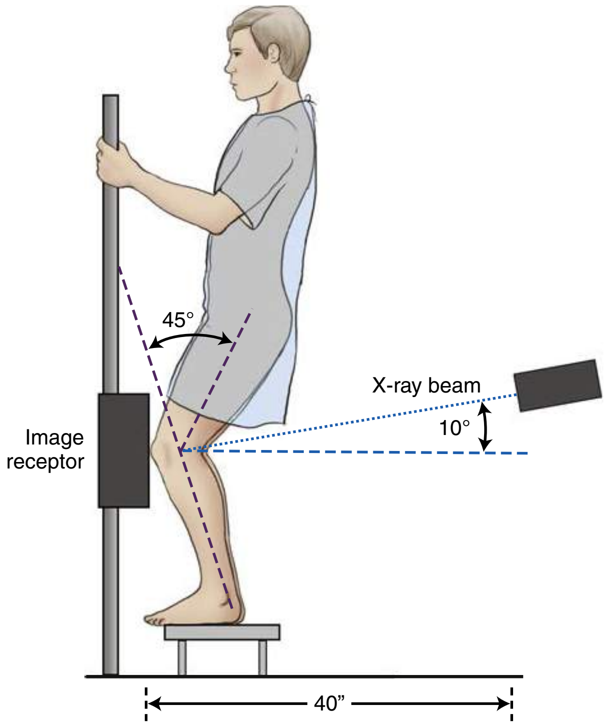


Fig 6.115 Rosenberg method—bilateral PA axial projection with 10° caudad.

Evaluation Criteria

Anatomy Demonstrated: • Distal femur, proximal tibia and fibula, femorotibial joint spaces, and intercondylar fossa are demonstrated bilaterally or unilaterally (Figs. 6.116 and 6.117).

Position: • **No rotation** of both knees is evident in symmetric appearance of femoral and tibial condyles. • Intercondylar fossa should be open. • Knee joint spaces should appear open if CR angle was correct and tibia was flexed 45°.

Exposure: • Optimal exposure should visualize intercondylar fossa and proximal tibia with open joint space • Trabecular markings of all bones should appear clear and sharp, indicating **no motion.**

Fig 6.116 Normal bilateral knee radiograph performed using the Rosenberg method. Both medial and lateral compartments show no significant narrowing.

Fig 6.117 Abnormal bilateral knee radiograph performed using the Rosenberg method. Note the obliterated lateral compartment of the left knee with associated joint space narrowing medially. (Reprinted with permission from Hobbs, DL: Osteoarthritis and the Rosenberg method, *Radiol Technol* 77:181, 2006.)

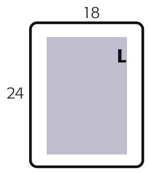
PA AND AP AXIAL PROJECTIONS ("TUNNEL VIEWS"): INTERCONDYLAR FOSSA

CAMP COVENTRY METHOD, HOLMBLAD METHOD (AND VARIATIONS), AND BÉCLERE METHOD

Knee—Intercondylar Fossa Projections
ROUTINE
- PA axial

Clinical Indications

- Intercondylar fossa, femoral condyles, tibial plateaus, and intercondylar eminence demonstrated
- Evidence of bony or cartilaginous pathology, osteochondral defects, or narrowing of joint space



NOTE: Several methods are described for demonstrating these structures. The prone position (Fig. 6.118) is an easier position for the patient to assume. The Holmblad kneeling method provides another option with a slightly different projection of these structures (Fig. 6.119). The disadvantage is that this position is sometimes uncomfortable for the patient. As a result of the advent of x-ray tables that can be raised and lowered, several Holmblad variations can be used to alleviate the pain of kneeling on both knees. These methods do not require a complete kneeling position, but they do require a cooperative ambulatory patient.

Technical Factors

- Minimum SID—40 inches (100 cm)
- IR size—8 × 10 inches (18 × 24 cm), or 14 × 17 inches (35 × 43 cm) for bilateral studies, portrait
- Grid
- kVp range: 70–80

Shielding Place lead shield over gonadal area. Secure around waist in kneeling position and extend shield down to midfemur level.

Patient Position

1. Place patient prone; provide a pillow for patient's head (Camp Coventry method).
2. Have patient kneel on x-ray table (Holmblad method).
3. Have patient partially standing, straddling x-ray table with one leg (Holmblad variation, requires elevation of examination table).
4. Have patient partially standing with affected leg on a stool or chair (Holmblad variation).

Fig 6.118 Camp Coventry method—prone position (40° to 50° flexion). (Position credited to Rosenberg TD, Paulos LE, Parker RD, et al: The 45° posteroanterior flexion weight-bearing radiograph of the knee, *J Bone Joint Surg Am* 70:1479, 1988.)

Fig 6.119 Holmblad method—kneeling position (60° to 70° flexion).

Evaluation Criteria

Anatomy Demonstrated: • Intercondylar fossa, articular facets (tibial plateaus), and knee joint space are demonstrated clearly (Figs. 6.120 and 6.121).

Position: • Intercondylar fossa should appear in profile, open without superimposition by patella. • **No rotation** is evidenced by symmetric appearance of distal posterior femoral condyles and superimposition of approximately half of fibular head by tibia. • Articular facets and intercondylar eminence of tibia should be well visualized without superimposition.

Exposure: • Optimal exposure should visualize soft tissue in knee joint space and an outline of the patella through the femur. • Trabecular markings of femoral condyles and proximal tibia should appear clear and sharp, with no motion.

Fig 6.120 PA axial projection.

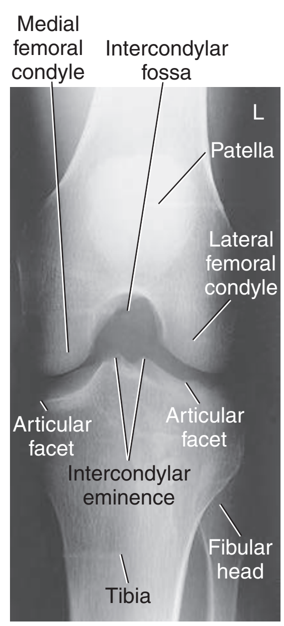


Fig 6.121 PA axial projection.

Part Position

1. Prone (Camp Coventry method) (see Fig. 6.118)
- Flex knee **40° to 50°;** place support under ankle.
- Center IR to knee joint, considering projection of CR angle.

2. Kneeling (Holmblad method) (see Fig. 6.119)
- With patient kneeling on "all fours," place IR under affected knee and center IR to popliteal crease.
- Ask patient to support body weight primarily on opposite knee.
- Place padded support under ankle and leg of affected limb to reduce pressure on injured knee.
- Ask patient to **lean forward slowly 20° to 30°** and to hold that position (results in 60° to 70° knee flexion).

3. Partially standing, straddling table (Holmblad variation)
- Lower examination table to a comfortable height for the patient, which is usually at the height of the knee joint.
- Ask patient to support body weight primarily on unaffected knee.
- Place affected knee over the bucky or IR.
- Ask patient to **lean forward slowly 20° to 30°** and to hold that position (results in 60° to 70° knee flexion).

4. Partially standing, affected leg on stool or chair (Holmblad variation) (Figs. 6.122 and 6.123)
- Adjust stool height to a comfortable height for the patient, which is usually at the height of the knee joint.
- Ask patient to support body weight primarily on the unaffected knee. **Provide a step stool for support.**
- Place the affected knee on the IR, while resting on the stool or chair.
- Ask patient to **lean forward slowly 20° to 30°** and to hold that position (results in 60° to 70° knee flexion).

CR

1. Prone: Direct CR **perpendicular to lower leg** (40° to 50° caudad to match degree of flexion).

2. Kneeling: Direct CR **perpendicular to IR and lower leg.**
- Direct CR to midpopliteal crease.

Recommended Collimation Collimate on four sides to knee joint area.

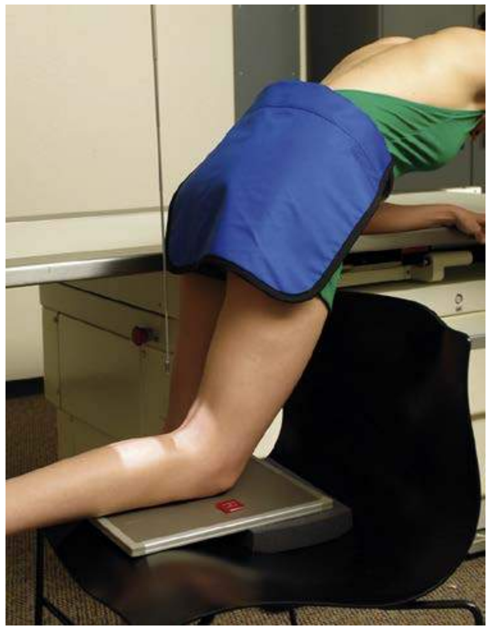


Fig 6.122 Holmblad variation—partially standing by examination table knee position (60° to 70° flexion).

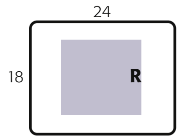
Fig 6.123 Holmblad variation—wheelchair version (60° to 70° flexion).

AP AXIAL PROJECTION: KNEE—INTERCONDYLAR FOSSA

BÉCLERE METHOD

Knee—Intercondylar Fossa
ROUTINE
- **PA axial**

SPECIAL
- **AP axial**



Clinical Indications

- Intercondylar fossa, femoral condyles, tibial plateaus, and intercondylar eminence demonstrated to look for evidence of bony or cartilaginous pathology
- Osteochondral defects, or narrowing of the joint space

NOTE: This is a reversal of the PA axial projection for patients who cannot assume the prone position. However, this is **not** a preferred projection because of distortion from the CR angle and increased part-IR distance. This projection also increases exposure for the gonadal region.

Technical Factors

- Minimum SID—40 inches (100 cm)
- IR size—8 × 10 inches (18 × 24 cm), landscape
- Grid
- kVp range: 65–80

Shielding Place lead shield over pelvic area, extending to midfemur.

Patient Position Place patient in supine position. Provide support under partially flexed knee with entire leg in anatomic position with no rotation.

Part Position

- Flex knee **40° to 45°,** and position support under IR as needed to place IR firmly against posterior thigh and lower leg, as shown in Figs. 6.124 and 6.125.
- Adjust IR as needed to center IR to midknee joint area.

CR

- Direct CR **perpendicular to lower leg** (approximately 40° to 45° cephalad).
- Direct CR to a point ½ inch (1.25 cm) distal to apex of patella.

Recommended Collimation Collimate on four sides to knee joint area.

Evaluation Criteria

Anatomy Demonstrated: • Intercondylar fossa, femoral condyles, tibial plateaus, and intercondylar eminence.
Position: • Center of four-sided collimation field should be to midknee joint area. • Intercondylar fossa should appear in profile, open without superimposition by patella. • Intercondylar eminence and tibial plateau and distal condyles of femur should be clearly visualized. • **No rotation** is evidenced by symmetric appearance of distal posterior femoral condyles and superimposition of approximately half of fibular head by tibia (Fig. 6.126).
Exposure: • Optimal exposure should visualize soft tissue in knee joint space and outline of patella through femur. • Trabecular markings of femoral condyles and proximal tibia should appear clear and sharp, with no motion.

Fig 6.124 AP axial—(40° flexion, CR perpendicular to lower leg approximately 40°cephalad).

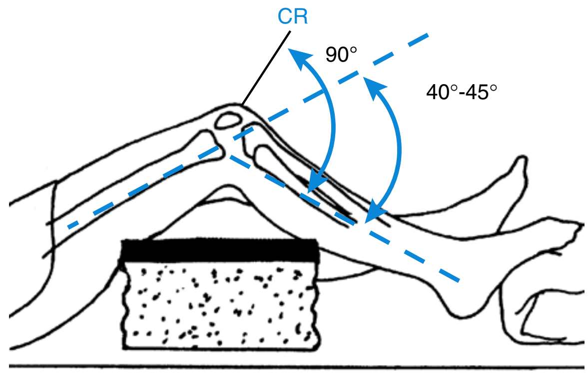


Fig 6.125 8 × 10 inches (18 × 24 cm) IR.

Fig 6.126 AP axial—40° flexion and CR angle.

PA PROJECTION: PATELLA AND PATELLOFEMORAL JOINT

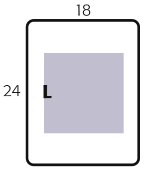
Patella
- PA
- Lateral
- Tangential

Clinical Indications

- Evaluation of patellar fractures before knee joint is flexed for other projections

Technical Factors

- Minimum SID—40 inches (100 cm)
- IR size—8 × 10 inches (18 × 24 cm), portrait
- Grid
- Nongrid for knee thickness <10 cm
- kVp range: 70–80 (increase 4 to 6 kVp from PA knee)



Shielding Shield radiosensitive tissues outside region of interest.

Patient Position Place patient in prone position, legs extended; provide a pillow for patient's head; place support under ankle and lower leg, with smaller support under femur above knee to prevent direct pressure on patella.

Part Position

- Align and center long axis of leg and knee to midline of table or IR (Fig. 6.127).
- **True PA:** Align interepicondylar line parallel to plane of IR. (This usually requires about **5° internal rotation of anterior knee.**)

CR

- CR is **perpendicular** to IR.
- Direct CR to **midpatella area** (which is usually at approximately the midpopliteal crease).

Recommended Collimation Collimate closely on four sides to include just the area of the patella and knee joint.

NOTE: With potential fracture of the patella, extra care should be taken **not to flex knee** and **provide support under thigh** (femur) so as not to put direct pressure on patellar area.

The projection also may be taken as an AP projection positioned similar to an AP knee if patient cannot assume a prone position.

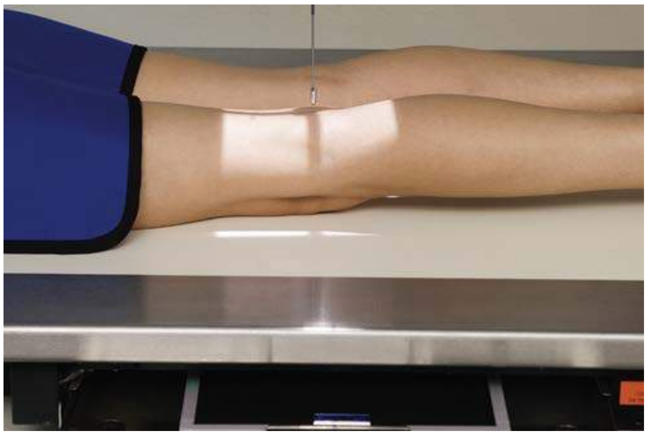


Fig 6.127 PA patella—CR 0° to midpatella.

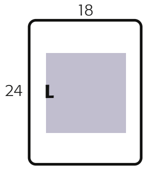
Fig 6.128 PA patella. (Courtesy Joss Wertz, DO.)

Evaluation Criteria

Anatomy Demonstrated: • Knee joint and patella are shown, with optimal recorded detail of patella because of decreased OID if taken as PA projection (Fig. 6.128).
Position: • **No rotation** is present, as evidenced by symmetric appearance of condyles. • Patella is centered to femur with correct slight internal rotation of anterior knee. • Patella is in center of collimated field.
Exposure: • Optimal exposure **without motion** visualizes soft tissue in joint area and clearly visualizes sharp bony trabecular markings and outline of patella as seen through distal femur.

LATERAL—MEDIOLATERAL PROJECTION: PATELLA

Patella
- PA
- Lateral
- Tangential



Clinical Indications
- Evaluation of patellar fractures in conjunction with the PA
- Abnormalities of patellofemoral and femorotibial joints

Technical Factors
- Minimum SID—40 inches (100 cm)
- IR size—8 × 10 inches (18 × 24 cm), portrait
- Grid
- Nongrid for knee thickness <10 cm
- kVp range: 70–80

Shielding Shield radiosensitive tissues outside region of interest.

Patient Position Place patient in lateral recumbent position, affected side down; provide a pillow for patient's head; provide support for knee of opposite limb placed behind affected knee.

Part Position
- Adjust rotation of body and leg until knee is in **true lateral** position (femoral epicondyles directly superimposed and plane of patella perpendicular to plane of IR).
- Flex knee **only 5° or 10°.** (Additional flexion may separate fracture fragments if present.)
- Align and center long axis of patella to CR and to centerline of table or IR (Fig. 6.129).

CR
- CR is **perpendicular** to IR.
- Direct CR to mid-patellofemoral joint.

Recommended Collimation Collimate closely on four sides to include just the area of the patella and knee joint.

NOTE: This also can be taken as a **horizontal beam lateral** with no knee flexion on a patient with severe trauma, as described in Chapter 15.

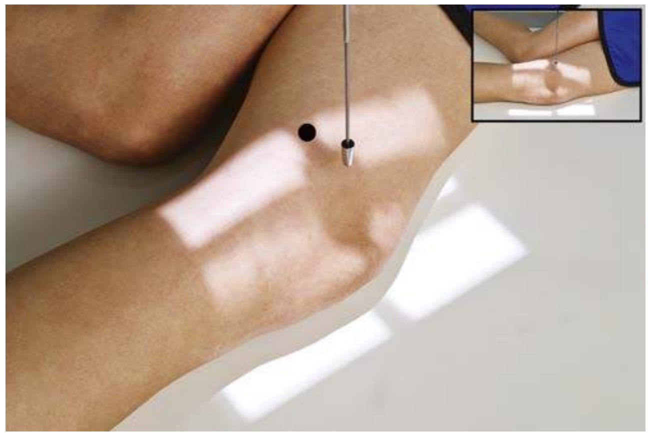


Fig 6.129 Lateral patella.

Fig 6.130 Lateral patella. (Courtesy Joss Wertz, DO.)

Evaluation Criteria
Anatomy Demonstrated: • Profile images of patella, patellofemoral joint, and femorotibial joint are demonstrated (Fig. 6.130).
Position: • **True lateral:** Anterior and posterior borders of medial and lateral femoral condyles should be directly superimposed, and patellofemoral joint space should appear open. • Centering and angulation are correct if patella is in center of the film and collimated field with joint spaces open.
Exposure: • Optimal exposure visualizes soft tissue detail and patella well without overexposure. • Trabecular markings of patella and other bones should appear clear and sharp.

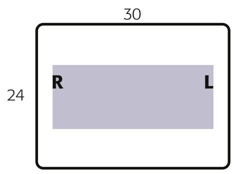
TANGENTIAL—AXIAL OR SUNRISE/SKYLINE PROJECTION: PATELLA

MERCHANT BILATERAL METHOD

Patella
- Tangential

Clinical Indications

- Subluxation of patella and other abnormalities of the patella and patellofemoral joint



Technical Factors

- SID—48 to 72 inches (120 to 180 cm) (increased SID reduces magnification)
- IR size—10 × 12 inches (24 × 30 cm), or 14 × 17 inches (35 × 43 cm) for bilateral studies of the knees; landscape
- Nongrid (grid is not required because of air gap caused by increased OID)
- kVp range: 70–80
- Some type of leg support and cassette holder should be used

Shielding Shield radiosensitive tissues outside region of interest.

Patient Position Place patient in the supine position with knees flexed 40° over the end of the table, resting on a leg support. Patient must be comfortable and relaxed for quadriceps muscles to be relaxed (see NOTE).

Part Position

- Place support under knees to raise distal femurs as needed so that they are parallel to tabletop.
- Place knees and feet together and secure lower legs together to prevent rotation and to allow patient to be totally relaxed.
- Place IR on edge against legs approximately 12 inches (30 cm) below the knees, **perpendicular** to x-ray beam (Figs. 6.131 and 6.132).

CR

- Angle CR caudad, **30° from horizontal plane** (CR 30° to femur). Adjust CR angle if needed for true tangential projection of patellofemoral joint spaces.
- Direct CR to a point midway between patellae.

Recommended Collimation Collimate **tightly** on all sides to patellae.

NOTE: Patient comfort and total relaxation are essential. The quadriceps femoris muscles must be relaxed to prevent subluxation of the patellae, wherein they are pulled into the intercondylar sulcus or groove, which may result in false readings.[6]

Fig 6.131 Merchant method—bilateral tangential, knees flexed 40°.

Fig 6.132 Adjustable-type leg support and IR holder. (Courtesy St. Joseph's Hospital and Medical Center, Phoenix, AZ.)

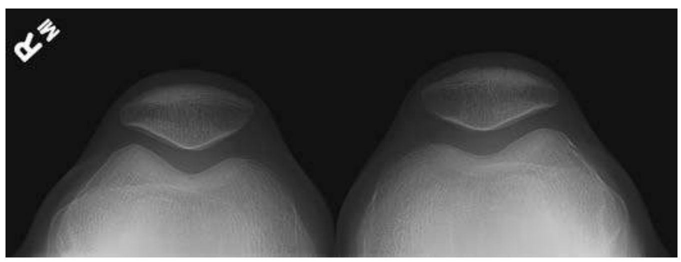


Fig 6.133 Merchant method—bilateral tangential. (Courtesy Joss Wertz, DO.)

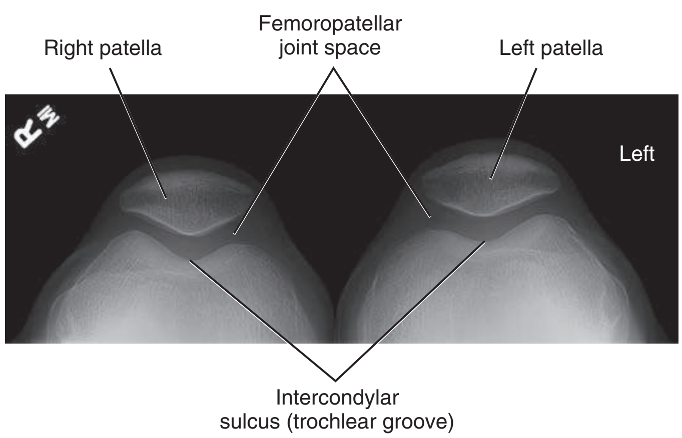


Fig 6.134 Merchant method—bilateral tangential. (Courtesy Joss Wertz, DO.)

Evaluation Criteria

Anatomy Demonstrated: • Intercondylar sulcus (trochlear groove) and patella of each distal femur should be visualized in profile with patellofemoral joint space open (Figs. 6.133 and 6.134).

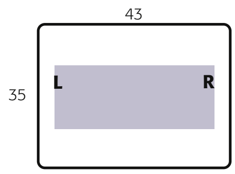
Position: • No rotation of knee is present, as evidenced by symmetric appearance of patella, anterior femoral condyles, and intercondylar sulcus. • Correct CR angle and centering are evidenced by open patellofemoral joint spaces.

Exposure: • Optimal exposure should clearly visualize soft tissue and joint space margins and trabecular markings of patellae. • Femoral condyles appear underexposed with only anterior margins clearly defined.

TANGENTIAL—AXIAL OR SUNRISE/SKYLINE PROJECTIONS: PATELLA

INFEROSUPERIOR AND HUGHSTON METHODS

Patella
- Tangential



Summary Three additional methods for tangential projections of the patellae and patellofemoral joints are described. Advantages and disadvantages of each are noted. **Both sides** generally are taken for comparison.

Technical Factors

- SID—40 to 48 inches (100 to 120 cm)
- IR size—14 × 17 inches (35 × 43 cm), landscape, for bilateral studies of the knees, or unilateral study 8 × 10 inches (18 × 24 cm), portrait
- Nongrid
- kVp range: 70–80

Inferosuperior Projection

- Place patient in supine position, legs together, with sufficient-sized support placed under knees for 40° to 45° knee flexion (legs relaxed).
- Ensure no leg rotation.
- Place IR on edge, resting on midthighs, tilted to be perpendicular to CR. Use sandbags and tape as shown, or use other methods to stabilize IR in this position. It is *not* recommended that patient be asked to sit up to hold IR in place because this may place patient's head and neck region in path of x-ray beam (Fig. 6.135).

CR

- Direct CR inferosuperiorly, at 10° to 15° angle from lower legs to be **tangential to patellofemoral joint.** Palpate borders of patella to determine specific CR angle required to pass through infrapatellar joint space.

NOTE 1: Major advantages of this method are that it does not require special equipment and a relatively comfortable position is required for the patient. Relaxation of the quadriceps muscle can be achieved with 40° to 45° knee flexion if properly sized support is placed under knees. The chief disadvantage is a potential problem with holding or supporting the IR in this position if the patient cannot cooperate fully.

Hughston Method[7]

This projection may be done bilaterally on one IR. Place patient in prone position, with IR placed under knee; slowly flex knee between 50° to 60° from full extension of lower leg (see NOTE 3); have patient hold foot with gauze, or rest foot on supporting device **(not on collimator)** (Fig. 6.136).

CR

- Angle CR 45° cephalad (CR tangential to patellofemoral joint).

NOTE 2: This is a relatively comfortable position for the patient, and relaxation of the quadriceps can be achieved. The major disadvantage is that this method requires the prone position, which is difficult for some patients. Additionally, image distortion is caused by awkward part alignment and difficulties encountered in angling the tube, which usually are caused by large collimators.

NOTE 3: Some authors suggest reduced flexion of only 20° to prevent the patella from being drawn into the patellofemoral groove, which may prevent detection of subtle abnormalities in alignment.[8]

Fig 6.135 Inferosuperior projection—40° to 45° flexion of knees.

Fig 6.136 Hughston method—50° to 60° flexion.

TANGENTIAL—AXIAL OR SUNRISE/SKYLINE PROJECTIONS: PATELLA

SETTEGAST METHOD

WARNING: This acute flexion of the knee should not be attempted until fracture of the patella has been ruled out by other projections.

- Place patient in prone position, with IR under knee; slowly flex knee to a **minimum of 90°;** have patient hold onto gauze or tape to maintain position (Fig. 6.137). An alternative seated variation is possible but with the risk of increased exposure to hands and thorax. Close collimation is required (Fig. 6.138).

CR

- Direct CR tangential to **patellofemoral joint space** (15° to 20° from lower leg).
- Minimum SID is 40 inches (100 cm).

NOTE 4: The major disadvantage of this method is that acute knee flexion tightens the quadriceps and draws the patella into the intercondylar sulcus, reducing the diagnostic value of this projection.[9]

Fig 6.137 Settegast prone method—90° flexion of knee.

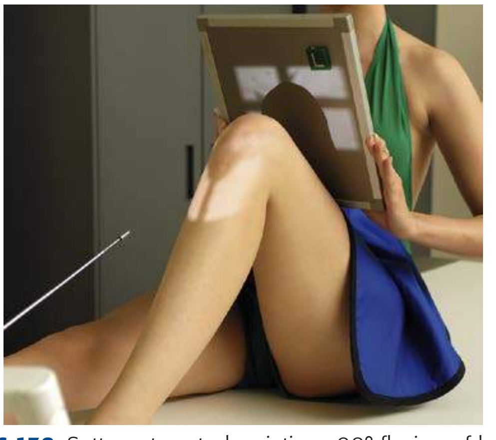


Fig 6.138 Settegast seated variation—90° flexion of knee.

SUPEROINFERIOR SITTING TANGENTIAL METHOD: PATELLA

HOBBS MODIFICATION

This method may be done bilaterally on one IR.

WARNING: This acute flexion of the knee should not be attempted until fracture of the patella has been ruled out by other projections.

- Place patient seated in a chair, with IR placed under knees resting on a step stool or support to help reduce OID; knees should be flexed with feet placed slightly underneath the chair (Fig. 6.139).

CR

- Align CR to be perpendicular to IR (tangential to patellofemoral joint).
- Direct CR to mid-patellofemoral joint.
- Minimum SID is 48 to 50 inches (120 to 125 cm) to reduce magnification because of increased OID.[10]

NOTE: The major advantage of this position is that the patient can be examined while sitting in a chair. This position also requires little manipulation of the x-ray tube. The major disadvantage is that it requires acute flexion of the knees.[10]

Evaluation Criteria

Anatomy Demonstrated: • Intercondylar sulcus (trochlear groove) and patella of each distal femur should be visualized in profile with patellofemoral joint space open. No superimposition of the patellae or tibial tuberosities[11] (Figs. 6.140 and 6.141).

Position: • No rotation of knee is present, as evidenced by symmetric appearance of patella, anterior femoral condyles, and intercondylar sulcus. • Correct CR angle and centering are evidenced by open patellofemoral joint space.

Exposure: • Optimal exposure should clearly visualize soft tissue and joint space margins and trabecular markings of patellae. • Femoral condyles appear underexposed with only anterior margins clearly defined.

Fig 6.139 Hobbs modification.

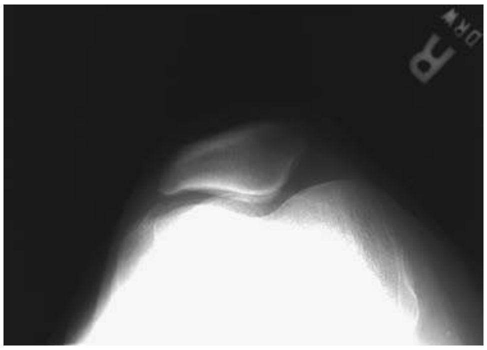


Fig 6.140 Hobbs modification—superoinferior sitting tangential method.

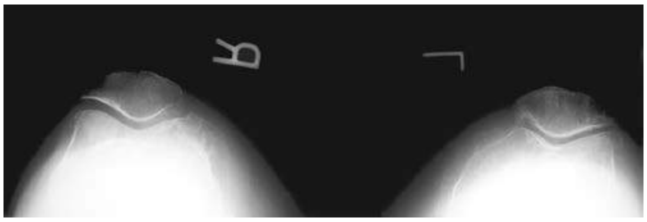


Fig 6.141 Superoinferior sitting tangential method.

RADIOGRAPHS FOR CRITIQUE

This section consists of an ideal projection (Image A) along with one or more projections that may demonstrate positioning and/or technical errors. Critique Figures C6.142 through C6.147. Compare Image A to the other projections and identify the errors. While examining each image, consider the following questions:

1. Is all essential anatomy demonstrated on the image?
2. What positioning errors are present that compromise image quality?
3. Are technical factors optimal?
4. Is there evidence of collimation and pre-exposure anatomic side markers visible on the image?
5. Do these errors require a repeat exposure?

Feedback for each set of images is located on the faculty Evolve site.

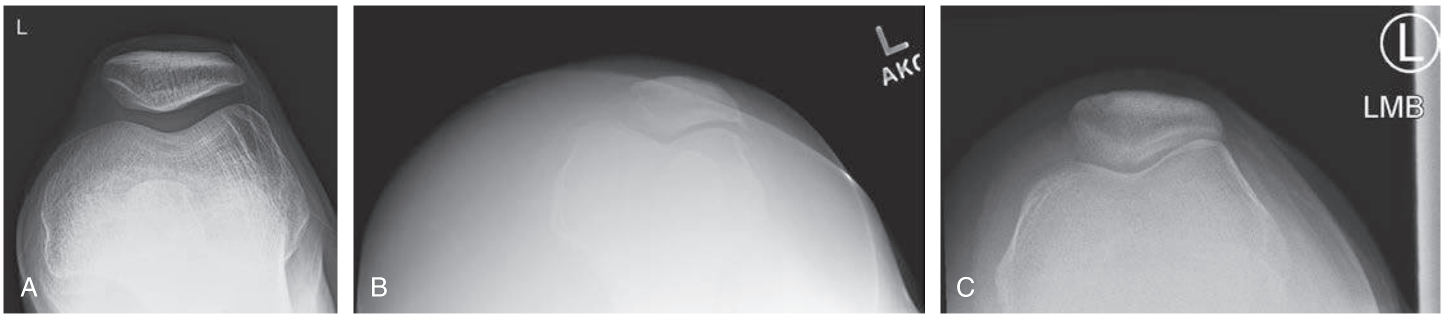


Fig C6.142 Bilateral tangential patella.

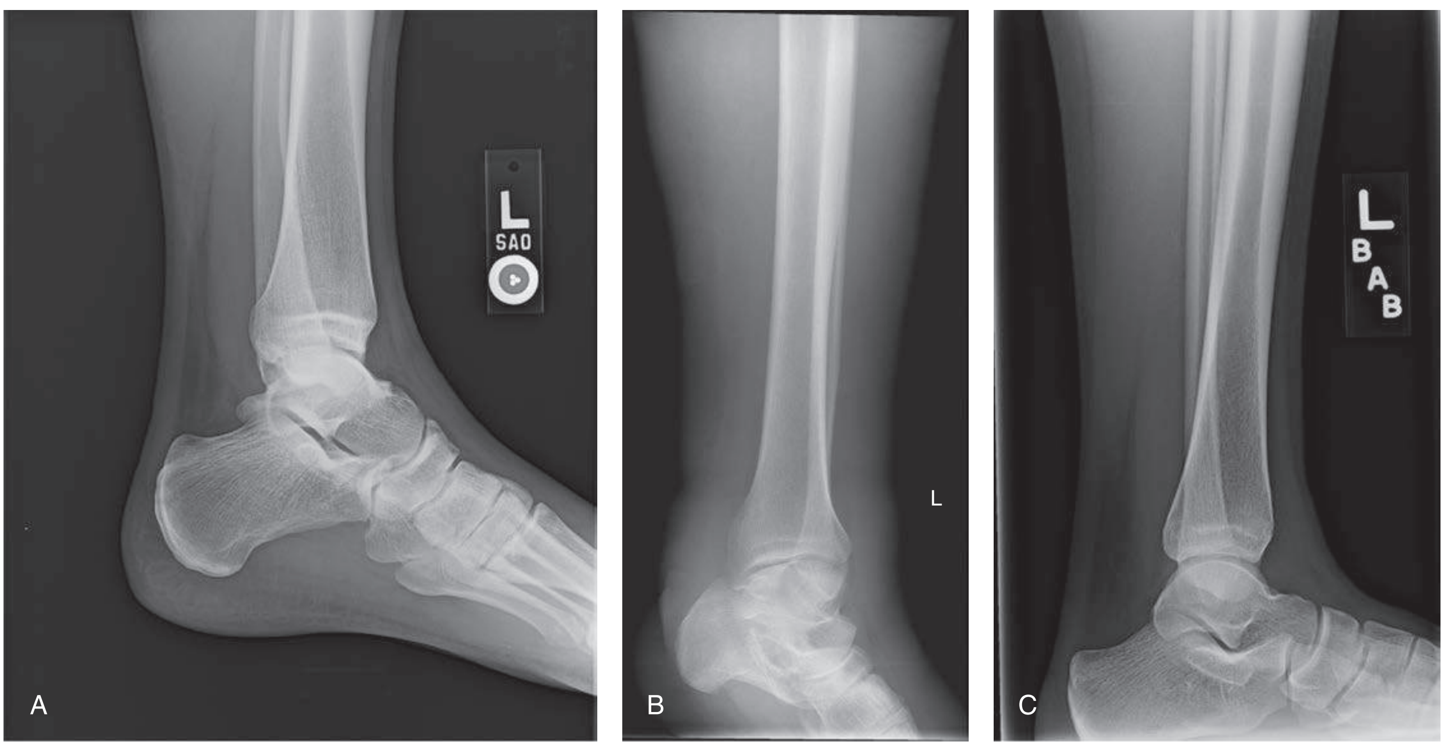


Fig C6.143 Lateral ankle.

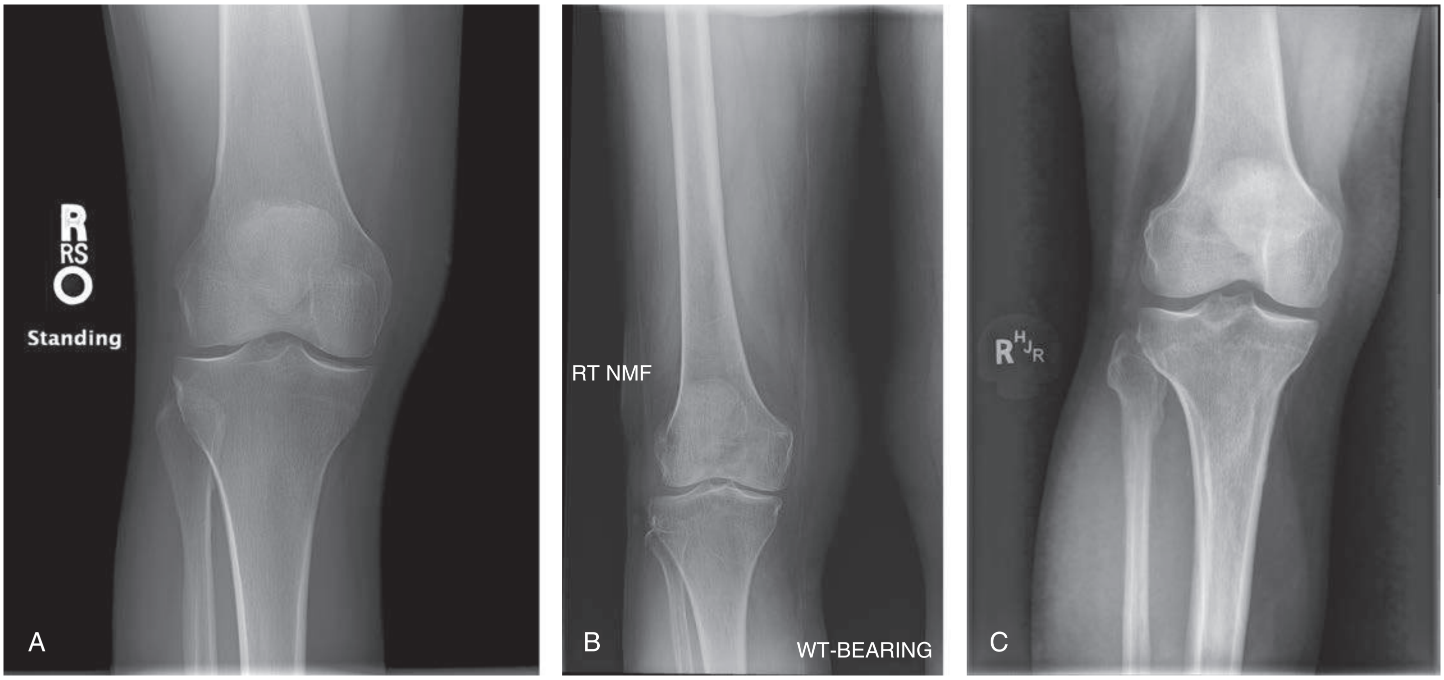


Fig C6.144 AP knee.

Fig C6.145 Lateral foot. (Courtesy Joss Wertz, DO.)

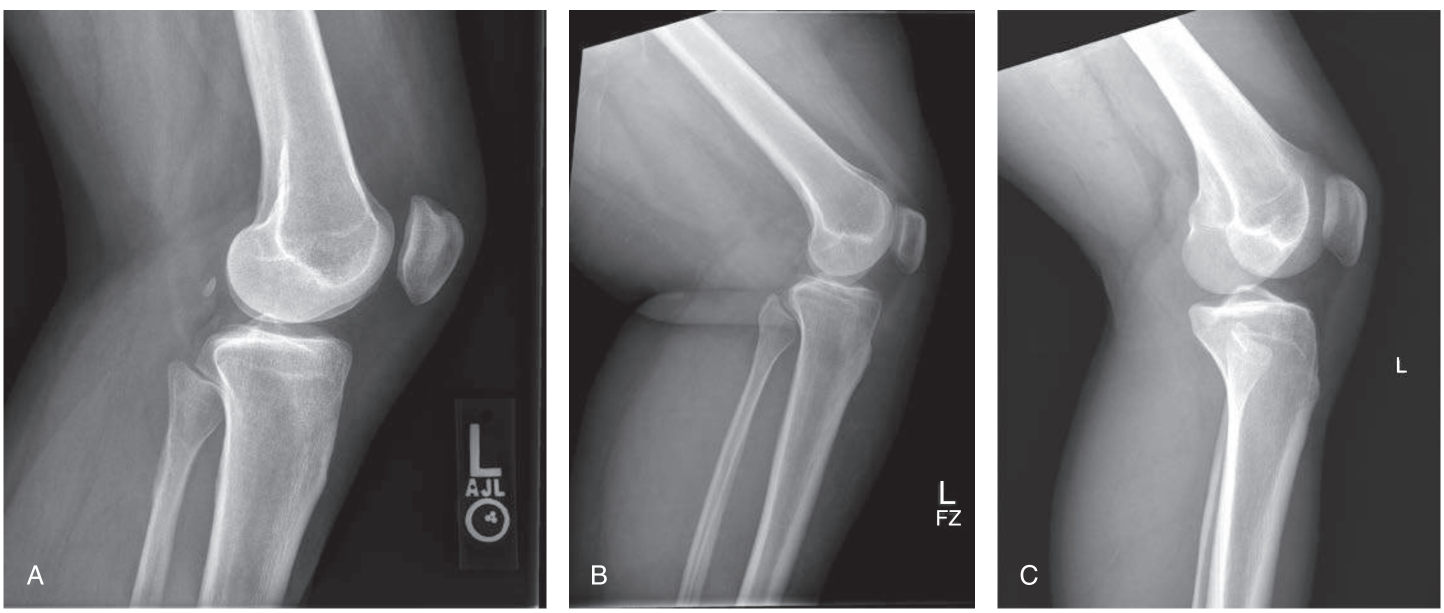


Fig C6.146 Mediolateral knee. (Courtesy Joss Wertz, DO.)

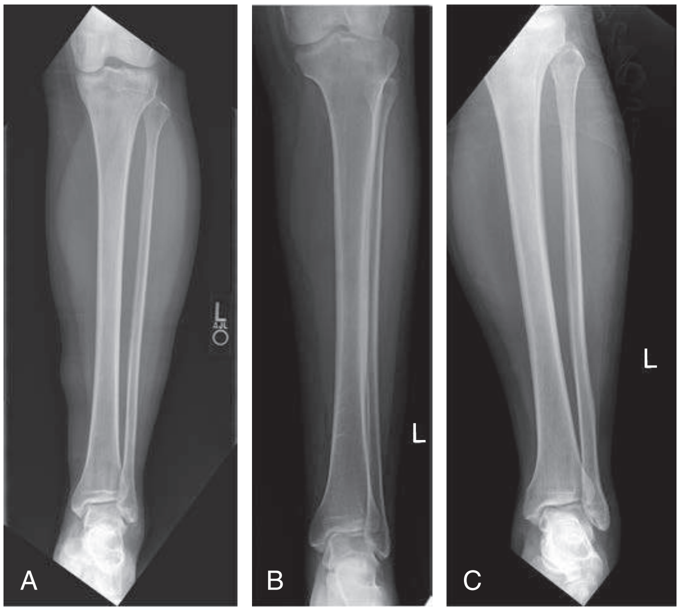


Fig C6.147 AP lower leg. (Courtesy Joss Wertz, DO.)